

Gamma and Zeta over the Surcomplex Numbers

Finite lifts, the Dirichlet algebra at infinity, Stirling and Hurwitz, phases on the horizontal tube, obstructions at infinite height, and transfer of the Riemann hypothesis

Merged research report from six manuscripts
AI-assisted, prepared for Vladimir Reshetnikov

September 2026

Abstract

Six manuscripts, written independently in September 2026, extend the Gamma and Riemann zeta functions to the surcomplex numbers $\mathbf{SC} = \mathbf{No}[z]$. This report merges them after a claim-by-claim comparison. No two of them contradict each other once four things are fixed: the domain, the exponential (phase), the summation notion and, for deformation results, the deformation class. The report states these conventions once and says, at every point where two sources seem to disagree, which of them separates the statements.

At finite arguments, Taylor–Laurent lifting preserves the exact zero and pole divisors. The finite Riemann hypothesis is therefore equivalent to the classical one. One unified stability theorem contains the deformation criteria of 04, 05 and 06: every finite zero of every J-symmetric real-infinitesimal deformation of the completed zeta function lies on the critical line if and only if RH holds and every nontrivial zero is simple. At positive infinite real part and finite imaginary part, every arithmetic function, of any growth, is realized injectively by a strongly summable Dirichlet series. Zeta has an exact Euler product there and no zeros. Its fibres over every nonzero infinitesimal value are found explicitly, and its jets are algebraically independent at dilations and at real shifts over fields containing Stirling Gamma values. An infinite-parameter Hurwitz function recovers the Stirling logarithm, and every phase gives Gamma, zeta and ξ on the whole horizontal tube $\{\text{Im } s \text{ finite}\}$, with reflection and the functional equation imposed as definitions on the left half of the tube (not a continuation or uniqueness theorem). The ξ divisor there is exactly the classical nontrivial divisor.

At infinite arguments, the report proves four obstructions. Reflection forces Gamma poles on the negative infinite periods of the phase, on the real axis. On the vertical strip of infinite height, reflection is incompatible with the Stirling modulus except at two exceptional abscissae for each height, and the Stirling Gamma itself, for every phase, fails reflection at every point of the strip. Coherent Dirichlet continuation replicates $\zeta^\#$ on resonant galaxies. Weak symmetric completions do not determine infinite zeros. Elementary extensions transfer RH and the classical zero theorems, and conditionally on RH and simplicity the nonreal de Bruijn–Newman zeros at negative infinitesimal time escape to infinite modulus. Several results needed for the reconciliation were established during the merge; they are marked as such and listed in the appendix.

Status. An unrefereed, AI-assisted research report. Complete arguments are given for the derived results, relative to the cited classical and surreal foundations. No theorem here has been verified in Lean or any other proof assistant. It does not prove the classical Riemann hypothesis, and it does not construct a canonical or unique global zeta or Gamma function on all of $\mathbf{No}[z]$. Priority is not certified for any consequence. The 2026 proportion results cited here are preprints.

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1 Scope, provenance and main results

1.1 The question has several answers

Replacing $\mathbb{C}$ by $\mathbf{SC} = \mathbf{No}[i]$ in a classical formula does not define Gamma or zeta. The classical formulas use limits, integrals, infinite products, analytic continuation, complex exponentiation and arithmetic counting. These operations need not have unique extensions to a non-Archimedean field, so the extension is part of the definition. Four constructions are kept separate throughout this report.

Construction	Domain and mechanism	What it establishes
Canonical finite lift	Finite arguments; classical germs evaluated at infinitesimals	Exact classical divisors; a finite RH equivalent to RH; a robust RH equivalent to RH plus simplicity
Series at infinity	Positive infinite real part; strong Hahn sums (Dirichlet, Stirling, Hurwitz)	An injective arithmetic algebra, Euler products, zero-freeness, inverse fibres, independence, exact Stirling and Lerch identities
A named global phase	Horizontal tube, right wing and galaxies; a phase Ψ_χ and a continuation rule	Gamma, zeta and ξ on the tube; forced infinite divisors; obstructions at infinite height
Elementary extension	A set-sized elementary model embedded in $\mathbf{No}[i]$	Transfer of RH and zero theorems at infinite height, with transported operations

1.2 Six sources

The report is built from six manuscripts, numbered 01–06 in order of arrival. All six date from September 2026 (06 gives only the month; the others are dated 22 September) and were written without knowledge of one another. Each cites a pinned commit of this repository.

No.	Title (subtitle shortened)	Pin	Distinctive contribution
01	<i>Gamma and Zeta over the Surcomplex Numbers: strong summation, a Hurwitz–Gamma bridge, phase choices, and exact inverse-zeta fibers</i> (31 pp.)	804c63c	inverse-zeta fibres (Section 9); the twisted family E_λ ; one-sided left zeta
02	<i>Gamma and Zeta at Surcomplex Scales</i> (29 pp.)	804c63c	independence at dilations with x , $L_{\text{St}}(x)$, $\Gamma_{\text{St}}(x)$ (Sections 10.1 and 10.2); phase families E_θ
03	<i>Gamma and Zeta on the Surcomplex Horizontal Tube</i> (34 pp.)	804c63c	Gamma, zeta, ξ on the whole tube for every phase (Part IV); independence at shifts (Section 10.3)
04	<i>Gamma, Zeta, and the Riemann Hypothesis over Surcomplex Numbers</i> (31 pp.)	905dd19	two-sign criterion; protected counts; flat Hurwitz modification; transported divisors; heat
05	<i>Gamma and Zeta over Surcomplex Fields</i> (28 pp.)	905dd19	reflection defect at $\frac{1}{2} + iT$; ratio cocycle; formal-series deformations; transfer
06	<i>Gamma and Zeta over the Surcomplex Numbers: canonical finite lifts, infinite Dirichlet series, phase obstructions, and the Riemann hypothesis</i> (28 pp.)	905dd19	the base text; robust RH; forced poles and resonant galaxies; weak completions; heat

The pins are $804c63c = 804c63c2c675be9fd4e5086460bfa795f17a6d86$ and $905dd19 = 905dd19954db43ac81f76f72036520d0eadc5710$. The six packages were placed in this directory in commit `e4f8848`. Manuscript 06 is the base text. It is the only source covering both the phase thread (Dirichlet algebra, Stirling and Hurwitz, phases, obstructions) and the RH thread (finite and robust RH, transfer, heat), and its exponential conventions are the most general. Where another source is strictly more general, its version is used: 03 for global objects on the tube, 01 and 02 for the Hurwitz parameter domain, 05 for formal-series deformations and the Stirling cocycle, and 04 for the one-parameter converse of the stability theorem. [Appendix A](#) records what each source contributed, where every result of every source is printed, where the merge had to choose, and every error found and fixed. The code and data of all six are shipped under the prefixes `01-inverse-zeta-`, `02-scale-separation-`, `03-horizontal-tube-`, `04-rh-transfer-`, `05-surcomplex-fields-` and `06-phases-and-rh-`; their manuscripts are not.

1.3 How results are attributed

Every result carries the sources that prove it in square brackets, for example $[01, 03]$. A result proved by several sources is printed once, with one proof. A genuinely different proof from another source is kept as a marked *second route*. The tag `[merge]` marks a statement established while reconciling the sources, not taken from any one of them. Each such statement has a complete proof here, and it was checked again for this report, re-derived in the same AI-assisted process; it is not refereed. Classical theorems and results of other reports in this collection are cited, not reproved, unless a source gives a different proof. *Derived theorem* describes a proof status. It does not certify historical novelty.

1.4 The main results

- (1) *Finite divisor rigidity and finite RH* ([Theorems 6.3](#) and [7.2](#)). The canonical Taylor–Laurent lift of a classical meromorphic function has exactly the classical zeros and poles on the finite plane $\mathcal{O}$, with no infinitesimally displaced zeros. Hence RH_{fin} is equivalent to RH.
- (2) *One stability theorem* ([Theorem 7.9](#)). RH together with simplicity of every nontrivial zero is equivalent to the statement that all finite zeros of every J-symmetric formal deformation of ξ , at every real infinitesimal parameter, lie on the critical line. It is also equivalent to the same statement for $\xi^\# \pm \eta$ with one positive infinitesimal η . The criteria of 04, 05 and 06 are special cases.
- (3) *The Dirichlet algebra on $\mathcal{T}_+$* ([Theorems 8.1](#) and [8.5](#)). On $\mathcal{T}_+ = \{\text{Re } s > \mathbb{R}, \text{Im } s \text{ finite}\}$, every complex arithmetic function gives a strongly summable Dirichlet series, and the result is an injective algebra homomorphism from the Dirichlet-convolution algebra. So ζ_D has an exact Euler product and no zeros.
- (4) *Inverse zeta* ([Theorems 9.4](#) and [9.8](#)). For every nonzero infinitesimal u , the solutions of $\zeta_D(s) = 1 + u$ in $\mathcal{T}_+$ form one explicit strongly summable branch for each ordinary integer.
- (5) *Algebraic independence* ([Theorems 10.8](#) and [10.14](#)). At a positive infinite real x , the values x , $L_{\text{St}}(x)$, $\Gamma_{\text{St}}(x)$ and all $\zeta_D^{(j)}(kx)$ are algebraically independent over $\mathbb{C}$. At a positive infinite monomial X , the shifted jets $\zeta_D^{(k)}(X + a)$ are independent over a field containing all real powers of X , its full slow Hahn field, the individual $e^{-\lambda X}$, and every $\Gamma_{\text{St}}(X + b)$ with $b \in \mathbb{R}$.
- (6) *Stirling and Hurwitz at infinity* ([Proposition 11.1](#) and [Theorems 11.2](#), [12.4](#) and [12.6](#)). These give exact recurrence, Gauss, translation and cocycle identities, Bernoulli values, and the

Lerch identity $\partial_s \zeta_H(s, A)|_{s=0} = L_{\text{St}}(A) - \frac{1}{2} \log 2\pi$.

- (7) *Phases on the horizontal tube* (Theorems 15.1, 16.2 and 16.4). For every phase Ψ_χ there are Gamma, zeta and ξ on $\mathcal{T}_h = \{\text{Im } s \text{ finite}\}$ with recurrence, reflection, Gauss and the functional equation (imposed as definitions on $\mathcal{T}_-$; not a continuation or uniqueness theorem). Gamma has poles on the negative infinite part of the period group $\mathcal{I}(\Psi_\chi)$, zeta has extra zeros on its double, and the ξ divisor is exactly the classical nontrivial divisor.
- (8) *Obstructions at infinite arguments* (Theorems 14.1, 17.1, 18.2 and 19.1). There are four. Reflection forces Gamma poles at negative infinite real points for every phase. On the vertical strip of infinite height, reflection is incompatible with the Stirling modulus except at two exceptional abscissae $\sigma_\pm(T)$ for each height, and $E \circ L_{\text{St}}$ fails reflection at every point of the strip, for every exponential E extending E_{fp} . Coherent Dirichlet continuation on resonant galaxies replicates $\zeta^\#$ and breaks the pointwise functional equation, for every phase with a resonance and every Gamma factor field-valued at $-2 - iT$. Local analyticity and the two ξ symmetries do not determine the infinite zeros.
- (9) *Transfer* (Theorems 20.3 and 22.2). RH in an elementary extension is equivalent to RH. The classical zero-free regions, counts, proportions and density bounds transfer with their quantifiers. Under RH and simplicity, the nonreal zeros of the transferred heat family at negative infinitesimal time have infinite modulus.

1.5 Relation to the collection

The collection had no zeta material before these sources. A search of docs/ at the placement commit finds no Riemann zeta function, Dirichlet series, Euler product, Hurwitz zeta or de Bruijn–Newman constant. Every source confirmed this independently. Everything about zeta in this report is therefore new to the collection.

Gamma functions over the surreals [26] constructs the Taylor–Stirling baseline L_0 (sec: baseline, eq:L0-finite, eq:L0-small, eq:L0-infinite). It proves the recurrence, formal uniqueness and Gauss formula (lem:formal-shift, eq:baseline-recurrence, lem:periodic, prop:gauss0), the complex log-Gamma on its tube $\{x > 0, y \text{ finite}\}$ (thm:complex-log, eq:complex-infinite, eq:log-infinite-complex), the phase lemma and the exact finite-phase domain Ω (lem:phase, thm:phase-domain, eq:omega-domain-intro), and the convex gauge family with the sharp classification (cor:headline, eq:explicit-gauge, thm:main, thm:independent). These results are cited here, not claimed. Where a source reproves one of them by a different route, the route is printed and marked. The report’s warning at its phase domain declines to choose a global exponential: “Such choices define a different problem.” This report takes up that different problem. It names the phase (Section 5), shows that outside Ω_∞ every value of Gamma is phase data (Proposition 11.6), and gives the vertical-strip obstruction (Theorem 17.1). The warning was a non-claim of that report, not an open question in it, so nothing there is re-scoped.

Trigonometry on the surcomplex plane [27] contains the Ehrlich–Kaplan canonical exponential Exp with kernel $2\pi i \mathbf{Oz}$ (trigonometry:def:globaltrig, trigonometry:thm:globalexp), the zero classes of Sin (trigonometry:cor:globalzeros), the classification of all phases Ψ_χ (trigonometry:thm:characters, trigonometry:per:prop:exponentials), the family at ω (trigonometry:ex:phases), unavoidable infinite periods (trigonometry:thm:infiniteperiods, trigonometry:per:thm:cofinal) and the normalized period group $\mathcal{I}(\Psi)$ (trigonometry:per:thm:dictionary). Sources 01, 02, 04 and 05 do not cite these results. 03 cites Ehrlich–Kaplan but not the report. 06 cites Ehrlich–Kaplan and the trigonometry report as a whole, for cis, the classification of phases and their infinite periods, but none of its labels. Four identifications follow. 01’s E_0 and 02’s $E_{\ell=0}$ are Exp; 03’s admissible phases are the Ψ_χ ;

03's period group Z_ϕ is $\mathcal{I}(\Psi_\chi)$; and the forced-pole theorems of 03 and 06 are consequences of `trigonometry.thm:infiniteperiods`.

Surcomplex analysis [28] contains the halo (`a:def:halo`), the canonical lift of holomorphic germs (`c:p4:def-lift`), the no-displaced-zeros proposition for holomorphic germs (`c:p4:liftzeros`), coherent sections (`c:p4:def-HU`), the canonical exponential (`e:def-canonicalexp`, `e:thm-exp`), the twisted family E_λ (`e:thm-twisted`) and the derivation obstruction (`b:derivation-obstruction`). 01's family E_λ is literally `e:thm-twisted`, with the same symbol and formula. According to the ledger `docs/FORMALIZATION.md`, some imported infrastructure is Lean-proved: the finite-variable and univariate clauses of the substitution lemma `a:cor:complexsub`, near-one binomial powers, and the fine derivative of evaluated ordinary power series. No theorem of this report is itself formalized.

Stale statements corrected. Sources 01–03 were written at 804c63c and 04–06 at 905dd19; both pins are ancestors of the placement commit. Since both pins the gamma-functions and analysis reports are unchanged. The trigonometry report had no section on noncanonical exponentials at either pin (`trigonometry:per:`); it now has one, and this report cites it. The sources' statements that the collection has no zeta material were true at their pins and remain true apart from this report. 06 itself adds that an empty search is not evidence of absence. The sources' descriptions of the neighbouring reports are accurate, and no source quotes a label or line number of the collection inaccurately. The one stale claim concerns a source's *own* construction: the remark after 01's left-zero theorem, corrected in [Remark 16.8](#).

2 Conventions and the table of notions

This section fixes one meaning for every symbol and word that the sources use in different ways. [Table 1](#) lists every renaming.

2.1 Scalars, orders and normal forms

We work in NBG with global choice, or in a universe-relative version of it. Every normal form, support, family and workspace used in a construction is set-sized, no sum over a proper class occurs, and every derivative order, multiplicity and summation index is an ordinary integer. The surcomplex field is $\mathbf{SC} = \mathbf{No}[i] = \{x + iy : x, y \in \mathbf{No}\}$ (01, 02 and 04 write $\mathbf{No}(i)$, which is the same field). The letter K is reserved for set-sized Hahn workspaces $K_\Gamma = \mathbb{C}((t^\Gamma))$, as in `docs/NOTATION.md`; 04's workspace is written K_Λ .

The modulus is $|x + iy| = (x^2 + y^2)^{1/2}$. Put $\mathcal{O} = \{z : |z| < r \text{ for some } r \in \mathbb{R}_{>0}\}$ and $\mathbf{m} = \{z : |z| < r \text{ for all } r \in \mathbb{R}_{>0}\}$, with real versions $\mathcal{O}_\mathbb{R}$ and $\mathbf{m}_\mathbb{R}$. Every $z \in \mathcal{O}$ is $z = \text{st } z + \varepsilon$ with $\text{st } z \in \mathbb{C}$ and $\varepsilon \in \mathbf{m}$. We write $x > \mathbb{R}$ for “ $x > r$ for every $r \in \mathbb{R}$ ”, $a \prec b$ for $a/b \in \mathbf{m}$, and $a \sim b$ for $a/b \in 1 + \mathbf{m}$. The *halo* of an ordinary set $U \subseteq \mathbb{C}$ is $U^\# = \text{st}^{-1}(U)$ (`a:def:halo`). The *monad* of a point b is $b + \mathbf{m}$, and its *galaxy* is $b + \mathcal{O}$, also for infinite b .

Monomials follow the collection's convention: $z = \sum_\gamma c_\gamma t^\gamma$ with $t^\gamma = \omega^{-\gamma}$, well-ordered support, and $v(z) = \min \text{supp } z$, so that larger valuation means smaller magnitude. Every real surreal splits as $x = \text{pp}(x) + \text{fin}(x)$. Here $\text{pp}(x)$ is the *purely infinite part*, the terms with positive Conway exponent, and $\text{fin}(x) \in \mathcal{O}_\mathbb{R}$ is the rest. Note that fin keeps the infinitesimal terms and is not the standard part. The purely infinite surreals form the real vector space Π (03's $\mathbb{J}$, 02's $\mathbf{No}^\uparrow$), and the omnific integers are $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$ ([29], `found:eq:omnific`). The unit circle is $\mathbb{T}(\mathbf{No}) = \{w \in \mathbf{SC} : w\bar{w} = 1\}$, and $S^1(\mathbb{C})$ is the ordinary circle inside it.

The real surreal exponential $\exp_{\mathbf{No}}$ is Gonshor's, with inverse log on $\mathbf{No}_{>0}$ [2, 4]. We use the standard fact that $\exp_{\mathbf{No}}$ maps Π isomorphically onto the group of Conway monomials [6,

Proposition 3.2]. In particular $e^{-X \log n}$ is a monomial for every $X \in \Pi$ and ordinary $n \geq 1$.

2.2 Three summation notions

- (a) An *ordinary sum* in $\mathbb{C}$.
- (b) A *strong Hahn sum* $\sum^H$ (Definition 4.1). A strong product $\prod^H$ means the strong sum of its expanded family of finite choices.
- (c) A *coefficientwise sum* $\sum^{\text{coef}}$: ordinary complex sums inside each Hahn coefficient, followed by a strong sum.

The three are different operations. At ordinary $\sigma > 1$ the family $(n^{-\sigma})$ is not strongly summable, since every member sits at exponent zero, but its ordinary sum is $\zeta(\sigma)$. At a positive infinite X the family (n^{-X}) is strongly summable. Every use of (c) in this report is labelled: the coefficientwise Dirichlet and Mellin formulas at finite s (Section 6.4), and the galaxy-wise summation of Theorem 18.2. Fine-topological limits are never used.

2.3 Exponentials and phases

The following table is the most important convention of the report. The sources use six names for four objects, and the word “canonical” in two senses.

Symbol here	Definition	Source names
$\text{Exp}_{\mathfrak{m}}, \text{Log}_{\mathfrak{m}}$	strong exponential on $\mathfrak{m}$, logarithm on $1 + \mathfrak{m}$	01, 03 $\text{exp}_{\mathfrak{m}}$; 04 $\text{exp}^\#$; 05 Exp_{fin} on $\mathfrak{m}$; 02 Exp on $\mathfrak{m}$
cis	$\text{cis}(r + \varepsilon) = e^{ir} \text{Exp}_{\mathfrak{m}}(i\varepsilon)$ on $\mathcal{O}_{\mathbb{R}}$, kernel $2\pi\mathbb{Z}$	all six; 04 cis_{fin} ; trigonometry:thm:polar
E_{fp} (finite-phase exponential)	$E_{\text{fp}}(x + iy) = \exp_{\mathbf{No}}(x) \text{cis}(y)$ for $x \in \mathbf{No}$, $y \in \mathcal{O}_{\mathbb{R}}$; kernel $2\pi i\mathbb{Z}$	01 Exp_{can} (“the canonical exponential”); 02 E_{f} ; 05 Exp_{fp} ; 06 E_{fp} ; gamma-functions E (eq:finite-exp)
Ψ_{χ}, E_{χ} (a phase)	$\Psi_{\chi}(p + f) = \chi(p) \text{cis } f$ for a character $\chi : \Pi \rightarrow \mathbb{T}(\mathbf{No})$; $E_{\chi}(x + iy) = \exp_{\mathbf{No}}(x) \Psi_{\chi}(y)$	03 admissible phase ϕ, E_{ϕ} ; 06 U ; trigonometry:thm:characters
Exp, Φ (canonical exponential)	$\chi = 1$: $\text{Exp}(x + iy) =$ $\exp_{\mathbf{No}}(x) \text{cis}(\text{fin } y)$; kernel $2\pi i\mathbf{Oz}$	Ehrlich–Kaplan; 01 E_0 ; 02 $E_{\ell=0}$; 03 E_{ϕ_0} ; 06 E_0 (“phase-forgetting”); trigonometry:thm:globalexp, e:thm-exp
E_{λ}	$\chi_{\lambda}(p) = e^{i\lambda c_{\omega}(p)}$, c_{ω} the coefficient of ω	01 E_{λ} ; e:thm-twisted ; trigonometry:ex:phases (χ_{τ})
$E_{\lambda, M}$	$\chi(p) = e^{i\lambda c_M(p)}$ for a fixed positive infinite monomial M	03 ϕ_{λ}
E_{θ}	$\chi(p) = \text{cis}(\theta(p))$, $\theta : \Pi \rightarrow \mathbb{R}$ real-linear	02 E_{ℓ} (renamed: 02’s ℓ collides with trigonometry’s $\ell = c_{\omega}$)
E	an additive-to-multiplicative $E : \mathbf{SC} \rightarrow \mathbf{SC}^{\times}$ extending E_{fp} ; the modulus law is <i>not</i> assumed	06 Proposition 6.1
$\mathcal{I}(\Psi)$	$\{x \in \mathbf{No} : \Psi(2\pi x) = 1\}$, the normalized period group; $\mathcal{I}(\Phi) = \mathbf{Oz}$	03 Z_{ϕ} ; trigonometry:per:thm:dictionary . Not trigonometry’s Z_E , which is the kernel-multiplier ring
$* \text{exp}, * \zeta, \dots$	operations transported from an elementary extension	04 exp_* ; 05 $* \text{exp}$; 06 M^*

Every E with $|E(z)| = \exp_{\mathbf{No}}(\operatorname{Re} z)$ that agrees with E_{fp} at finite imaginary arguments is E_χ for exactly one χ (`trigonometry:per:prop:exponentials`). All the families above are phases in this sense, and $E_\lambda = E_{\lambda,\omega}$.

Two meanings of “canonical”. “Canonical finite lift” refers to the explicit Taylor–Laurent rule ([Definition 6.1](#)). It is not a uniqueness claim among all functions agreeing with f on $\mathbb{C}$. The *canonical exponential* Exp is canonical in the sense of Ehrlich and Kaplan [6, §11]: it comes from the unique initial integer part $\mathbf{Oz}$. It is not unique under the analytic axioms, since every E_χ satisfies them (`trigonometry:thm:characters`), and initiality of the kernel-multiplier ring does not force it (`trigonometry:per:thm:robustness`). 01 calls E_{fp} “the canonical exponential”, and 02 calls its $\ell = 0$ member merely one choice. Both usages are replaced here by the table above. The attribute “phase-forgetting” (06) is kept only as a gloss for the property $\text{Exp}(iY) = 1$ for all $Y \in \Pi$.

Powers. For a positive surreal a and $s \in \mathbf{SC}$ we write $a^{-s} = E_{\text{fp}}(-s \log a)$ whenever $(\operatorname{Im} s) \log a$ is finite; this value does not depend on the phase. Otherwise $a^{-s} = E_\chi(-s \log a)$, and the phase χ is always displayed.

Logarithm. $\log z = \log |z| + i \operatorname{Arg} z$ with finite $\operatorname{Arg} z$. 01 and 05 take $\operatorname{Arg} \in (-\pi, \pi]$, and 06 takes $(-\pi/2, \pi/2)$ on $\operatorname{Re} z > 0$. The two conventions agree wherever both are used, namely on $\operatorname{Re} z > 0$. There $\operatorname{Log}(x + iy) = \log x + \operatorname{Log}_{\mathbf{m}}(1 + iy/x)$ when $y/x \in \mathbf{m}$, and $E_\chi(\operatorname{Log} z) = z$ for every phase, since $\operatorname{Arg} z$ is finite.

2.4 Domains

Symbol	Definition	Source names
$\mathcal{O}$	finite plane	all
$U^\#, c + \mathbf{m}, b + \mathcal{O}$	halo, monad, galaxy	02, 04 $U^\#$; 06 galaxy
$\mathcal{T}_+$	$\{\operatorname{Re} s > \mathbb{R}, \operatorname{Im} s \in \mathcal{O}_{\mathbb{R}}\}$	01 $\mathcal{D}_\infty$; 02 $\mathcal{R}_\infty = \mathcal{T}_\infty$; 03 $\mathcal{T}_+$; 04 Dirichlet chart; 05 $\mathcal{D}_{\mathbb{R}}$; 06 $\mathcal{D}_+$
$\mathcal{T}_-$	$\{\operatorname{Re} s < -\mathbb{R}, \operatorname{Im} s \in \mathcal{O}_{\mathbb{R}}\}$	01 $\mathcal{D}_{-\infty}$; 03 $\mathcal{T}_-$
$\mathcal{T}_h$	$\mathcal{T}_- \sqcup \mathcal{O} \sqcup \mathcal{T}_+ = \{\operatorname{Im} s \in \mathcal{O}_{\mathbb{R}}\}$, the horizontal tube	03 $\mathcal{T}$. <i>Not</i> gamma-functions’ tube $\mathcal{T} = \{x > 0, y \in \mathcal{O}_{\mathbb{R}}\}$
$\mathcal{W}$	$\{\operatorname{Re} s > \mathbb{R}\}$, any imaginary part	01 $\mathcal{R}_\infty$; 02 “ $\operatorname{Re} s > \mathbb{R}$ ”; 03 “canonical wing”
$\mathcal{S}_\infty$	$\{\operatorname{Re} z > 0, z \text{ infinite}\}$, the Stirling half-plane at infinity	05 “right infinite domain”; 06 “infinite right half-plane”
Ω_∞	$\{x + iy : x > \mathbb{R}, y \in \mathcal{O}_{\mathbb{R}}, y \log x \in \mathcal{O}_{\mathbb{R}}\} = \Omega \cap \{x > \mathbb{R}\}$	01 “horn”; 02 Ω_∞ ; gamma-functions Ω
$\mathcal{S}_A$	$\{s \in \mathcal{O} : (\operatorname{Im} s) \log A \in \mathcal{O}_{\mathbb{R}}\}$, for real $A > \mathbb{R}$	02 $\mathcal{S}_a$
$\mathcal{V}$	$\{s : 0 < \operatorname{Re} s < 1, \operatorname{Re} s \text{ finite}, \operatorname{Im} s \notin \mathcal{O}_{\mathbb{R}}\}$, the vertical strip of infinite height	04 “infinite critical heights”; 05 $\frac{1}{2} + iT$; 06 (galaxies $\pm iT + \mathcal{O}$ meet $\mathcal{V}$ but are not contained in it)

Note that $\mathcal{T}_+ \subset \mathcal{W} \subset \mathcal{S}_\infty$, that $\mathcal{T}_h \cap \mathcal{S}_\infty = \mathcal{T}_+$, and that $\mathcal{V}$ is disjoint from $\mathcal{T}_h$ and contained in $\mathcal{S}_\infty$. 05 used “right infinite” for both $\mathcal{S}_\infty$ and $\mathcal{T}_+$; the two names are now separated.

2.5 Functions and normalizations

Symbol	Definition, domain	Source names
$f^\#, \Gamma^\#, \zeta^\#, \xi^\#$	Taylor–Laurent lift on $\mathcal{O}$ (Definition 6.1)	01, 05 $f^\#$; 03 f^H
ξ	$\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, entire, $\xi(0) = \xi(1) = \frac{1}{2}$	all six
Ξ	$\Xi(w) = \xi(\frac{1}{2} + iw)$	04
H_t	Rodgers–Tao heat family, $H_0(z) = \frac{1}{8}\xi(\frac{1}{2} + \frac{iz}{2}) = \frac{1}{8}\Xi(z/2)$	04, 05, 06
$S_{\text{St}}, L_{\text{St}}$	$S_{\text{St}}(z) = \sum_{k \geq 1}^H b_k z^{1-2k}$, $b_k = \frac{B_{2k}}{2k(2k-1)}$; $L_{\text{St}}(z) = (z - \frac{1}{2}) \text{Log } z - z + \frac{1}{2} \log 2\pi + S_{\text{St}}(z)$ on $\mathcal{S}_\infty$	01, 05, 06 L_{St} ; 02 L_∞ ; 03 L, S ; 04 L_S ; gamma-functions L_0 , $\tilde{L}_0$
Γ_{St}	$\exp_{\mathbf{No}} L_{\text{St}}(x)$ at real $x > \mathbb{R}$	01 Γ_{St} ; 02 Γ_∞ ; 04 Γ_S ; gamma-functions Γ_0
$\Gamma_\chi, \Gamma_{\text{Exp}}$	$E_\chi \circ L_{\text{St}}$ on $\mathcal{S}_\infty$, extended to $\mathcal{T}_h$ in Section 15	01 $\Gamma_{\text{St}, \lambda}$; 02 Γ_ℓ ; 03 $\Gamma_\phi^+, \Gamma_\phi$
Γ_{fp}	$E_{\text{fp}} \circ L_{\text{St}}$ on Ω_∞	01 Γ_{St} on the horn; 02 Γ_{can}
$\Gamma_{E, \text{St}}$	$E \circ L_{\text{St}}$ on $\mathcal{S}_\infty$, E as in the last table	06
ψ_{St}	$\text{Log } z - \frac{1}{2z} - \sum_{k \geq 1}^H \frac{B_{2k}}{2k} z^{-2k}$	01 ψ_{St} ; 02 ψ_∞ ; 03 ψ_+
ζ_{D}	$\sum_{n \geq 1}^H n^{-s}$ on $\mathcal{T}_+$	01 Z_∞ ; 02 ζ_∞ ; 03 ζ_+ ; 04, 05, 06 ζ_{D}
ζ_χ, ξ_χ	zeta and ξ on $\mathcal{T}_h$ for the phase χ	03 ζ_ϕ, ξ_ϕ ; 01 $Z_{-, \lambda}$ on $\mathcal{T}_-$
$\zeta_\chi^{\mathcal{W}}$	$\sum_n^H E_\chi(-s \log n)$ on $\mathcal{W}$, where summable	02 ζ_ℓ ; 03 canonical wing
$\zeta_{\text{H}}(s, A)$	$\frac{A^{1-s}}{s-1} + \frac{A^{-s}}{2} + \sum_{k \geq 1}^H \frac{B_{2k}}{(2k)!}(s)_{2k-1} A^{1-s-2k}$	02 $\mathcal{H}_\infty$; 01 H ; 03 $\mathcal{Z}_\phi$; 04 Z_H ; 06 $Z_\infty(s, A)$
$\hat{\zeta}_{\text{H}}(s, A)$	$A^{s-1} \zeta_{\text{H}}(s, A) =$ $\frac{1}{s-1} + \frac{1}{2A} + \sum_{k \geq 1}^H \frac{B_{2k}}{(2k)!}(s)_{2k-1} A^{-2k}$	01 $\hat{H}$; 03 H
$Q_{a,b}, R_{a,b}$	ratio cocycle (Theorem 11.3); $Q_a := Q_{a,0}$	05; 03 R_a ; 02 eq. 8.6

The Bernoulli numbers satisfy $u/(e^u - 1) = \sum_n B_n u^n/n!$, so $B_1 = -\frac{1}{2}$, and $ue^{au}/(e^u - 1) = \sum_n B_n(a) u^n/n!$. The Stirling constant is $\frac{1}{2} \log 2\pi$ in every source. The Hurwitz letter is the only normalization trap: 03's H is 01's $\hat{H}$, not 02's $\mathcal{H}_\infty$ (which is ζ_{H}). Neither H nor Z_∞ is used for Hurwitz here, since H is the heat family and Z_∞ is 01's ζ_{D} . Likewise Γ_0 is not used: it means the gauge $a = 0$ in gamma-functions and the phase $\ell = 0$ in 02.

2.6 Tempting false readings

Each line pairs a false reading with the true statement it resembles.

- *False:* Exp is the unique global exponential. *True:* Exp is canonical in the integer-part sense. Every E_χ satisfies the same group, modulus and local laws ([Section 2.3](#)).
- *False:* Gamma is undefined outside Ω_∞ . *True:* the phase-free exponential $E_{\text{fp}} \circ L_{\text{St}}$ is defined exactly on Ω_∞ . Outside Ω_∞ every phase gives a value, and every such value is phase data ([Proposition 11.6](#)).
- *False:* zeta has a single pole at 1 everywhere. *True:* on $\mathcal{T}_h$, ζ_χ has a single pole at 1 ([Theorem 16.2](#)). On resonant galaxies outside $\mathcal{T}_h$, coherent Dirichlet continuation has poles at $1 \pm iT$ ([Theorem 18.2](#)).

Table 1: Renamings of source symbols. Normalizations are unchanged unless stated.

Source	Renamings
01	$K = \mathbf{No}(i) \rightarrow \mathbf{SC}; \mathcal{O}_K, \mathbf{m}_K \rightarrow \mathcal{O}, \mathbf{m}; \text{Exp}_{\text{can}} \rightarrow E_{\text{fp}}; E_0 \rightarrow \text{Exp} (E_\lambda \text{ kept}); \sin_\lambda \rightarrow \text{Sin}_{\chi_\lambda};$ $\text{horn} \rightarrow \Omega_\infty; \mathcal{D}_\infty \rightarrow \mathcal{T}_+; \mathcal{D}_{-\infty} \rightarrow \mathcal{T}_-; \mathcal{R}_\infty \rightarrow \mathcal{W} \text{ (for } L_{\text{St}}: \mathcal{S}_\infty); Z_\infty(s) \rightarrow \zeta_{\text{D}}(s);$ $Z_{-, \lambda} \rightarrow \zeta_{\chi_\lambda} \text{ on } \mathcal{T}_-; \widehat{H} \rightarrow \widehat{\zeta}_H, H(s, A) \rightarrow \zeta_H(s, A); H_A(S) \rightarrow \zeta_H(S, A) \text{ (a formal germ); } P_\lambda$ $\text{kept}; A_\omega \text{ kept (a constant, not a Hurwitz parameter); } \Gamma_{\text{St}, \lambda} \rightarrow \Gamma_{\chi_\lambda}; \Gamma_{\text{St}} \text{ on the horn}$ $\rightarrow \Gamma_{\text{fp}}; R(z) \rightarrow S_{\text{St}}(z); c_\Gamma \rightarrow \frac{1}{2} \log 2\pi; f^\# \rightarrow f^\#; [\omega]b \rightarrow c_\omega(b); \mathcal{E}_s \rightarrow \mathcal{D}_s.$
02	$\mathbf{No}(i) \rightarrow \mathbf{SC}; \mathcal{O}_{\mathbb{C}}, \mathbf{m}_{\mathbb{C}} \rightarrow \mathcal{O}, \mathbf{m}; E_{\text{f}} \rightarrow E_{\text{fp}}; \mathbf{No}^\uparrow \rightarrow \Pi; \ell \rightarrow \theta, E_\ell \rightarrow E_\theta, \pi_\ell \rightarrow \pi_\theta,$ $\Gamma_\ell \rightarrow \Gamma_{\chi_\theta}, \zeta_\ell \rightarrow \zeta_{\chi_\theta}^{\mathcal{W}}; \Gamma_0, \zeta_0 \rightarrow \Gamma_{\text{Exp}}, \zeta_{\text{Exp}}^{\mathcal{W}}; \mathcal{R}_\infty = \mathcal{T}_\infty \rightarrow \mathcal{T}_+; \Omega_\infty \text{ kept}; \mathcal{S}_a \rightarrow \mathcal{S}_A; \zeta_\infty \rightarrow \zeta_{\text{D}};$ $L_\infty, \Gamma_\infty, \psi_\infty \rightarrow L_{\text{St}}, \Gamma_{\text{St}}, \psi_{\text{St}}; \mathcal{H}_\infty \rightarrow \zeta_H; \Gamma_{\text{can}} \rightarrow \Gamma_{\text{fp}}; \mathcal{F}_x \rightarrow \mathcal{F}_x^{\text{Dir}}; \mathcal{D}_s, \mathcal{A} \text{ kept.}$
03	$\mathbf{SC} = \mathbf{No}[i] \text{ kept}; \mathbb{J} \rightarrow \Pi; \phi \rightarrow \Psi_\chi, E_\phi \rightarrow E_\chi; \phi_0 \rightarrow \Phi (E_{\phi_0} \rightarrow \text{Exp}); \phi_\lambda \rightarrow \Psi_{\chi_\lambda, M};$ $Z_\phi \rightarrow \mathcal{I}(\Psi_\chi); \sin_\phi, \cos_\phi \rightarrow \text{Sin}_\chi, \text{Cos}_\chi; \mathcal{T}, \mathcal{T}_\pm, \mathcal{O}_{\mathbb{C}} \rightarrow \mathcal{T}_h, \mathcal{T}_\pm, \mathcal{O}; f^H \rightarrow f^\#; \zeta_+ \rightarrow \zeta_{\text{D}};$ $L, S \rightarrow L_{\text{St}}, S_{\text{St}}; \Gamma_\phi^+ \rightarrow \Gamma_\chi \text{ on } \mathcal{T}_+; g_\phi \rightarrow g_\chi = 1/\Gamma_\chi; \chi_\phi(s) \rightarrow \mathcal{X}_\chi(s) \text{ (the letter } \chi \text{ is the}$ $\text{character}); R_a \rightarrow Q_a; H(s, A) \rightarrow \widehat{\zeta}_H, \mathcal{Z}_\phi(s, A) \rightarrow \zeta_H(s, A) \text{ with } A^w = E_\chi(w \log A);$ $F_X \rightarrow F_X^{\text{Hahn}}; M_X, E_X, E_X^\Gamma, T_X \text{ kept}; \psi_+ \rightarrow \psi_{\text{St}}.$
04	$\mathbf{No}(i) \rightarrow \mathbf{SC}; \mathbb{K} = \mathbb{C}((t^A)) \rightarrow K_\Lambda; \text{cis}_{\text{fin}} \rightarrow \text{cis}; \text{Dirichlet chart} \rightarrow \mathcal{T}_+; L_S, \Gamma_S \rightarrow L_{\text{St}}, \Gamma_{\text{St}};$ $Z_H \rightarrow \zeta_H; \exp_*, \Gamma_*, \zeta_*, \xi_*, N_* \rightarrow * \exp, * \Gamma, * \zeta, * \xi, * N; \Lambda_{\text{DN}} \text{ kept}; \mathcal{E}_q \text{ kept}; \chi \text{ of its}$ $\text{Proposition 9.1 is a set-sized character in the sense of } \text{trigonometry:thm:characters}.$
05	$\mathbb{K} \rightarrow \mathbf{SC}; \text{Exp}_{\text{fin}} \rightarrow \text{Exp}_{\mathbf{m}} \text{ on } \mathbf{m} \text{ and cis on } \mathcal{O}_{\mathbb{R}}; \text{Exp}_{\text{fp}} \rightarrow E_{\text{fp}}; \text{“right infinite domain”}$ $\rightarrow \mathcal{S}_\infty; \mathcal{D}_{\mathbb{R}} \rightarrow \mathcal{T}_+; \mathcal{E}_s \rightarrow \mathcal{D}_s; \text{gauge } \Gamma_\lambda = e^{h_\lambda} \Gamma \rightarrow e^{h_\lambda^{05}} \Gamma_{\text{St}} \text{ with } h_\lambda^{05} = \lambda e^{-\text{pp}}; \Xi \text{ of its}$ $\text{Theorem 9.1} \rightarrow \text{a symmetric function } F; \mathcal{A}, * \mathcal{A} \rightarrow \mathfrak{A}, \mathfrak{M}^*; \text{the four-point set } A \text{ in its proof}$ $\text{of Theorem 9.1} \rightarrow \mathcal{Q}.$
06	$\mathbb{K} \rightarrow \mathbf{SC}; E_0 \rightarrow \text{Exp}; U \rightarrow \Psi_\chi; \text{Sin}_U \rightarrow \text{Sin}_\chi; \mathcal{D}_+ \rightarrow \mathcal{T}_+; Z_\infty(s, A) \rightarrow \zeta_H(s, A); \Gamma_{E, \text{St}}$ $\text{kept}; \text{gauge } h_\lambda \rightarrow h_{a_\lambda} \text{ with } a_\lambda(m) = \lambda e^{-m} \text{ (gamma-functions } \text{cor:headline}); \text{class}$ $S \rightarrow \mathcal{S}_{\text{sym}}; \mathcal{M}^*, \xi^* \rightarrow \mathfrak{M}^*, * \xi; \text{parametric } \kappa \text{ of its Corollary 10.2} \rightarrow c_*.$
Local	Letters that are reserved above keep one meaning; local objects get their own: the Lagrange function Θ and Rodgers–Tao kernel Φ_{RT} (not the phase Φ); the de Bruijn–Newman constant Λ_{DN} (not von Mangoldt’s Λ or the group of K_Λ); word length ν, tuple length d and a finite index set $\mathcal{K}$ (not a workspace K); $F(x)$ written out in Lemma 10.10; the Platt–Trudgian height T_{PT} and the factor P_U of Theorem 19.1 (not the heat family H); monomials $\mathfrak{t}_n$, the map $\nu(P)$ and the prime projection pr with derivation δ_Z (not Möbius μ or zeros ρ); the Hartogs number $\mathfrak{h}$ and constants $\kappa_0, \kappa_s, c, c_*$; the field map ι_c and the $\mathbb{Q}$-linear maps $\mathfrak{s}, \mathfrak{s}'$ (not τ or σ); the imaginary part y in Theorem 9.4; the index letters i, b (not ℓ); the structure $\mathfrak{A}$ and the set $\mathcal{Q}$ (not $\mathcal{A}$); the Hermitian matrix Δ (not E).

- *False*: an infinite period is a resonance. *True*: every phase has infinite periods ([Theorem 5.7](#)). A simultaneous logarithmic resonance is stronger. It exists for every phase with ordinary-circle values, and fails for some other phases ([Proposition 5.10](#)).
- *False*: reflection holds for Stirling Gamma off the real axis. *True*: it holds on $\mathcal{T}_h$ by construction ([Theorem 15.1](#)). On the vertical strip a general prescription with Stirling moduli contradicts it except at two exceptional abscissae for each height, and $E \circ L_{St}$ itself fails it at every point of the strip, for every E extending E_{ip} ([Theorem 17.1](#)).
- *False*: $\zeta_H(s, A) = \sum_{n \geq 0} (A + n)^{-s}$ as a strong sum. *True*: that family is not strongly summable ([Warning 12.2](#)).
- *False*: a strong sum is a limit of partial sums. *True*: no set-indexed sequence that is not eventually constant converges in the fine topology ([Remark 4.7](#)).

3 How the six sources fit together

A claim-by-claim comparison of the six sources found no genuine contradiction. Every apparent conflict comes from a different domain, exponential, summation notion or deformation class. [Table 2](#) lists each apparent conflict and what separates the two sides. The same reconciliation is also printed at the point of use.

Table 2: Apparent conflicts and their resolution.

Apparent conflict	Resolution	Where
Divisor preservation (all six)	Identical statements; printed once	Theorem 6.3
Three stability criteria: 04 linear J-symmetric, one fixed $\eta > 0$; 05 formal series, both symmetries, $\tau > 0$; 06 linear, both symmetries, both signs	Subclasses of one class $\mathcal{D}_J$ (06's lies in 05's; 04's and 05's are incomparable); one theorem, with the one- η converse	Theorem 7.9
04: exact and shadow finite RH are “inequivalent in strength”	The exact form implies the shadow form, and is equivalent to it together with simplicity of every nontrivial zero. They can differ in truth value only if RH holds and some nontrivial zero is multiple, which is not known to happen	Remark 7.12
02 Warning 6.4 : zero-freeness is confined to $\mathcal{T}_+$; 02 Prop. 11.3 (E_θ) and 03 Prop. 13.1 (Exp only): zero-free on $\mathcal{W}$	The warning concerns the raw ζ_D ; the wing sums use a named phase	Section 13
03: Gamma with reflection on $\mathcal{T}_h$; 06: no Gamma with only ordinary poles satisfies reflection	The same fact: poles are forced on $\mathcal{I}(\Psi_\chi)$, and 03's pole set satisfies 06's requirement	Theorem 14.1
03: ζ_χ has a single pole at 1; 06: poles at $1 \pm iT$	Different domains ($\mathcal{T}_h$ versus resonant galaxies outside it) and different axioms	Theorem 18.2
03: reflection holds; 05: reflection fails at $\frac{1}{2} + iT$	$\frac{1}{2} + iT \notin \mathcal{T}_h$; 05's obstruction concerns $\mathcal{V}$, and 06's $\Gamma_{E,St}$ fails reflection at every point of $\mathcal{V}$	Theorem 17.1
01: $Z_{-, \lambda}(-\omega)$ zero or not, “nothing canonical decided”; 03: $\zeta_{Exp}(-\omega) = 0$	Phase choice. With Ehrlich–Kaplan's Exp ($\lambda = 0$) the zero is present; no functional equation forces it	Proposition 16.7 and Remark 16.8
03: poles on $\mathbf{Oz}_{\leq 0}$; 04, 06: poles at internal integers $-^*\mathbf{N}$	Different constructions: the period group of a phase versus the transported integer sort	Section 20.2

Apparent conflict	Resolution	Where
“Canonical exponential” means E_{fp} in 01 and Exp in 03, 06; 02: “no canonical choice”	Canonical in the integer-part sense is not unique	Section 2.3
Gamma non-uniqueness: 01, 02, 03, 06 (phase); 04, 05 (flat gauges); 06, gamma-functions (convex gauges)	Three independent mechanisms; the flat gauges of 04 and 05 break Gauss, the convex gauges keep it	Section 11.5
Hurwitz H in 01, 02, 03	03’s H is 01’s $\widehat{H}$; 02’s $\mathcal{H}_\infty$ is 01’s H ; all are ζ_H or $\widehat{\zeta}_H$	Section 2.5
Hurwitz domains $\mathcal{S}_A$ (01, 02), all finite s (03), germs (04, 06)	ζ_H is phase-free exactly on $\mathcal{S}_A$; elsewhere it depends on the phase and all identities hold for every phase	Section 12
Two distribution identities (01, 03)	The same identity after $A \mapsto A/q$	Theorem 12.4
02: independence at dilations over $\mathbb{C}$ and $\mathcal{F}_x^{\text{Dir}}$; 03: at shifts over E_X^Γ	Incomparable families and base fields; both kept; the combined statement is open	Section 10.4
05 “right infinite” twice	$\mathcal{S}_\infty$ versus $\mathcal{T}_+$	Section 2.4
Arg in $(-\pi, \pi]$ (01, 05) or $(-\pi/2, \pi/2)$ (06)	The conventions coincide on $\text{Re } z > 0$	Section 2.3
05’s cocycle $R_{a,b}$ as a Gamma ratio	A formal identity; the classical reading holds only away from the negative real direction	Remark 11.4
Heat variables: $\Xi(w) = \xi(\frac{1}{2} + iw)$ (04 stability) versus H_0 (all)	$H_0(z) = \Xi(z/2)/8$	Section 22
04: the heat theorem’s nonreal zero under RH; 05, 06: unconditional	Hypotheses attached clause by clause	Theorem 22.2
Platt–Trudgian simplicity: 05 uses it; 06 “no simplicity inferred”	The published statement includes simplicity; 06’s argument does not need it, and Corollary 21.2 (from 05) uses it	Section 21.2
$N(\sigma, T)$ counts $ \text{Im } \rho \leq T$ (Guth–Maynard, 04) or $0 < \text{Im } \rho \leq T$ (05, 06)	The one-sided count is smaller, so the bounds hold for it	Section 21.5
Guth–Maynard: arXiv (04, 06) versus <i>Annals</i> (05)	Published: <i>Ann. of Math.</i> 203 (2026)	Section 21.5
Proportions 0.417293, 0.407511, $> 5/12$, C_0	Different quantities (on the line versus simple on the line) and statuses (published versus preprint)	Section 21.4
Γ_0 in 02 and in gamma-functions	Renamed; neither is used	Section 2.5
01’s E_0 , 02’s $E_{\ell=0}$, 03’s E_{ϕ_0} , 06’s E_0	All equal Exp	Section 2.3
06: the transported phase is not Exp	A first-order two-frequency test separates them	Proposition 5.8
06 Remark 7.4: an infinite period need not give a resonance (unproved)	True: Proposition 5.10(c) constructs, using choice (a basis of $\mathbb{R}$ over $\mathbb{Q}(\log 3/\log 2)$), a phase with no resonance	Proposition 5.10
01: $E_{\text{fp}}(L_{\text{St}}(z))$ defined iff $y \log x$ finite; Exp is defined everywhere	The statement concerns E_{fp} ; the phase-free locus of all phases is exactly Ω_∞	Proposition 11.6

Part I

Foundations

4 Strong summation and infinitesimal evaluation

4.1 Strong summability

Normal forms are Conway's [1, 2].

Definition 4.1 (Strong summability [01–06]). A set-indexed family $(f_j)_{j \in I}$ of Hahn series (or of normal forms) is *strongly summable* if $\bigcup_j \text{supp } f_j$ is well ordered and every exponent lies in the support of only finitely many f_j . Its strong sum $\sum_j^H f_j$ is formed by finite coefficient addition at each exponent.

The second condition is substantive: a convergent ordinary series of nonzero complex constants is not strongly summable, because every member contributes at exponent zero. For $\sum_{n \geq 1} 2^{-n}$ the ordinary value 1 comes from ordinary analysis, not from Definition 4.1. Conversely $\sum_{n \geq 0}^H n! \varepsilon^n$ exists for every $\varepsilon \in \mathfrak{m}$, although the classical series has radius zero [01, 02]. The converse fails: nothing here licenses substituting a finite non-infinitesimal value into an arbitrary formal series [02].

Lemma 4.2 (Neumann's support lemma [3]). *Let S be a well-ordered subset of the positive cone of an ordered abelian group. The additive monoid $\langle S \rangle$ is well ordered, and each of its elements has only finitely many representations as a finite sum of elements of S , over all word lengths. If A, B are well ordered, so is $A + B$, and every element of $A + B$ has finitely many representations $a + b$.*

The lemma is imported; the collection proves it in the analysis report (`a:sec:neumann`) and in Lean (`Surreal/HahnSeries/Neumann.lean`, as recorded in `docs/FORMALIZATION.md`).

Lemma 4.3 (Infinitesimal substitution [01–06]). *Let $\varepsilon_1, \dots, \varepsilon_d \in \mathfrak{m}$ and $F \in \mathbb{C}[[X_1, \dots, X_d]]$, with no condition on the growth of its coefficients. Then $F(\varepsilon_1, \dots, \varepsilon_d)$ is defined by a strongly summable family. Evaluation respects sums and products, and it commutes with formal composition when every inner series has zero constant term. A series with nonzero constant term evaluates to a unit of $\mathcal{O}$. The same holds for a Laurent series with finitely many negative powers in one nonzero infinitesimal.*

Proof. Let S be the union of the supports of the nonzero ε_j , a finite union of well-ordered subsets of the positive cone. Every exponent in ε^α is a sum of $|\alpha|$ elements of S , so $\text{supp } \varepsilon^\alpha \subseteq \langle S \rangle$. By Lemma 4.2, a fixed exponent has finitely many representations over all word lengths. Assigning each letter of a word to one of the d variables leaves finitely many possibilities, so each exponent receives finitely many contributions. Products and compositions are regrouped exponent by exponent, and each coefficient identity is a finite algebraic identity. The constant term of the evaluation is $F(0)$. A finite principal part multiplies the evaluation by a fixed nonzero power of ε . $\square$

The finite-variable clauses are the analysis report's `a:cor:complexsub`, whose finite-variable part is recorded as Lean-proved in `docs/FORMALIZATION.md`. The strong exponential and logarithm

$$\text{Exp}_{\mathfrak{m}}(u) = \sum_{n \geq 0}^H \frac{u^n}{n!} \quad (u \in \mathfrak{m}), \quad \text{Log}_{\mathfrak{m}}(1 + u) = \sum_{n \geq 1}^H \frac{(-1)^{n+1} u^n}{n}$$

are mutually inverse bijections $\mathbf{m} \leftrightarrow 1 + \mathbf{m}$. They satisfy the exponential laws and commute with conjugation. For $a \in \mathbb{C}$ the near-one power is $(1 + u)^a = \text{Exp}_{\mathbf{m}}(a \text{Log}_{\mathbf{m}}(1 + u))$, equivalently the binomial series.

Lemma 4.4 (Mixed supports [02, 03, 04]). *Let (t_n) be distinct monomials whose valuations form a well-ordered set B , let $\varepsilon_1, \dots, \varepsilon_r \in \mathbf{m}$, and let $F_n \in \mathbb{C}[[T_1, \dots, T_r]]$ be arbitrary. Then the family $(t_n [T^\alpha] F_n \varepsilon^\alpha)_{n, \alpha}$ is strongly summable.*

Proof. With S as in the previous proof, $\text{supp } \varepsilon^\alpha$ lies in the $|\alpha|$ -fold sums of elements of S , hence in $\langle S \rangle$. Every exponent of the family lies in $B + \langle S \rangle$, which is well ordered by Lemma 4.2. A fixed exponent has finitely many decompositions $\beta + \gamma$ with $\beta \in B$ and $\gamma \in \langle S \rangle$. Distinctness of the t_n gives at most one n for each β , and for each γ only finitely many words, hence finitely many α . Complex coefficients do not change supports. $\square$

This spells out the “words” argument of 02’s Lemma 2.2(b), which treated ε^α as a family of monomials; the fix is the inclusion $\text{supp } \varepsilon^\alpha \subseteq \langle S \rangle$ just used (error 8 of Table 3).

Lemma 4.5 (Locally finite real exponents [01]). *Let $A \subseteq \mathbb{R}_{\geq 0}$ meet every bounded interval in a finite set, let $0 < u \in \mathbf{m}_{\mathbb{R}}$, and let $c_a \in \mathbb{C}$. Then $(c_a u^a)_{a \in A}$ is strongly summable, where $u^a = \exp_{\mathbf{No}}(a \log u)$. The same holds for a family with finitely many members at each exponent. For a nonzero $w \in \mathbf{m}$ put $w^a = E_{\text{fp}}(a \text{Log } w) = |w|^a \text{cis}(a \text{Arg } w)$; the family $(c_a w^a)_{a \in A}$ is again strongly summable.*

Proof. Write $u = c t(1 + \eta)$ with real $c > 0$, a Conway monomial $t < 1$ and $\eta \in \mathbf{m}_{\mathbb{R}}$. Then $u^a = c^a t^a (1 + \eta)^a$. The set $\{t^a : a \in A\}$ is reverse well ordered, with the order type of A . The binomial expansion of $(1 + \eta)^a$ has support in $\langle \text{supp } \eta \rangle$, independently of a . Lemma 4.2 gives both conditions of Definition 4.1, and finite multiplicities change nothing. For complex w write $\text{Arg } w = \theta_0 + \delta$ with $\theta_0 \in \mathbb{R}$ and $\delta \in \mathbf{m}_{\mathbb{R}}$. Then $w^a = |w|^a e^{ia\theta_0} \text{Exp}_{\mathbf{m}}(ia\delta)$, and the extra factor has support in $\langle \text{supp } \delta \rangle$, independently of a . $\square$

Lemma 4.6 (Evaluation of weighted series [merge]). *Let $(x_i)_{i \in I}$ be a set of formal variables with real weights $w_i > 0$ such that for every R only finitely many finitely supported multi-indices ν have $w(\nu) = \sum_i \nu_i w_i \leq R$. Let $\mathbb{C}[[x]]$ be the ring of all formal series $F = \sum_{\nu} f_{\nu} x^{\nu}$. Fix constants $c_i \in \mathbb{C}$ and $0 < u \in \mathbf{m}_{\mathbb{R}}$, or a nonzero $u \in \mathbf{m}$ with the powers of Lemma 4.5. Then*

$$\text{ev}(F) = \sum_{\nu}^H f_{\nu} c^{\nu} u^{w(\nu)}$$

is defined, and ev is a ring homomorphism. It is strongly additive: if (F_j) is a family in which each monomial x^{ν} has nonzero coefficient in only finitely many F_j , then the expanded family of all ev-terms is strongly summable and $\text{ev}(\sum_j F_j) = \sum_j^H \text{ev}(F_j)$. In particular $\text{ev}((1 + W)^a) = (1 + \text{ev } W)^a$ for every W with zero constant term and every $a \in \mathbb{C}$.

Proof. The exponents $w(\nu)$ form a set meeting each bounded interval finitely, with finitely many ν at each exponent, so Lemma 4.5 gives the first claim. The product of two strongly summable families is strongly summable, and its strong sum is the product of the sums. Grouping the product family by $\nu = \nu' + \nu''$, a finite grouping, gives $\text{ev}(FG) = \text{ev}(F) \text{ev}(G)$. For a formally summable family (F_j) , the terms at a fixed exponent come from finitely many ν and, for each ν , from finitely many j . So the doubly indexed family is strongly summable and may be regrouped either way. Finally $(1 + W)^a = \sum_j \binom{a}{j} W^j$ is formally summable because W^j has only monomials of weight at least $j \min_i w_i$. Applying the two previous properties and Lemma 4.3 to the infinitesimal $\text{ev } W$ gives $\text{ev}((1 + W)^a) = \sum_j^H \binom{a}{j} (\text{ev } W)^j = (1 + \text{ev } W)^a$. $\square$

This is the lemma that 01’s existence argument for the near-one solution of the inverse-zeta equation uses without statement (error 3 of Table 3). It is used in Lemma 9.3.

Remark 4.7 (Strong sums are not fine limits [01, 03, 05, 06]). The fine topology on $\mathbf{SC}$ has balls of every positive surreal radius (`a: def: fine`). A set-indexed net that is not eventually constant cannot converge in it, since a positive surreal smaller than every nonzero distance in a set of values exists by the cut property [01, 03]. In particular a sequence of ordinary complex partial sums converges to a finite surcomplex number only if it is eventually constant. Fine convergence forces the standard parts to converge ordinarily, and then an infinitesimal radius separates every other term [05]. A fixed set-sized Hahn field with its own valuation topology need not behave this way. No strong sum in this report is a fine limit, and convergence in one topology is never used to justify summation in another. The same holds for strong products.

4.2 Derivatives

All six sources differentiate *in the argument*: $f'(a)$ is the coefficient of h in the local expansion of $f(a + h)$, and $D_z \omega = 0$. This is not the Berarducci–Mantova derivation of scalars [8], which has $\partial \omega = 1$. The two notions are incompatible, not just normalized differently (`b: derivation-obstruction`): $\text{Exp}(i\omega) = 1$ cannot be differentiated with $\partial \omega = 1$. Every local germ in this report is a series in $\mathbf{SC}[[h]]$ that is strongly evaluable on some fine ball. By the analysis report’s germ theorem (`b: realization, b: germalgebra`), its formal derivative is its fine derivative on a small enough ball, and sums, products, quotients and chain rules hold. This settles the outline argument of 03’s Section 2.3 (error 19). The admissible ball can be much smaller than $\mathbf{m}$: near an infinite scalar A , an increment may need $h \log A \in \mathbf{m}$ [02, 03]. No source, and no statement here, asserts compatibility with the Berarducci–Mantova derivation.

5 Exponentials, phases and period groups

5.1 Finite angles and the finite-phase exponential

Proposition 5.1 (Polar form and finite-phase exponential [01, 02, 03, 05, 06]). *The map $\text{cis}(r + \varepsilon) = e^{ir} \text{Exp}_{\mathbf{m}}(i\varepsilon)$ ($r \in \mathbb{R}$, $\varepsilon \in \mathbf{m}_{\mathbb{R}}$) is a homomorphism from $(\mathcal{O}_{\mathbb{R}}, +)$ onto $\mathbb{T}(\mathbf{No})$ with kernel $2\pi\mathbb{Z}$. Every $w \in \mathbb{T}(\mathbf{No})$ is $\text{cis}\theta$ for a finite θ , unique modulo $2\pi\mathbb{Z}$. The finite-phase exponential $E_{\text{fp}}(x + iy) = \exp_{\mathbf{No}}(x) \text{cis}(y)$, for $x \in \mathbf{No}$ and $y \in \mathcal{O}_{\mathbb{R}}$, is a homomorphism with kernel $2\pi i\mathbb{Z}$ that commutes with conjugation and satisfies $|E_{\text{fp}}(z)| = \exp_{\mathbf{No}}(\text{Re } z)$. Every $z \neq 0$ has a finite principal argument, and $E_{\text{fp}}(\text{Log } z) = z$.*

Proof. If $w\bar{w} = 1$, then w is finite and $|\text{st } w| = 1$. Choose an ordinary argument θ_0 of $\text{st } w$ and put $v = e^{-i\theta_0} w \in 1 + \mathbf{m}$. Then $\text{Log}_{\mathbf{m}} v + \overline{\text{Log}_{\mathbf{m}} v} = \text{Log}_{\mathbf{m}}(v\bar{v}) = 0$, so $\text{Log}_{\mathbf{m}} v = i\delta$ with $\delta \in \mathbf{m}_{\mathbb{R}}$, and $w = \text{cis}(\theta_0 + \delta)$. A kernel element of cis has standard part in $2\pi\mathbb{Z}$, and its infinitesimal part is zero because $\text{Exp}_{\mathbf{m}}$ is injective. The statements about E_{fp} follow from the real exponential laws. For $z \neq 0$ apply the first part to $z/|z|$. $\square$

This is the trigonometry report’s `trigonometry:thm:polar`. The finite-phase exponential is the gamma-functions report’s E (`eq:finite-exp`) and agrees with the trigonometry report’s strip exponential (`trigonometry:thm:stripexp`). It exponentiates exactly the arguments with finite imaginary part.

5.2 Phases

Definition 5.2 (Phase [03, 06]). A *phase* is an additive-to-multiplicative homomorphism $\Psi : (\mathbf{No}, +) \rightarrow \mathbb{T}(\mathbf{No})$ that agrees with cis on $\mathcal{O}_{\mathbb{R}}$. Its exponential is $E_{\Psi}(x + iy) = \exp_{\mathbf{No}}(x) \Psi(y)$, with $\text{Sin}_{\Psi} z = (E_{\Psi}(iz) - E_{\Psi}(-iz))/(2i)$ and $\text{Cos}_{\Psi} z = (E_{\Psi}(iz) + E_{\Psi}(-iz))/2$.

Since $\mathbf{No} = \Pi \oplus \mathcal{O}_{\mathbb{R}}$, every phase is $\Psi_{\chi}(p+f) = \chi(p) \text{cis } f$ for a unique character $\chi : (\Pi, +) \rightarrow \mathbb{T}(\mathbf{No})$, and every character occurs (`trigonometry:thm:characters`). We write E_{χ} , Sin_{χ} , Cos_{χ} . 03's admissible phases and 06's phase homomorphisms U are exactly the Ψ_{χ} .

Proposition 5.3 (Exponential calculus [01, 02, 03]). *Each E_{χ} is a surjective homomorphism $(\mathbf{SC}, +) \rightarrow (\mathbf{SC}^{\times}, \cdot)$ with $|E_{\chi}(z)| = \exp_{\mathbf{No}}(\text{Re } z)$ that commutes with conjugation. It agrees with E_{fp} at finite imaginary part, and $E_{\chi}(z+h) = E_{\chi}(z) \text{Exp}_{\mathbf{m}}(h)$ for $h \in \mathbf{m}$. So E_{χ} is locally analytic with $E'_{\chi} = E_{\chi}$, and Sin_{χ} , Cos_{χ} satisfy the addition identities and have the usual derivatives. Conversely, every homomorphism E that agrees with E_{fp} at finite imaginary arguments and satisfies $|E(z)| = \exp_{\mathbf{No}}(\text{Re } z)$ is some E_{χ} .*

Proof. The group law and modulus are immediate. Conjugation follows from $\Psi(-y) = \Psi(y)^{-1} = \overline{\Psi(y)}$. For surjectivity write $w = |w| \text{cis } \theta$ by [Proposition 5.1](#). An infinitesimal increment lies in the finite summand, which gives the local law. For the converse, the modulus law makes $y \mapsto E(iy)$ a phase (`trigonometry:per:prop:exponentials`). $\square$

5.3 The canonical exponential and the explicit families

Definition 5.4 (The canonical exponential). The character $\chi = 1$ gives $\Phi(y) = \text{cis}(\text{fin } y)$ and

$$\text{Exp}(x + iy) = \exp_{\mathbf{No}}(x) \text{cis}(\text{fin } y), \quad \ker \text{Exp} = 2\pi i \mathbf{Oz}.$$

This is the canonical surcomplex exponential of Ehrlich and Kaplan [6, Theorem 11.1], obtained by reducing angles modulo the initial integer part $\mathbf{Oz}$. In the collection it is `trigonometry:def:globaltrig`, `trigonometry:thm:globalexp`, and in the analysis report `e:def-canonicalexp`, `e:thm-exp`. It satisfies $\text{Exp}(iY) = 1$ for every $Y \in \Pi$, and $\{z : \text{Sin } z = 0\} = \pi \mathbf{Oz}$ (`trigonometry:cor:globalzeros`).

06 and 03 use Exp and credit Ehrlich–Kaplan. 01's E_0 and 02's $E_{\ell=0}$ are also Exp , although neither source says so (errors 1 and 14). 04 mentions Ehrlich–Kaplan without using it, and 05 uses only E_{fp} (error 38). With Exp fixed, $\Gamma_{\text{Exp}} = \text{Exp} \circ L_{\text{St}}$ is the canonical-exponential Gamma on all of $\mathcal{S}_{\infty}$ ([Section 11.4](#)). It is canonical in the integer-part sense, not by uniqueness ([Section 2.3](#)).

Proposition 5.5 (Explicit phase families [01, 02, 03]). *For a positive infinite monomial M let $c_M(p) \in \mathbb{R}$ be the coefficient of M in $p \in \Pi$; for example c_{ω} is the coefficient of ω .*

- (a) [01] *For $\lambda \in \mathbb{R}$, the character $\chi_{\lambda}(p) = e^{i\lambda c_{\omega}(p)}$ gives $E_{\lambda}(a + ib) = \exp_{\mathbf{No}}(a) \text{cis}(P_{\lambda}(b))$ with $P_{\lambda}(b) = \text{fin } b + \lambda c_{\omega}(b)$. Then $E_0 = \text{Exp}$ and $E_{\lambda}(i\omega) = e^{i\lambda}$.*
- (b) [03] *For fixed M , $\chi_{\lambda,M}(p) = e^{i\lambda c_M(p)}$ gives $E_{\lambda,M}$. Its normalized period group is $\mathcal{I} = \{x : \text{fin } x + \lambda c_M(x) \in \mathbb{Z}\}$.*
- (c) [02] *For a real-linear $\theta : \Pi \rightarrow \mathbb{R}$, $\chi_{\theta} = \text{cis} \circ \theta$ gives $E_{\theta}(a + ib) = \exp_{\mathbf{No}}(a) \text{cis}(\text{fin } b + \theta(\text{pp } b))$. Coefficient functionals $\theta = r c_M$ ($r \in \mathbb{R}$) are explicit and need no choice.*

In each case the character takes values in the ordinary circle $S^1(\mathbb{C})$.

Proof. c_M and fin are additive and real-linear, so each formula defines a character. The value at $i\omega$ follows from $\text{fin } \omega = 0$ and $c_{\omega}(\omega) = 1$. For (b), $\Psi(2\pi x) = \text{cis}(2\pi(\text{fin } x + \lambda c_M(x)))$. $\square$

(a) is literally the analysis report's twisted family `e:thm-twisted`, with the same symbol and formula, and the trigonometry report's family at ω (`trigonometry:ex:phases`). All three families are subfamilies of `trigonometry:thm:characters`.

Proposition 5.6 (Prescribed prime phases on a set-sized subspace [04]). *Let T be positive infinite, and let $V \subset \mathbf{No}$ be a set-sized $\mathbb{Q}$ -subspace containing $\mathbb{R}$ and every $T \log p$ for ordinary primes p . Let V_0 be its subspace of finite elements. For arbitrary ordinary angles ϑ_p there is a homomorphism $\chi : V \rightarrow \mathbb{T}(\mathbf{No})$ with $\chi|_{V_0} = \text{cis}$, $\chi(T \log p) = e^{i\vartheta_p}$, and $\chi(y + h) = \chi(y) \text{cis } h$ for $h \in V_0$.*

Proof. The classes of the $T \log p$ in V/V_0 are $\mathbb{Q}$ -independent: a nonzero rational combination of the $\log p$ is a nonzero real by unique factorization, and multiplying it by T gives an infinite element. Define a $\mathbb{Q}$ -linear map on $V_0 \oplus \text{span}_{\mathbb{Q}}\{T \log p\}$ fixing V_0 and sending $T \log p$ to ϑ_p . Extend it by a basis of the quotient (choice) to $P : V \rightarrow V_0$, and set $\chi = \text{cis} \circ P$. $\square$

These are local-data extensions. 04 does not claim they satisfy Ehrlich–Kaplan’s initial-integer-part characterization, and they do not contradict it. The proposition shows that finite-angle data and finite translation laws leave the phases at infinite arguments undetermined.

5.4 Period groups and forced infinite periods

Theorem 5.7 (Period groups [03]). *For a phase $\Psi = \Psi_\chi$ let $\mathcal{I}(\Psi) = \{x \in \mathbf{No} : \Psi(2\pi x) = 1\}$.*

- (a) *$\mathcal{I}(\Psi)$ is an additive subgroup with $\mathcal{I}(\Psi) \cap \mathcal{O}_{\mathbb{R}} = \mathbb{Z}$. Every coset $x + \mathcal{O}_{\mathbb{R}}$ meets it in exactly one coset of $\mathbb{Z}$. In particular it has positive and negative infinite elements.*
- (b) *$\text{Sin}_\chi(\pi z) = 0 \iff z \in \mathcal{I}(\Psi)$, and $\text{Sin}_\chi(\pi z/2) = 0 \iff z \in 2\mathcal{I}(\Psi)$. These zeros are real and simple, and $\text{Cos}_\chi(\pi x) = \pm 1$ for $x \in \mathcal{I}(\Psi)$.*
- (c) *$\mathcal{I}(\Phi) = \mathbf{Oz}$, and for 01’s family, $\mathcal{I}(\Psi_{\chi_\lambda}) = \{x : P_\lambda(x) \in \mathbb{Z}\}$.*

Proof. (a) The finite part is the kernel statement of Proposition 5.1. For $x \in \mathbf{No}$ choose a finite θ with $\text{cis } \theta = \Psi(2\pi x)$; then $x - \theta/2\pi \in \mathcal{I}(\Psi)$ differs from x by a finite amount. Two elements of $\mathcal{I}(\Psi)$ in one finite coset differ by a finite period, hence by an ordinary integer. (b) $\text{Sin}_\chi(\pi z) = 0$ means $E_\chi(2\pi iz) = 1$. For $z = x + iy$ the modulus is $\exp_{\mathbf{No}}(-2\pi y)$, which equals 1 only if $y = 0$, and then $x \in \mathcal{I}(\Psi)$. There $E_\chi(i\pi x)^2 = 1$, so $\text{Cos}_\chi(\pi x) = \pm 1$ and the derivative $\pi \text{Cos}_\chi(\pi x)$ is nonzero. Replace z by $z/2$ for the second statement. (c) $\Phi(2\pi x) = \text{cis}(2\pi \text{fin } x)$, so $\mathcal{I}(\Phi) = \{x : \text{fin } x \in \mathbb{Z}\} = \Pi \oplus \mathbb{Z} = \mathbf{Oz}$; the second claim is Proposition 5.5(b) with $M = \omega$. $\square$

Part (a) is `trigonometry:thm:infiniteperiods` in normalized form, and $\mathcal{I}(\Psi)$ is the normalized period group of `trigonometry:per:thm:dictionary`. (03 writes Z_ϕ ; the name Z_E in the trigonometry report denotes the kernel-multiplier ring, a different object.) By `trigonometry:per:thm:cofinal`, every phase on $\mathbf{No}$ even has cofinally large periods all of whose rational multiples are periods. (b) for Φ is `trigonometry:cor:globalzeros`.

5.5 Resonances

Proposition 5.8 (Two-frequency test [06]). *The statement $\text{cis}(t \log 2) = 1 \wedge \text{cis}(t \log 3) = 1 \Rightarrow t = 0$ holds for ordinary real t . It remains true in every elementary extension of the real field with its cosine and sine. It fails for the phase Φ of Exp: every nonzero $T \in \Pi$ satisfies both antecedents.*

Proof. Over $\mathbb{R}$, $t \log 2 = 2\pi k$ and $t \log 3 = 2\pi l$ give $3^k = 2^l$, so $k = l = 0$ by unique factorization. The implication is a first-order sentence in the named functions, so it transfers. For Φ , $\Phi(T \log n) = \text{cis}(\text{fin}(T \log n)) = 1$. $\square$

So Exp cannot be the full complex exponential of an elementary extension. The structure behind this is the following notion.

Definition 5.9 (Resonance). A nonzero $T \in \mathbf{No}$ is a *resonance* of the phase Ψ if $\Psi(T \log n) = 1$ for every ordinary integer $n \geq 1$.

Every nonzero $T \in \Pi$ is a resonance of Φ . A resonance is a nonzero element of $\bigcap_{n \geq 2} (2\pi / \log n) \cdot \mathcal{I}(\Psi)$, and it is much rarer than a period. The following three facts are established in this merge. They settle the last sentence of 06's Remark 7.4 ("The existence of an infinite period of U alone does *not* imply such a simultaneous logarithmic resonance", where 06's U is Ψ_χ), which 06 stated without proof (error 39).

Proposition 5.10 (Existence and failure of resonances [merge]).

- (a) If $\chi(\Pi) \subseteq S^1(\mathbb{C})$, then Ψ_χ has resonances in Π of arbitrarily large modulus. This covers Exp , every E_λ , every $E_{\lambda, M}$ and every E_θ .
- (b) If $\Psi(p + f) = \text{cis}(\varphi(p) + f)$ with $\varphi : \Pi \rightarrow \mathcal{O}_\mathbb{R}$ real-linear, then $T = p - \varphi(p)$ is a resonance for every nonzero $p \in \Pi$. Its galaxy is $iT + \mathcal{O} = ip + \mathcal{O}$.
- (c) Assume choice for a basis of $\mathbb{R}$ over $\mathbb{Q}(\log 3 / \log 2)$. There is a phase Ψ with no nonzero $T \in \mathbf{No}$ such that $\Psi(T \log 2) = \Psi(T \log 3) = 1$. In particular Ψ has no resonance. It has infinite periods, as every phase does.

Proof. (a) Let $B \in \mathbf{No}$. Let $\mathfrak{h}$ be the Hartogs number of the set $S^1(\mathbb{C})^\mathbb{P}$, where $\mathbb{P}$ is the set of ordinary primes; $\mathfrak{h}$ is an ordinal that does not inject into that set. Choose a positive surreal g with $\omega^g > \omega(|B| + 1)$, and let $G \subset \Pi$ be the real span (finite combinations) of the monomials $\omega^{g+\beta}$, $\beta < \mathfrak{h}$. A nonzero element of G has a leading term $r\omega^{g+\beta}$ with $r \in \mathbb{R}^\times$ and $\beta \geq 0$, so its modulus exceeds $\frac{1}{2}|r|\omega^g > \frac{1}{2}|r|\omega(|B| + 1) > |B|$. The map $G \rightarrow S^1(\mathbb{C})^\mathbb{P}$, $P \mapsto (\chi(P \log p))_p$, is a group homomorphism, because χ is a character and $P \mapsto P \log p$ is additive into Π . It cannot be injective, since $\beta \mapsto \omega^{g+\beta}$ injects $\mathfrak{h}$ into G . So there is a nonzero $P \in G$ with $\chi(P \log p) = 1$ for every prime p . For every $n = \prod p^{v_p}$, $\chi(P \log n) = \prod_p \chi(P \log p)^{v_p} = 1$. Since $P \log n \in \Pi$, $\Psi_\chi(P \log n) = \chi(P \log n) = 1$, and $\pm P$ is a positive resonance with $|P| > |B|$. The families of [Proposition 5.5](#) take values in $S^1(\mathbb{C})$.

(b) For every n , $T \log n = p \log n - \varphi(p) \log n$ with $p \log n \in \Pi$ and $\varphi(p) \log n$ finite. By real-linearity, $\Psi(T \log n) = \text{cis}(\varphi(p \log n) - \varphi(p) \log n) = \text{cis}(0) = 1$. Also $T - p = -\varphi(p)$ is finite.

(c) Let $c = \log 3 / \log 2$. It is irrational by unique factorization, and transcendental by the Gelfond–Schneider theorem, since $2^c = 3$. Hence $\iota_c : \mathbb{Q}(c) \rightarrow \mathbb{Q}(c)$, $g(c) \mapsto g(c^2)$, is an injective field homomorphism. Choose a basis (e_i) of $\mathbb{R}$ over $\mathbb{Q}(c)$ and define $\mathfrak{s}'(\sum_i g_i(c) e_i) = \sum_i \iota_c(g_i(c)) e_i$. Then $\mathfrak{s}'$ is $\mathbb{Q}$ -linear and injective, and $\mathfrak{s}'(cr) = c^2 \mathfrak{s}'(r)$. Hence $\mathfrak{s}'(cr) - c \mathfrak{s}'(r) = (c^2 - c) \mathfrak{s}'(r) \neq 0$ for $r \neq 0$. Put $\mathfrak{s}(x) = \log 2 \cdot \mathfrak{s}'(x / \log 2)$. For $P = \sum_y r_y \omega^y \in \Pi$ (all $y > 0$) define

$$\nu(P) = \sum_y \mathfrak{s}(r_y) \omega^{-1/y} \in \mathbf{m}_\mathbb{R}.$$

Since $y \mapsto -1/y$ is increasing on positive surreals, the exponents form a reverse well-ordered set of negative surreals, so this is a normal form. The map ν is additive because $\mathfrak{s}$ is. Hence $\Psi(P + f) = \text{cis}(\nu(P) + f)$ is a phase, and it has infinite periods by [Theorem 5.7](#).

Suppose $T = P + f \neq 0$ ($P \in \Pi$, $f \in \mathcal{O}_\mathbb{R}$) satisfies $\Psi(T \log 2) = \Psi(T \log 3) = 1$. Then $\nu(P \log 2) + f \log 2 \in 2\pi\mathbb{Z}$ and $\nu(P \log 3) + f \log 3 \in 2\pi\mathbb{Z}$. Taking standard parts, $\text{st}(f) \log 2 = 2\pi k$ and $\text{st}(f) \log 3 = 2\pi l$. If $\text{st } f \neq 0$, then $\log 2 / \log 3 = k/l$, which is impossible. So $k = l = 0$ and $f = -\nu(P \log 2) / \log 2 = -\nu(P \log 3) / \log 3$. Comparing coefficients at each $\omega^{-1/y}$ gives $\mathfrak{s}(r_y \log 2) / \log 2 = \mathfrak{s}(r_y \log 3) / \log 3$, that is, $\mathfrak{s}'(r_y) = \mathfrak{s}'(cr_y) / c$. By the displayed property of $\mathfrak{s}'$ this forces $r_y = 0$ for every y . So $P = 0$, then $f = 0$ and $T = 0$, a contradiction. $\square$

Part (a) is the pigeonhole argument of `trigonometry:per:thm:cofinal`, applied to the prime values of a character. Part (b) was observed in the review of 06, and it covers every family of [Proposition 5.5](#); for E_λ it gives $T = \omega - \lambda$. Part (c) needs characters whose values are $\text{cis } \delta$ with nonzero infinitesimal δ , values in $1 + \mathfrak{m}$ off $S^1(\mathbb{C})$. By (a) no ordinary-circle phase can serve. Consequence: 06's resonance theorem applies verbatim to every phase with a resonance ([Theorem 18.2](#)), as 06's Remark 7.4 anticipates. This includes every explicitly specified phase in the sources and the collection's explicit ordinary-circle families (Exp , E_λ , $E_{\lambda,M}$, E_θ). 03's statements for arbitrary phases, 04's prescribed prime phases ([Proposition 5.6](#), partial characters on a set-sized subspace), and the collection's exponentials whose characters take values in $1 + \mathfrak{m}$ (`trigonometry:per:thm:robustness`) are covered only when the phase in question has a resonance. The theorem does not apply to the phase in (c), and whether a coherent galaxy zeta exists for that phase is open ([Question 23.1](#)).

Part II

The finite plane

6 Canonical finite lifts and divisor rigidity

6.1 The lift

Definition 6.1 (Canonical finite lift [01–06]). Let f be an ordinary meromorphic function near $c \in \mathbb{C}$, with Laurent germ $f(c + u) = \sum_{n \geq -m} a_n u^n$. For $\varepsilon \in \mathfrak{m}$, $\varepsilon \neq 0$, put

$$f^\#(c + \varepsilon) = \sum_{n=-m}^{-1} a_n \varepsilon^n + \sum_{n \geq 0}^H a_n \varepsilon^n.$$

At $\varepsilon = 0$ use the ordinary value if f is regular at c . At a pole no field value is assigned and the meromorphic germ is retained. Removable singularities are removed in the ordinary germ before lifting.

[Lemma 4.3](#) makes this well defined, and the centre $c = \text{st } z$ is unique. A pole does not delete its monad: every nonzero infinitesimal displacement has a value, always of infinite modulus, since it is $g(0)\varepsilon^m(1 + \eta)$ with $m \leq -1$ ([Theorem 6.3](#); 05 and 06 say “usually”, error 53). This is the analysis report's canonical lift (`c:p4:def-lift`), extended here to meromorphic germs. 03 writes f^H , and 01 and 05 write $f^\sharp$.

Proposition 6.2 (Functoriality [01–06]). *Lifting respects sums, products, reciprocals where nonzero, analytic composition where the ordinary germs compose (expanded around the ordinary value of the inner germ), and coordinate differentiation. Every identity of ordinary meromorphic germs becomes an identity of lifted germs on every common finite domain. For conjugation, put $\bar{f}(z) = \overline{f(\bar{z})}$. Then $(\bar{f})^\#(\bar{s}) = \overline{f^\#(s)}$, and so $f^\#(\bar{s}) = \overline{f^\#(s)}$ when f is real-symmetric ($\bar{f} = f$), as Γ , ζ and ξ are.*

Proof. The identities are identities of Taylor–Laurent series. After finitely many negative powers are factored out, all operations are covered by [Lemma 4.3](#). A unit has nonzero constant term, so its inverse is a geometric series in an infinitesimal. Composition is expanded after subtracting the ordinary value of the inner germ. Differentiation is coefficientwise, and it is the fine derivative by `b:germalgebra`. For conjugation, the Laurent coefficients of $\bar{f}$ at $\bar{c}$ are $\overline{a_n}$, and conjugation acts coefficientwise on normal forms. $\square$

The conjugation clause corrects 05's Proposition 3.1, which states it without the bar on f (error 35).

Theorem 6.3 (Divisor rigidity [01–06]). *Let f be a nonzero ordinary meromorphic function on a connected ordinary domain, and write $f(c + u) = u^m g(u)$ with $m \in \mathbb{Z}$ and $g(0) \neq 0$. For every nonzero $\varepsilon \in \mathfrak{m}$,*

$$f^\#(c + \varepsilon) = g(0) \varepsilon^m (1 + \eta), \quad \eta \in \mathfrak{m}, \quad v(f^\#(c + \varepsilon)) = m v(\varepsilon).$$

In particular $f^\#(c + \varepsilon) \neq 0$. The zeros and poles of $f^\#$ on the finite plane are exactly the embedded ordinary zeros and poles of f , with the same multiplicities and orders.

Proof. $g(\varepsilon)/g(0) = 1 + \eta$ with $\eta \in \mathfrak{m}$ by Lemma 4.3, and this is a unit of valuation zero. Multiply by $g(0)\varepsilon^m$. At $\varepsilon = 0$ the assertion is the ordinary one. Every finite point has a unique standard part, so these local statements exhaust $\mathcal{O}$. $\square$

All six sources prove this (01 Thm 3.2; 02 Thm 3.1, Cor 3.2; 03 Thm 3.2; 04 Thm 3.1, Cor 3.2; 05 Thm 4.1; 06 Thm 3.3). In the holomorphic case it is also the analysis report's `c:p4:liftzeros` with `c:p4:rk-liftzeros`. The theorem is stronger than preservation of standard parts. A fixed classical function acquires no cloud of displaced zeros by adjoining infinitesimals, not even at a multiple zero. It concerns the *unchanged* classical germ: a Hahn-coefficient deformation such as $f^\# + \eta h^\#$ can move zeros within their monads (Section 7; analysis `d:thm:zerocluster`).

Remark 6.4 (The word canonical [01, 02, 04, 06]). The lift is canonical as the evaluation of a specified germ. It is not a uniqueness theorem among all finely differentiable functions that agree with f on $\mathbb{C}$, and agreement at ordinary points does not imply the germ law.

The classical facts used below are in [13, 15].

Corollary 6.5 (Finite Gamma, zeta and ξ [01–06]). $\Gamma^\#$ has no zeros. Its poles are exactly the ordinary integers $0, -1, -2, \dots$, simple, with residues $(-1)^m/m!$, and $(1/\Gamma)^\#$ has zeros exactly there. $\zeta^\#$ has one simple pole, at 1, with residue 1. Its zeros are exactly the ordinary zeros of ζ with their multiplicities, and the trivial zeros $-2, -4, \dots$ are simple. $\xi^\#$, the lift of the entire function ξ , has exactly the nontrivial zeros with their multiplicities.

6.2 Explicit germs

For nonzero $\varepsilon \in \mathfrak{m}$, with Euler's constant γ , the Stieltjes constants γ_n ($\gamma_0 = \gamma$) and $H_m = \sum_{j \leq m} 1/j$ ($H_0 = 0$) [01–06]:

$$\Gamma^\#(\varepsilon) = \varepsilon^{-1} - \gamma + \left(\frac{\gamma^2}{2} + \frac{\pi^2}{12}\right)\varepsilon + O(\varepsilon^2), \quad (6.1)$$

$$\Gamma^\#(-m + \varepsilon) = \frac{(-1)^m}{m! \varepsilon} (1 + (H_m - \gamma)\varepsilon + O(\varepsilon^2)), \quad (6.2)$$

$$\zeta^\#(1 + \varepsilon) = \varepsilon^{-1} + \sum_{n \geq 0}^H \frac{(-1)^n \gamma_n}{n!} \varepsilon^n, \quad (6.3)$$

$$\text{Log}_m \Gamma^\#(1 + \varepsilon) = -\gamma \varepsilon + \sum_{n \geq 2}^H \frac{(-1)^n \zeta(n)}{n} \varepsilon^n. \quad (6.4)$$

Here $O(\varepsilon^k)$ means ε^k times a finite strong evaluation of the indicated germ. It is not a limit along a sequence. The next coefficient in (6.1) is $-(\gamma^3/6 + \gamma\pi^2/12 + \zeta(3)/3) \approx -0.9075$. At a zero ρ of ζ of order m , $\zeta^\#(\rho + \varepsilon) = \frac{\zeta^{(m)}(\rho)}{m!} \varepsilon^m (1 + \eta)$. At the trivial zeros [01, 02, 04],

$$\zeta'(-2r) = \frac{(-1)^r (2r)! \zeta(2r+1)}{2(2\pi)^{2r}} \neq 0, \quad \zeta^\#(-2r + \varepsilon) = \zeta'(-2r) \varepsilon (1 + \eta), \quad (6.5)$$

so $\zeta'(-2) = -\zeta(3)/(4\pi^2)$. For a simple zero ρ the formal Newton map satisfies $N(\rho + \varepsilon) - \rho = \frac{\zeta''(\rho)}{2\zeta'(\rho)}\varepsilon^2 + O(\varepsilon^3)$ [05]: the valuation at least doubles. At $\varepsilon = \omega^{-1}$ [02, 04, 05],

$$\Gamma^\#(\omega^{-1}) = \omega - \gamma + \left(\frac{\gamma^2}{2} + \frac{\pi^2}{12}\right)\omega^{-1} + O(\omega^{-2}), \quad \zeta^\#(1 + \omega^{-1}) = \omega + \gamma_0 - \gamma_1\omega^{-1} + \frac{\gamma_2}{2}\omega^{-2} - \dots,$$

and at a simple zero, $\zeta^\#(\rho + \omega^{-1}) \sim \zeta'(\rho)\omega^{-1}$. At a hypothetical double zero the scale would be ω^{-2} . Small residuals measure multiplicity. They are not further exact zeros.

6.3 Identities

On finite arguments, as identities of lifted meromorphic germs [01–06]:

$$\Gamma^\#(z+1) = z\Gamma^\#(z), \quad \Gamma^\#(z)\Gamma^\#(1-z) = \frac{\pi}{\sin^\#(\pi z)}, \quad (6.6)$$

$$\prod_{j=0}^{q-1} \Gamma^\#\left(z + \frac{j}{q}\right) = (2\pi)^{(q-1)/2} q^{1/2-qz} \Gamma^\#(qz) \quad (q \in \mathbb{N}_{\geq 1}), \quad (6.7)$$

$$\zeta^\#(s) = 2^s \pi^{s-1} \sin^\#(\pi s/2) \Gamma^\#(1-s) \zeta^\#(1-s), \quad (6.8)$$

$$\xi^\#(s) = \xi^\#(1-s), \quad \xi^\#(\bar{s}) = \overline{\xi^\#(s)}, \quad \Xi^\#(-w) = \Xi^\#(w), \quad (6.9)$$

where $\Xi(w) = \xi(\frac{1}{2} + iw)$ is real entire. Powers of ordinary positive constants use E_{fp} , and the index q is an ordinary integer. At points where a pole meets a zero, the germs are multiplied first. $0 \cdot \Gamma^\#(0)$ is not a product of field values. No product with a surreal number of factors and no recurrence iterated a surreal number of times is used in the surreal constructions. Internal products occur only inside transported models (Section 20.2).

6.4 Coefficientwise formulas and the failure of the raw series

For $a \in \mathbb{C}$ with $\text{Re } a > 1$ and $\varepsilon \in \mathfrak{m}$ [02, 04, 06],

$$\zeta^\#(a + \varepsilon) = \sum_{k \geq 0}^H \frac{(-\varepsilon)^k}{k!} \left(\sum_{n \geq 1} (\log n)^k n^{-a} \right), \quad (6.10)$$

and for $\text{Re } a > 0$ [02, 04],

$$\Gamma^\#(a + \varepsilon) = \sum_{j \geq 0}^H \frac{\varepsilon^j}{j!} \int_0^\infty e^{-t} t^{a-1} (\log t)^j dt. \quad (6.11)$$

The inner sums and integrals are ordinary, absolutely convergent, and computed in $\mathbb{C}$; only the outer sums are strong. This is summation notion (c) of Section 2.2. Likewise $\Gamma(c)\zeta(c) = \int_0^\infty x^{c-1}(e^x - 1)^{-1} dx$ gives the coefficients of $(\Gamma\zeta)^\#$ at $\text{Re } c > 1$ [04].

Warning 6.6 (The raw finite Dirichlet family [01, 02, 04, 05, 06]). At a finite $s = a + \varepsilon$, the family $(n^{-s})_{n \geq 1}$ is not strongly summable, even when $\text{Re } a > 1$: every member has the nonzero standard part n^{-a} at exponent zero. The double family behind (6.10) is not strongly summable either. Raw Euler, Weierstrass and Hadamard products are equally inadmissible at finite arguments.

Corollary 6.7 (Zero-free halos [06]). *If an ordinary set $U \subseteq \mathbb{C}$ contains no zero of f , then $f^\#$ has no zero on $U^\#$. In particular the zero-free half-plane $\text{Re } s \geq 1$, the localization of nontrivial zeros in $0 < \text{Re } s < 1$, and every numerically verified zero-free region hold exactly for $\zeta^\#$, with no infinitesimal boundary layer of new zeros.*

Proof. A zero of $f^\#$ is an ordinary zero by Theorem 6.3, and its standard part is itself. $\square$

7 The finite Riemann hypothesis and its robust form

7.1 Finite RH

Definition 7.1 ([04, 05, 06]). RH_{fin} : every $s \in \mathcal{O}$ with $\xi^\#(s) = 0$ has $\text{Re } s = \frac{1}{2}$. Its *shadow* form [04] replaces the conclusion by $\text{Rest } s = \frac{1}{2}$.

Theorem 7.2 (Finite RH [01, 02, 04, 05, 06]). *The following are equivalent: classical RH; RH_{fin} ; its shadow form; “every zero of $\zeta^\#$ with $0 < \text{Re } s < 1$ has $\text{Re } s = \frac{1}{2}$ ” (01); and “all zeros of $\xi^\#$ have real part exactly $\frac{1}{2}$ ” (02).*

Proof. By [Theorem 6.3](#) the zeros of $\xi^\#$ and the zeros of $\zeta^\#$ in the finite strip are exactly the classical nontrivial zeros, at the embedded ordinary points. For an ordinary point, $\text{Re } s = \text{Rest } s$. Conversely, restrict each statement to $\mathbb{C}$. $\square$

The equivalence neither strengthens nor weakens RH. What it rules out is spurious “infinitesimal counterexamples”: $\xi^\#(\rho + \varepsilon) \sim \xi'(\rho)\varepsilon \neq 0$ at a simple zero, and an infinitesimal residual is not a zero. 03’s horizontal form ([Corollary 16.5](#)) is the same statement, because ξ_χ has no zeros on $\mathcal{T}_+ \cup \mathcal{T}_-$. The transferred form is [Theorem 20.3](#).

7.2 Local lemmas

Lemma 7.3 (Formal implicit function [05]). *Let $A(X, u) \in \mathbb{C}[[X, u]]$ satisfy $A(0, 0) = 0$ and $\partial_X A(0, 0) \neq 0$. There is a unique $d(u) \in u\mathbb{C}[[u]]$ with $A(d(u), u) = 0$, and $A = (X - d(u))U(X, u)$ with $U(0, 0) = \partial_X A(0, 0)$. For every infinitesimal τ , the only infinitesimal ε with $A(\varepsilon, \tau) = 0$ is $d(\tau)$, including solutions outside the subfield generated by τ .*

Proof. Solve for the coefficients of d recursively; at each stage the new coefficient is multiplied by $\partial_X A(0, 0) \neq 0$. Formal division by $X - d(u)$, using $X^k - d^k = (X - d) \sum_j X^{k-1-j} d^j$, gives U . After strong evaluation $U(\varepsilon, \tau)$ is a unit. $\square$

Lemma 7.4 (Constant perturbation of a zero [04]). *Let f be holomorphic near c with $f(c+u) = u^m a(u)$, $m \geq 1$, $a(0) = a_0 \neq 0$. Let $\eta > 0$ be infinitesimal and $\delta \in \{\pm 1\}$. Then $f^\# + \delta\eta$ has exactly m zeros in $c + \mathfrak{m}$, all simple: $z_j = c + rv_j(r)$ with $r = \eta^{1/m}$, where the v_j are convergent ordinary germs with $a_0 v_j(0)^m = -\delta$ and distinct nonzero $v_j(0)$.*

Proof. $F(r, v) = v^m a(rv) + \delta$ has m simple nonzero roots at $r = 0$. The ordinary implicit-function theorem gives the germs v_j , and evaluation at r gives zeros. Conversely, let $z = c + u$ be a zero in the monad. Then $u \neq 0$ and $mv(u) = v(\eta)$ by [Theorem 6.3](#), so $v = u/r$ has valuation zero. Reducing $F(r, v) = 0$ modulo $\mathfrak{m}$ shows that $\text{st } v$ is one of the $v_j(0)$. The local factorization $F = (v - v_j(r))U_j$ with $U_j(0, v_j(0)) \neq 0$ becomes a unit after evaluation, so $v = v_j(r)$. Finally $f'(c+u) = u^{m-1}(ma_0 + \text{infinitesimal}) \neq 0$. $\square$

Proposition 7.5 (Local multiplicity conservation [04]). *Let $f \neq 0$ and h be ordinary entire functions, η a nonzero infinitesimal, and $F_\eta = f^\# + \eta h^\#$. In the monad of a zero c of f of order m , F_η has exactly m zeros counted with multiplicity. No finite zero lies outside the monads of zeros of f . If f and h are real entire, η is real and c is a real simple zero of f , the zero of F_η in $c + \mathfrak{m}$ is unique, real and simple, and $c_\eta = c - \eta h(c)/f'(c) + O(\eta^2)$ [04].*

Proof. The last claim follows by taking standard parts. If $h \equiv 0$ near c there is nothing to prove. Otherwise write $h(c+u) = u^k b(u)$ with $b(0) \neq 0$. If $k \geq m$, then $F_\eta = u^m[a(u) + \eta u^{k-m}b(u)]$ and the bracket is a unit on the monad. If $k < m$, then $F_\eta = u^k\{u^{m-k}a(u) + \eta b(u)\}$. Put

$r = \eta^{1/(m-k)}$ (a root in $\mathbf{SC}$) and $u = rv$. The argument of [Lemma 7.4](#) gives exactly $m - k$ simple nonzero roots of the bracket, so the total is m . For the last clause, apply [Lemma 7.3](#) to $A(X, u) = f(c + X) + u h(c + X)$, with $\partial_X A(0, 0) = f'(c) \neq 0$: the unique zero in the monad is $c + d(\eta)$ with $d'(0) = -h(c)/f'(c)$, and the evaluated factorization $A = (X - d(u))U$ with U a unit shows that it is simple. Since f, h, c, η are real, conjugation maps this zero to a zero of F_η in the same monad, so uniqueness makes it real. $\square$

7.3 Protection of a simple critical-line zero

Definition 7.6 (J-symmetric formal deformations). Let $J(s) = 1 - \bar{s}$. The class $\mathcal{D}_J$ consists of formal deformations $F(s, u) = \xi(s) + \sum_{n \geq 1} f_n(s)u^n$ with ordinary entire f_n satisfying $f_n(1 - \bar{s}) = \overline{f_n(s)}$. No convergence in u is required. For real $\tau \in \mathbf{m}$ and $s = a + \varepsilon \in \mathcal{O}$ put

$$F_\tau(s) = \xi^\#(s) + \sum_{n \geq 1, k \geq 0}^H \frac{f_n^{(k)}(a)}{k!} \tau^n \varepsilon^k,$$

a strongly summable double family by [Lemma 4.3](#).

The three source classes are subclasses of $\mathcal{D}_J$:

- 04: $f^\# + \eta h^\#$ with h real entire in the variable w of $s = \frac{1}{2} + iw$. In s this is exactly $h(1 - \bar{s}) = \overline{h(s)}$.
- 05: formal series whose coefficients satisfy both $f_n(1 - s) = f_n(s)$ and $f_n(\bar{s}) = \overline{f_n(s)}$, which together imply J-symmetry, with $\tau > 0$.
- 06: $\xi^\# + \varepsilon h^\#$ with $h \in \mathcal{S}_{\text{sym}}$, the real vector space of entire h with both symmetries, with ε real of either sign.

Every F_τ has standard part $\xi(\text{st } s)$, so its finite zeros lie in monads of zeros of ξ . For real τ , $F_\tau(J(s)) = \overline{F_\tau(s)}$. To see this at a point ρ with $\text{Re } \rho = \frac{1}{2}$, write $f_n(\rho + X) = \sum_k c_{n,k} X^k$. J-symmetry gives $\overline{c_{n,k}} = (-1)^k c_{n,k}$, and $J(\rho + \varepsilon) = \rho - \bar{\varepsilon}$.

Theorem 7.7 (Protection of simple critical-line zeros [04, 05, 06]). *Let ρ be a simple zero of ξ with $\text{Re } \rho = \frac{1}{2}$, let $F \in \mathcal{D}_J$, and let τ be a real infinitesimal. Then F_τ has exactly one zero ρ_τ in $\rho + \mathbf{m}$, and $\text{Re } \rho_\tau = \frac{1}{2}$ exactly. Moreover $\rho_\tau = d(\tau) + \rho$ for a formal series d with purely imaginary coefficients, and*

$$\rho_\tau = \rho - \tau \frac{f_1}{\xi'} + \tau^2 \left(\frac{f_1 f_1'}{\xi'^2} - \frac{\xi'' f_1^2}{2 \xi'^3} - \frac{f_2}{\xi'} \right) + O(\tau^3),$$

all functions evaluated at ρ . If $f_1(\rho) \neq 0$ and $\tau \neq 0$, the displacement has valuation $v(\tau)$.

Proof. Apply [Lemma 7.3](#) to $A(X, u) = F(\rho + X, u)$, for which $\partial_X A(0, 0) = \xi'(\rho) \neq 0$. This gives the unique zero $\rho + d(\tau)$ in the monad. The function $d^*(u) = -\bar{d}(u)$, with conjugated and negated coefficients, also solves $A(d^*(u), u) = 0$, because $\overline{c_{n,k}} = (-1)^k c_{n,k}$. By uniqueness $d^* = d$, so every coefficient of d is purely imaginary. Since τ is real, $d(\tau)$ is purely imaginary, so $\text{Re } \rho_\tau = \frac{1}{2}$. Equivalently, J maps the zero to a zero in the same monad, and uniqueness forces $J(\rho_\tau) = \rho_\tau$. The expansion comes from comparing the coefficients of u and u^2 in $A(d(u), u) = 0$. The valuation claim holds because the first coefficient is a nonzero ordinary number. $\square$

06 gives the second-order formula for linear deformations ($f_2 = 0$) and checks it symbolically. 04's Appendix A gives the same coefficients in the w -plane, as $B = h_0 h_1 / a^2 - b h_0^2 / a^3$ with $b = f''/2$ (checked for this merge, [Appendix B](#)). 05 states that every displacement coefficient is purely imaginary but gives no argument. The involution argument above supplies the missing step (error 33). The hypotheses that τ is real and the deformation J-symmetric are essential. Without them, even a simple critical-line zero moves off the line at first order.

7.4 One stability theorem

Theorem 7.8 (Two-sign reality criterion [04]). *Let f be a nonzero real entire function ($f(\bar{w}) = \overline{f(w)}$) and fix one positive infinitesimal η . The following are equivalent: (i) every ordinary zero of f is real and simple; (ii) every finite zero of $f^\# + \eta$ and of $f^\# - \eta$ is real.*

Proof. Every finite zero of $f^\# \pm \eta$ lies in the monad of a zero of f . Assume (i). In the monad of a real simple zero, [Lemma 7.4](#) gives a unique zero for each sign. Conjugation fixes the equation and the monad, so the zero is real. Conversely, a nonreal zero of f gives nonreal perturbed zeros. Let c be a real zero of order $m \geq 2$ with real leading coefficient a_0 . For $m = 2$ choose δ with the sign of a_0 ; then $a_0 v^2 = -\delta$ has nonreal roots. For $m \geq 3$ the m distinct roots of $a_0 v^m = -\delta$ are not all real. In both cases [Lemma 7.4](#) gives a zero $c + rv_j(r)$ whose imaginary part has the nonzero leading coefficient $\text{Im } v_j(0)$ at scale $\eta^{1/m}$. $\square$

Theorem 7.9 (Unified stability theorem [04, 05, 06]; unified form [merge]). *The following are equivalent.*

- (i) *Classical RH holds and every nontrivial zero of ζ is simple.*
- (ii) *For every $F \in \mathcal{D}_J$ and every real infinitesimal τ , every finite zero of F_τ lies on $\text{Re } s = \frac{1}{2}$.*
- (iii) *For one positive infinitesimal η , every finite zero of $\xi^\# + \eta$ and of $\xi^\# - \eta$ lies on $\text{Re } s = \frac{1}{2}$.*

If “lies on $\text{Re } s = \frac{1}{2}$ ” is replaced by “has standard part on $\text{Re } s = \frac{1}{2}$ ” in (ii) or (iii), each becomes equivalent to RH alone.

Proof. (i) $\Rightarrow$ (ii): a finite zero has as standard part a zero ρ of ξ , which is simple and on the line by (i). Apply [Theorem 7.7](#). (ii) $\Rightarrow$ (iii): take $F = \xi \pm u$, which lie in $\mathcal{D}_J$, and $\tau = \eta$. (iii) $\Rightarrow$ (i): with $s = \frac{1}{2} + iw$, $\xi^\#(\frac{1}{2} + iw) = \Xi^\#(w)$ by functoriality, and $\text{Re } s = \frac{1}{2}$ means w is real. Apply [Theorem 7.8](#) to the real entire Ξ : all zeros of Ξ are real and simple, which is (i). For the shadow forms, every finite zero of F_τ has as standard part a zero of ξ . Every zero of ξ is the standard part of a zero of $\xi^\# \pm \eta$ ([Lemma 7.4](#)), and of F_τ with $F = \xi$. $\square$

Corollary 7.10 (The source criteria [04, 05, 06]). *Each of the following is equivalent to RH plus simplicity.*

- (a) [04] *For one fixed positive infinitesimal η , all finite zeros of $\Xi^\# \pm \eta$ are real. Equivalently, for every ordinary real entire h (in w), all finite zeros of $\Xi^\# + \eta h^\#$ are real; testing $h = \pm 1$ suffices.*
- (b) [05] *For every formal deformation with coefficients having both symmetries and every positive infinitesimal τ , all finite zeros lie on the line; testing ξ and $\xi \pm u$ suffices.*
- (c) [06] *For every ordinary real c and positive infinitesimal ε , all finite zeros of $\xi^\# + c\varepsilon$ lie on the line. Equivalently, the same holds for every $h \in \mathcal{S}_{\text{sym}}$ and every real infinitesimal ε , with $\xi^\# + \varepsilon h^\#$.*

Proof. Each class lies in $\mathcal{D}_J$ and contains the test family of [Theorem 7.9](#)(iii), or its w -plane form. 06’s converse uses the constant $c = b_m$, where $b_m = i^m \xi^{(m)}(\rho)/m! \in \mathbb{R} \setminus \{0\}$ is the leading coefficient of $\xi(\rho + iu)$ at an m -fold zero. This is the sign choice of [Theorem 7.8](#) (a second route to the converse). $\square$

Remark 7.11 (General real entire functions [04]). *For a nonzero real entire f and one fixed positive infinitesimal η , all zeros of f are real and simple if and only if, for every ordinary real entire h , all finite zeros of $f^\# + \eta h^\#$ are real; testing $h = \pm 1$ suffices (04 Corollary 6.2).*

The forward direction is the last clause of [Proposition 7.5](#), and the converse is [Theorem 7.8](#). [Corollary 7.10\(a\)](#) is the case $f = \Xi$.

Remark 7.12 (Exact versus shadow [04]). For constant perturbations every zero in the monad of ρ has standard part ρ , so the shadow statement is RH, while the exact statement for both signs is RH plus simplicity. 04 calls the two “inequivalent in strength” and the shadow form “weaker”. The correct wording is *formally stronger*: the exact statement implies the shadow statement, and it is equivalent to the shadow statement together with simplicity of every nontrivial zero. The two can differ in truth value only if RH holds and some nontrivial zero is multiple, which is not known to happen (error 22).

Corollary 7.13 (Splitting scale [04, 06]). *At an m -fold critical-line zero, the off-line zero constructed in the proof has displacement of valuation $v(\eta)/m$. A symmetric perturbation of size η can produce a much larger transverse displacement. For example, $u^2 + \varepsilon = 0$ gives $u = \pm i\sqrt{\varepsilon}$, so $s = \rho + iu$ moves horizontally by $\mp\sqrt{\varepsilon}$, and the two zeros are exchanged by J instead of being fixed by it. For u^3 , the zeros of $u^3 + \eta$ are $-\eta^{1/3}$ and $\eta^{1/3}(\frac{1}{2} \pm \frac{\sqrt{3}}{2}i)$.*

Both signs are needed in [Theorem 7.9\(iii\)](#): a real double zero with positive quadratic coefficient splits into two real zeros under $-\eta$ and into two nonreal zeros under $+\eta$ [04]. The perturbed functions keep both symmetries but not the Euler product, and they are not proposed as new zeta functions. What the theorem shows is that the functional-equation symmetries alone do not prevent infinitesimal off-line splitting [04]. It proves neither RH nor simplicity [04, 05, 06]. It concerns infinitesimal parameters, not a uniform real neighbourhood of 0 valid at all heights [05]. The class $\mathcal{D}_J$ is not claimed to be maximal [merge].

7.5 Multiplicative deformations and protected counts

A multiplicative perturbation $f^\#(1 + \eta h^\#)$ has exactly the finite divisor of $f^\#$, because the second factor is a unit at every finite point [04]. Additive and multiplicative deformations must not be conflated.

Let $N(T)$ count the nontrivial zeros with $0 < \text{Im } \rho \leq T$, with multiplicity, and $N_0^s(T)$ those that are simple and on the line. For ordinary $T > 0$ that is not a zero ordinate, put $D_T = \{w : 0 < \text{Re } w < T, |\text{Im } w| < 1\}$. Its boundary contains no zero of Ξ , since zeros have $|\text{Im } w| = |\frac{1}{2} - \beta| < \frac{1}{2}$, and it contains every zero counted by $N(T)$.

Theorem 7.14 (Protected counts [04]). *Let h be an ordinary real even entire function and $\eta > 0$ infinitesimal. Let $N_\eta(T)$ count the zeros of $\Xi_\eta = \Xi^\# + \eta h^\#$ in $D_T^\#$ with multiplicity, and $N_{\eta, \mathbb{R}}(T)$ the real ones. Then*

$$N_\eta(T) = N(T), \quad N_{\eta, \mathbb{R}}(T) \geq N_0^s(T).$$

Hence, for every ordinary c , $\liminf N_0^s(T)/N(T) \geq c$ implies $\liminf N_{\eta, \mathbb{R}}(T)/N_\eta(T) \geq c$, the $\liminf$ taken over ordinary T that are not zero ordinates.

Proof. Every finite zero has as standard part a zero of Ξ . Apply [Proposition 7.5](#) to the finitely many zero monads in $D_T^\#$ and add. Each simple real zero gives one real deformed zero, by [Theorem 7.7](#) in the w -plane. The last statement compares ordinary integer-valued functions of ordinary T . $\square$

Evenness preserves $\Xi_\eta(-w) = \Xi_\eta(w)$; the local theorem needs only realness. If h is not real, a simple critical-line zero can move off the line. The theorem proves neither RH nor simplicity. Admissible values of c are lower bounds for κ_s , not for κ_0 ; they and their statuses are in [Section 21.4](#).

Call an ordinary critical-line zero a *uniformly stable centre* if the conclusion of [Theorem 7.7](#) holds at it for every $h \in \mathcal{S}_{\text{sym}}$ and every real infinitesimal parameter [06]. Every simple critical-line zero is one, and the proof of [Theorem 7.8](#) shows that a multiple critical-line zero is not. Proportions of stable centres are centre-indexed counts with the classical total count as denominator. They are not counts of displaced roots below a surcomplex height.

Part III

Positive infinite real part

8 The Dirichlet algebra on $\mathcal{T}_+$

8.1 Separated monomials

Every $s \in \mathcal{T}_+$ has a unique decomposition

$$s = X + c + \varepsilon, \quad X = \text{pp}(\text{Re } s) \in \Pi, \quad X > 0, \quad c \in \mathbb{C}, \quad \varepsilon \in \mathfrak{m}. \quad (8.1)$$

Since $\text{Im } s$ is finite, $(\text{Im } s) \log n$ is finite for every ordinary n , and $n^{-s} = E_{\text{fp}}(-s \log n)$ needs no phase: it is the same for every exponential of [Section 2.3](#). Explicitly,

$$n^{-s} = q_n n^{-c} \text{Exp}_{\mathfrak{m}}(-\varepsilon \log n), \quad q_n = \exp_{\mathbf{No}}(-X \log n). \quad (8.2)$$

Each q_n is a Conway monomial, since $X \log n \in \Pi$. They satisfy $q_m q_n = q_{mn}$, $1 = q_1 > q_2 > \dots$, and $q_n/q_m \in \mathfrak{m}$ for $n > m$, so their valuations form an increasing sequence of order type ω .

Theorem 8.1 (Arbitrary-coefficient Dirichlet realization [01–06]). *Let $s \in \mathcal{T}_+$, and let $a : \mathbb{N}_{\geq 1} \rightarrow \mathbb{C}$ be an arbitrary arithmetic function, with no growth condition. Then*

$$\mathcal{D}_s(a) = \sum_{n \geq 1}^H a(n) n^{-s}$$

*exists. The map $\mathcal{D}_s$ is an injective unital $\mathbb{C}$ -algebra homomorphism from arithmetic functions under Dirichlet convolution $(a * b)(n) = \sum_{d|n} a(d)b(n/d)$ to $\mathbf{SC}$. If $a \neq 0$ and n_0 is its least nonzero index, then $\mathcal{D}_s(a) = a(n_0) n_0^{-s} (1 + \eta)$ with $\eta \in \mathfrak{m}$.*

Proof. Expand every term by (8.2) and apply [Lemma 4.4](#) with $\mathfrak{t}_n = q_n$ and the one infinitesimal ε . This gives strong summability, including all the expansions inside the terms. Large ordinary coefficients $a(n)n^{-c}$ do not change supports. For a product, expand both families. A fixed q_k receives contributions only from the finitely many factorizations $k = mn$, and $m^{-s}n^{-s} = k^{-s}$ by the exponential laws, so regrouping gives the convolution. The function δ_1 maps to 1. For the leading term, divide by $a(n_0)n_0^{-s}$. Every remaining term with $n > n_0$ has leading monomial $q_n/q_{n_0} \in \mathfrak{m}$, and all its other monomials are smaller, so the normalized tail is a strong family of positive valuation. Its sum is infinitesimal, which gives the leading form and hence injectivity. $\square$

Remark 8.2 (Scope [02, 04, 06]). The theorem embeds the whole algebra of *formal* Dirichlet series, for example $a(n) = \exp(n^2)$ or $\exp(\exp n)$. It is a strong-normal-form evaluation, not a statement about classical analytic continuation of such data. It holds pointwise in s and is not effective: equality of values reduces to equality of all coefficients, and this gives no decision procedure [02]. The units of the arithmetic algebra are exactly the a with $a(1) \neq 0$, although every nonzero image is invertible in $\mathbf{SC}$ [04, 06]. 04 states the same theorem in a set-sized Hahn field: $\mathcal{E}_q(a) = \sum a(n)q^{\log n}$ with $q = t^\delta$ has valuation $\delta \log n_0$ and leading coefficient $a(n_0)$. The

realization above is the case $q = e^{-X}$, whose real powers are the monomials q_n . 02, 03 and 04 point out that the multiplication itself is the familiar formal Dirichlet-series algebra (compare [9, §2]). What is new is the exact evaluation in **SC**.

Proposition 8.3 (Local analytic dependence [01, 02, 03, 05, 06]). *For $s \in \mathcal{T}_+$, every arithmetic function a and every $h \in \mathfrak{m}$,*

$$\mathcal{D}_{s+h}(a) = \sum_{k \geq 0}^H \frac{h^k}{k!} \mathcal{D}_s(\delta^k a), \quad (\delta a)(n) = -(\log n) a(n),$$

the double family being strongly summable. Hence $D^k \mathcal{D}_s(a) = \sum_n^H a(n)(-\log n)^k n^{-s}$, the local derivatives are fine derivatives, and sum, product and quotient rules hold for every infinitesimal increment.

Proof. Add $\text{supp } h$ to the support set in the proof of [Theorem 8.1](#) and expand $\text{Exp}_{\mathfrak{m}}(-h \log n)$. [Lemma 4.4](#) with two infinitesimals justifies the regrouping. The derivation δ satisfies the Leibniz rule because $\log(mn) = \log m + \log n$. For a quotient, $\mathcal{D}_s(\delta^k b)/\mathcal{D}_s(b) = (-\log n_0)^k (1 + \eta_k)/(1 + \eta)$ is finite for every k . So $\mathcal{D}_{s+h}(b)/\mathcal{D}_s(b) - 1$ is a strong sum of terms of positive valuation, hence infinitesimal, for every $h \in \mathfrak{m}$. $\square$

02's Proposition 5.3 states the quotient rule only “for h sufficiently small”. As the proof shows, every infinitesimal h works (error 9).

Definition 8.4 (Dirichlet zeta [01–06]). $\zeta_D(s) = \mathcal{D}_s(\mathbf{1}) = \sum_{n \geq 1}^H n^{-s}$ on $\mathcal{T}_+$, and $\zeta_D^{(k)}(s) = \mathcal{D}_s(n \mapsto (-\log n)^k)$.

Theorem 8.5 (Euler identities and zero-freeness [01–06]). *For every $s \in \mathcal{T}_+$, with strong sums and the strong expansion of the product:*

$$\begin{aligned} \zeta_D(s) &= \prod_p^H (1 - p^{-s})^{-1}, & \zeta_D(s)^{-1} &= \sum_{n \geq 1}^H \mu(n) n^{-s}, \\ \text{Log}_{\mathfrak{m}} \zeta_D(s) &= \sum_p^H \sum_{k \geq 1}^H \frac{p^{-ks}}{k} = \sum_{n \geq 2}^H \frac{\Lambda(n)}{\log n} n^{-s}, & -\frac{\zeta_D'(s)}{\zeta_D(s)} &= \sum_{n \geq 1}^H \Lambda(n) n^{-s}. \end{aligned}$$

Moreover $\zeta_D(s) \in 1 + \mathfrak{m}$, so ζ_D has no zeros or poles on $\mathcal{T}_+$, and:

- (a) $\zeta_D(s) - \sum_{n \leq N} n^{-s} = (N+1)^{-s}(1 + \eta_N)$ with $\eta_N \in \mathfrak{m}$, for every ordinary $N \geq 1$. In particular $\zeta_D(s) - 1 \sim 2^{-s}$ and $\text{Log}_{\mathfrak{m}} \zeta_D(s) \sim 2^{-s}$.
- (b) $\zeta_D^{(k)}(s) = (-\log 2)^k 2^{-s}(1 + \eta_k)$ for $k \geq 1$.
- (c) For real $x > \mathbb{R}$: $\zeta_D(x) > 1$, $(-1)^k \zeta_D^{(k)}(x) > 0$ for every ordinary k , and $x \mapsto \zeta_D(x)$ is strictly decreasing on $\{x \in \mathbf{No} : x > \mathbb{R}\}$. Also $\zeta_D(x) - 1 \prec x^{-m}$ for every ordinary m .
- (d) For completely multiplicative a , $\mathcal{D}_s(a) = \prod_p^H (1 - a(p)p^{-s})^{-1}$. In particular Dirichlet L -series with character coefficients are defined on $\mathcal{T}_+$ and lie in $1 + \mathfrak{m}$.

Proof. The expanded Euler product has exactly one term for each ordinary integer, by unique factorization, and its family is the family of [Theorem 8.1](#). The identity $\mu * \mathbf{1} = \delta_1$ gives the reciprocal. $\zeta_D - 1$ is a strong sum of infinitesimals, so its near-one logarithm exists. In the formal logarithm of $1 + \sum_{n \geq 2} n^{-s}$, the coefficient at n receives contributions only from the finitely many factorizations of n into factors at least 2, of length at most $\log_2 n$. So the logarithm is computed coefficientwise in the arithmetic algebra, where it is $1/k$ at p^k and 0 elsewhere. Differentiation by [Proposition 8.3](#), or applying $\mathcal{D}_s$ to $\Lambda * \mathbf{1} = \log$, gives the von Mangoldt formula. (a) and (b) are the leading-term clause of [Theorem 8.1](#) applied to the tail, to the logarithm coefficients and to $(-\log n)^k$. (c) For real arguments every term beyond 1 is positive, and so is every term of

$(-1)^k \zeta_D^{(k)}$. If $x_2 > x_1 > \mathbb{R}$, the differences $n^{-x_1} - n^{-x_2}$ ($n \geq 2$) are positive and form a strong family. A nonempty strong family of positive elements has positive sum, because the leading coefficients at the largest occurring monomial are positive; no mean value theorem is used. Finally $2^{-x} x^m = \exp(-x \log 2 + m \log x) \in \mathfrak{m}$. (d) is the same factorization argument, since each $a(p)p^{-s}$ is infinitesimal however large $a(p)$ is. For a Dirichlet character ϖ , $a(1) = 1$. $\square$

All six sources prove these identities: 01 Thms 7.3–7.4; 02 Thm 6.1 and Cor 6.2; 03 Thm 7.3; 04 Thm 7.3; 05 Cor 8.3; 06 Thm 5.3 and its complete-monotonicity paragraph. In 05’s Corollary 8.3, ζ_D' is used before it is defined; here it is defined in [Proposition 8.3](#) first (error 31). 06 calls the zero-freeness theorem “a genuine new domain result”. It is a result on a new domain for the specified extension, and no priority is claimed (error 42).

Proposition 8.6 (Summability dichotomy [02]). *Let $\text{Im } s$ be finite. The raw family $(n^{-s})_{n \geq 1}$ is strongly summable if and only if $\text{Re } s$ is positive infinite.*

Proof. The positive infinite case is [Theorem 8.1](#). If $\text{Re } s$ is finite, every term has a nonzero coefficient at exponent zero. If $\text{Re } s$ is negative infinite, its purely infinite part is negative, and the leading monomials of n^{-s} strictly increase with n . So the union of supports is not well ordered in the valuation order. $\square$

Remark 8.7 (What the right-hand theory does not determine [01, 04, 05, 06]). At positive infinite X , $e^{-X^2} \prec n^{-X}$ for every ordinary $n \geq 2$, because $X^2 - X \log n$ is positive infinite [06]. So agreement with every fixed Dirichlet asymptotic order is weaker than equality with the exact strong sum. This is the zeta analogue of flat Gamma gauges. The finite prescription $\zeta^\#$ and ζ_D live on disjoint domains, and their union is an explicitly specified partial function. No path from the finite halo to $\mathcal{T}_+$ exists, and ζ_D is not claimed to be the analytic continuation of $\zeta^\#$ [01, 04, 05]. The L -series chart gives no surcomplex GRH, since the hard region would again have finite real part and infinite imaginary part [04]. $\mathcal{T}_+$ contains no point of the infinite-height strip $\mathcal{V}$, so zero-freeness on $\mathcal{T}_+$ says nothing about RH at infinite height [05, 06].

Example 8.8 (At ω [02, 03, 04, 05]). $\zeta_D(\omega) = 1 + 2^{-\omega} + 3^{-\omega} + 4^{-\omega} + \dots$, where the ellipsis denotes the complete strong family, and $(\zeta_D(\omega) - 1 - 2^{-\omega})/3^{-\omega} = 1 + \varepsilon$ with $\varepsilon \in \mathfrak{m}_{\mathbb{R}}$. Further,

$$\begin{aligned} \text{Log}_{\mathfrak{m}} \zeta_D(\omega) &= 2^{-\omega} + 3^{-\omega} + \frac{1}{2} 4^{-\omega} + 5^{-\omega} + 7^{-\omega} + \frac{1}{3} 8^{-\omega} + \frac{1}{2} 9^{-\omega} + \dots \quad (\text{no } 6^{-\omega}), \\ \zeta_D(s)^{-1} &= 1 - 2^{-s} - 3^{-s} - 5^{-s} + 6^{-s} - 7^{-s} + 10^{-s} - \dots \end{aligned}$$

Each $n^{-\omega}$ is smaller than every ω^{-N} . At $s = \omega + c + \varepsilon$ the first nonconstant scale is $2^{-c} 2^{-\omega} \text{Exp}_{\mathfrak{m}}(-\varepsilon \log 2)$.

9 Inverse zeta on $\mathcal{T}_+$

This section is source 01’s. Nothing else in the six sources treats inverse zeta. Lagrange inversion is classical [11], and inverse zeta has a literature [12]. The claim is the proved package with the surreal domain and support assertions stated here, not priority for the idea of inverting zeta.

For $n \geq 3$ put $\alpha_n = \log_2 n$ and $\beta_n = \alpha_n - 1 = \log_2(n/2)$, and write $\beta = \beta_3 = \log_2(3/2) \approx 0.584963$.

Theorem 9.1 (Inverse-zeta fibre [01]). *For every positive infinitesimal $u \in \mathbf{No}$, the fibre $\{s \in \mathcal{T}_+ : \zeta_D(s) = 1 + u\}$ consists of exactly one point $s_k(u)$ for each ordinary $k \in \mathbb{Z}$. Its imaginary standard part is $-2\pi k / \log 2$, and*

$$s_k(u) = -\frac{\log u}{\log 2} - \frac{2\pi i k}{\log 2} + \frac{e^{2\pi i k \beta}}{\log 2} u^\beta + \frac{u}{\log 2} - \frac{(2\beta + 1)e^{4\pi i k \beta}}{2 \log 2} u^{2\beta} + o(u^{2\beta}).$$

The full expansion is a strongly summable generalized power series, with coefficients and exponent collisions given in [Theorem 9.8](#). The branch $s_0(u)$ is real, $s_{-k}(u) = \overline{s_k(u)}$, and s_0 inverts the decreasing bijection $x \mapsto \zeta_D(x) - 1$ from $\{x > \mathbb{R}\}$ onto the positive infinitesimals.

Here $o(v)$ means an element whose quotient by v is infinitesimal, a dominance statement at the given argument. The proof occupies the rest of this section.

Lemma 9.2 (The weight monoid [01]). *For finitely supported $\nu = (\nu_n)_{n \geq 3}$ with $\nu_n \in \mathbb{N}$, put $|\nu| = \sum \nu_n$ and $B(\nu) = \sum \nu_n \beta_n$. For every real R only finitely many ν satisfy $B(\nu) \leq R$.*

Proof. Every generator is at least $\beta_3 > 0$, so $|\nu| \leq R/\beta_3$, and an occurring generator satisfies $\beta_n \leq R$, that is, $n \leq 2^{R+1}$. $\square$

The exponents are not distinct: $\beta_6 = \beta_3 + \beta_4$. It is finite multiplicity, not absence of collisions, that allows evaluation and regrouping.

Fix a positive infinitesimal u and $k \in \mathbb{Z}$, and put $c_{n,k} = e^{2\pi i k \alpha_n} = e^{2\pi i k \beta_n}$. Consider

$$V + \sum_{n \geq 3}^H c_{n,k} u^{\beta_n} V^{\alpha_n} = 1, \quad V \in 1 + \mathfrak{m}, \quad (9.1)$$

with near-one binomial powers of V .

Lemma 9.3 (Unique near-one solution [01]; existence step completed [merge]). *Equation (9.1) has exactly one solution $V_k(u)$ in $1 + \mathfrak{m}$. It is the strong evaluation of a generalized power series in u with exponents in the monoid generated by the β_n .*

Proof. Existence. Take formal variables x_n of weight β_n . These weights satisfy the hypothesis of [Lemma 4.6](#), by [Lemma 9.2](#). The equation

$$W = - \sum_{n \geq 3} x_n (1 + W)^{\alpha_n}$$

has a unique formal solution with zero constant term. The right side is a formally summable family, each of whose terms multiplies every coefficient of W by a positive-weight variable, so the coefficients are determined recursively in increasing weight. Its explicit coefficients are given in [Lemma 9.7](#). Apply the evaluation $x_n \mapsto c_{n,k} u^{\beta_n}$ of [Lemma 4.6](#). Being a strongly additive ring homomorphism with $\text{ev}((1 + W)^a) = (1 + \text{ev } W)^a$, it turns the formal equation into (9.1) for $V = 1 + \text{ev } W$, and $\text{ev } W \in \mathfrak{m}$. (01 asserted this step without a lemma; error 3.)

Uniqueness among all near-one elements. Let $V, V' \in 1 + \mathfrak{m}$ both solve (9.1). For $a \in \mathbb{C}$ the formal divided-difference identity $V^a - V'^a = (V - V')D_a(V - 1, V' - 1)$ holds with $D_a \in \mathbb{C}[[X, Y]]$ of constant term a , and all evaluated supports lie in the Neumann monoid of $\text{supp}(V - 1) \cup \text{supp}(V' - 1)$, independently of a . Subtracting the two equations gives $(V - V')(1 + \sum_n^H c_{n,k} u^{\beta_n} D_{\alpha_n}(V - 1, V' - 1)) = 0$. The bracket is 1 plus an infinitesimal strong sum, hence a unit, so $V = V'$. $\square$

Define

$$s_k(u) = -\frac{\log u}{\log 2} - \frac{2\pi i k}{\log 2} - \frac{\text{Log}_{\mathfrak{m}} V_k(u)}{\log 2}.$$

Its real part is positive infinite and its imaginary part finite, so $s_k(u) \in \mathcal{T}_+$.

Theorem 9.4 (Complete fibre [01]). *For every $0 < u \in \mathfrak{m}_{\mathbb{R}}$, $\{s \in \mathcal{T}_+ : \zeta_D(s) = 1 + u\} = \{s_k(u) : k \in \mathbb{Z}\}$, without repetition, and branch k is characterized by $\text{st}(\text{Im } s) = -2\pi k / \log 2$.*

Proof. By construction $2^{-s_k} = uV_k$, and $n^{-s_k} = c_{n,k}u^{\alpha_n}V_k^{\alpha_n}$ for $n \geq 3$. After division by u , $\zeta_D(s_k) - 1 = u$ is exactly (9.1). Conversely, let $s = x + iy \in \mathcal{T}_+$ with $\zeta_D(s) - 1 = u$. The first-tail formula gives $u = 2^{-s}(1 + \eta)$, so $V = 2^{-s}/u \in 1 + \mathfrak{m}$. Its modulus gives $\exp_{\mathbf{No}}(-x \log 2)/u \in 1 + \mathfrak{m}_{\mathbb{R}}$, and its phase gives $e^{-i \operatorname{st}(y) \log 2} = 1$, so $\operatorname{st} y = -2\pi k / \log 2$ for a unique $k \in \mathbb{Z}$. The kernel of cis and the near-one logarithm give $-s \log 2 = \log u + \operatorname{Log}_{\mathfrak{m}} V + 2\pi i k$. Substituting into every n^{-s} and dividing by u gives (9.1), and Lemma 9.3 forces $V = V_k(u)$, so $s = s_k(u)$. Distinct k have distinct $\operatorname{st} \operatorname{Im} s$. $\square$

The exhaustion step matters. A formal inverse in one power-series ring would not exclude solutions whose supports lie outside it. The divided-difference argument and the first-tail formula exclude them throughout $\mathcal{T}_+$.

Corollary 9.5 (Conjugation and the real inverse [01]). $V_{-k}(u) = \overline{V_k(u)}$ and $s_{-k}(u) = \overline{s_k(u)}$. The point $s_0(u)$ is the unique real solution, and $x \mapsto \zeta_D(x) - 1$ is a bijection from $\{x \in \mathbf{No} : x > \mathbb{R}\}$ onto $\mathfrak{m}_{\mathbb{R}, > 0}$.

Proof. Conjugation sends $c_{n,k}$ to $c_{n,-k}$ and preserves strong sums; uniqueness gives the first formula. A real solution has $k = 0$. Monotonicity and positivity are Theorem 8.5(c). $\square$

Corollary 9.6 (All nonzero infinitesimal values [01]). The image of $s \mapsto \zeta_D(s) - 1$ on $\mathcal{T}_+$ is exactly $\mathfrak{m} \setminus \{0\}$. For each such w the fibre has exactly one point for every $k \in \mathbb{Z}$, obtained by replacing u^a with $w^a = E_{\text{fp}}(a \operatorname{Log} w)$ and $\log u$ with $\operatorname{Log} w$. Replacing $\operatorname{Log} w$ by $\operatorname{Log} w + 2\pi i j$ relabels branch k as $k + j$.

Proof. Lemmas 4.5 and 4.6 cover the complex powers, and the existence and divided-difference arguments are unchanged. The condition $2^{-s}/w \in 1 + \mathfrak{m}$ again determines k . Conversely every value is a nonzero infinitesimal by Theorem 8.5. A change of $\operatorname{Log} w$ by $2\pi i j$ multiplies w^a by $e^{2\pi i j a}$ and shifts the base point by $-2\pi i j / \log 2$. $\square$

The statement concerns fibres and values. No covering-map claim in the fine topology is made [01]. The fibres over the right wing are in Corollary 13.4.

9.1 Coefficients

For a multi-index ν put $\nu! = \prod \nu_n!$. Rising factorials have ordinary length, and $(a)_0 = 1$.

Lemma 9.7 (Weighted Lagrange formula [01]). For the formal solution $V = 1 + W$,

$$V^r = 1 + \sum_{\nu \neq 0} \frac{(-1)^{|\nu|} r(r+B+1)_{|\nu|-1}}{\nu!} x^\nu, \quad \log V = \sum_{\nu \neq 0} \frac{(-1)^{|\nu|} (B+1)_{|\nu|-1}}{\nu!} x^\nu,$$

with $B = B(\nu)$.

Proof. Below a weight cutoff only finitely many variables occur. Write $W = t\Theta(W)$ with $\Theta(T) = -\sum_n x_n(1+T)^{\alpha_n}$ and apply Lagrange's formula $[t^d]F(W) = \frac{1}{d}[T^{d-1}]F'(T)\Theta(T)^d$ to $F = (1+T)^r$ [11]. Since Θ is linear in the x_n , a monomial x^ν occurs only in Θ^d with $d = |\nu|$, with coefficient $\frac{(-1)^d d!}{\nu!} (1+T)^{d+B}$, since $\sum \nu_n \alpha_n = d+B$. Extracting T^{d-1} from $r(1+T)^{r-1+d+B}$ gives $\frac{(-1)^d r(d-1)!}{\nu!} \binom{r+d+B-1}{d-1}$, which is the stated coefficient. Differentiate at $r = 0$ for $\log V$. The univariate identity is a universal polynomial identity in the coefficients, so it needs no analytic hypothesis. $\square$

Evaluating [Lemma 9.7](#) at $x_n = c_{n,k} u^{\beta_n}$ gives

$$s_k(u) = -\frac{\log u}{\log 2} - \frac{2\pi i k}{\log 2} + \frac{1}{\log 2} \sum_{\nu \neq 0}^H \frac{(-1)^{|\nu|+1} (B+1)_{|\nu|-1}}{\nu!} e^{2\pi i k B} u^B,$$

an exact strong value at every positive infinitesimal u , whatever its normal form, by [Lemmas 4.6](#) and [9.2](#) [01].

Let $\mathcal{M} = \{\prod_{j=1}^d (n_j/2) : d \geq 0, n_j \geq 3\} \subseteq \mathbb{Q}_{\geq 1}$, let $A_d(r)$ count the ordered d -tuples with product r , and for $r > 1$ put

$$C_r = \sum_{d \geq 1} \frac{(-1)^{d+1} A_d(r)}{d!} (1 + \log_2 r)_{d-1},$$

a finite sum, since $d \leq \log r / \log(3/2)$ and each $n_j \leq 2r$.

Theorem 9.8 (Arithmetic coefficient formula [01]). *For every positive infinitesimal u and $k \in \mathbb{Z}$,*

$$s_k(u) = -\frac{\log u}{\log 2} - \frac{2\pi i k}{\log 2} + \frac{1}{\log 2} \sum_{r \in \mathcal{M}, r > 1}^H C_r e^{2\pi i k \log_2 r} u^{\log_2 r}.$$

The exponents are locally finite in $\mathbb{R}_{>0}$, and C_r includes every cancellation between distinct factorizations. The same formula holds for nonzero complex infinitesimal w .

Proof. Since $2^{B(\nu)} = \prod_n (n/2)^{\nu_n}$, two multi-indices have the same exponent exactly when they have the same product r . For fixed $d = |\nu|$ and r , $\sum_{\nu} 1/\nu! = A_d(r)/d!$, and every term has the phase $e^{2\pi i k B} = e^{2\pi i k \log_2 r}$. Regroup the evaluated [Lemma 9.7](#), which is legitimate by strong summability. $\square$

r	3/2	2	9/4		5/2	3	27/8		7/2	15/4
C_r	1	1	$-\frac{1}{2}(1 + \log_2 \frac{9}{4})$		1	$-\log_2 3$	$\frac{1}{6}(1 + L)(2 + L)$		1	$-(1 + \log_2 \frac{15}{4})$

Here $L = \log_2(27/8)$, and $C_4 = -\frac{1}{2}$. The coefficient at $r = 3$ shows why the collision bookkeeping matters. The length-one factorization $6/2$ contributes 1, and the two ordered length-two factorizations $(3/2)(4/2)$ contribute $-(1 + \log_2 3)$, so $C_3 = -\log_2 3$. The 23 coefficients with $r < 8$ are recorded in `data/01-inverse-zeta-inverse-coefficients.csv`. The smallest elements of $\mathcal{M}$ above 1 are $3/2 < 2 < 9/4 < 5/2$, so the first exponents are $\beta < 1 < 2\beta < \log_2(5/2)$. With $C_{3/2} = C_2 = 1$, $C_{9/4} = -\frac{1}{2}(1 + 2\beta)$ and the phase $e^{2\pi i k \log_2 r}$, this gives the expansion displayed in [Theorem 9.1](#) [01].

Theorem 9.9 (The integer-power sector [01]). $C_{2^m} = (-1)^{m+1}/m$ for every ordinary $m \geq 1$, independently of the branch. The only rational exponents occurring are the positive integers, so the rational-exponent sector of the correction is exactly $\text{Log}_m(1 + u)/\log 2$.

Proof. Every member of $\mathcal{M}$ has denominator a power of 2. So discarding the indices with an odd prime factor is a multiplicative projection of the generalized series ring. Applied to (9.1), only $n = 2^a$ ($a \geq 2$) survive, with phase 1. The projected equation $V_2 + \sum_{a \geq 2} u^{a-1} V_2^a = V_2/(1 - uV_2) = 1$ gives $V_2 = (1 + u)^{-1}$ and $-\log V_2 = \text{Log}_m(1 + u)$. If $\log_2 r = p/q$ in lowest terms, then $r^q = 2^p$ forces $r = 2^m$ and $q = 1$, and every 2^m is a product of m factors $4/2$. $\square$

Equivalently $\sum_{d=1}^m \frac{(-1)^{d+1}}{d!} (m-1)_{d-1} (m+1)_{d-1} = \frac{(-1)^{m+1}}{m}$. The projection acts on the formal u -expansion. For a nonmonomial u it must not be confused with discarding monomials of the evaluated normal form [01].

Corollary 9.10 (The real branch is rightmost [01]). *For $k \neq 0$, $s_k(u) - s_0(u) + \frac{2\pi ik}{\log 2} \sim \frac{e^{2\pi ik\beta} - 1}{\log 2} u^\beta$ and $\operatorname{Re} s_k(u) - s_0(u) \sim \frac{\cos(2\pi k\beta) - 1}{\log 2} u^\beta < 0$. The fibre is not an exact vertical arithmetic progression.*

Proof. β is irrational, since otherwise a power of 3 would equal a power of 2. So $e^{2\pi ik\beta} \neq 1$. Subtract the two expansions: the first exponents are $\beta < 1 < 2\beta < \log_2(5/2)$. $\square$

Example 9.11 ([01]). $s_0(\omega^{-1}) = \frac{\log \omega}{\log 2} + \frac{\omega^{-\beta}}{\log 2} + \frac{\omega^{-1}}{\log 2} - \frac{2\beta+1}{2\log 2} \omega^{-2\beta} + o(\omega^{-2\beta})$, which exhibits the intermediate scale $\omega^{-\log_2(3/2)}$.

The ordinary-complex illustration of 01's suite takes $u = 10^{-8}$ and the terms with $r < 8$, and compares them with roots of $\zeta(s) = 1 + u$ computed at 100 digits:

$$\begin{aligned} s_0 &\approx 26.575454935879472, & s_{\pm 1} &\approx 26.575398807000944 \mp 9.064735632317692 i, \\ s_{\pm 2} &\approx 26.575439317457936 \mp 18.129414142729476 i, \end{aligned}$$

with errors of about 1.0×10^{-23} and zeta residuals below 7×10^{-32} . These numbers illustrate signs and coefficients. They are not error bounds or surreal proofs [01].

10 Algebraic independence at infinite arguments

Two sources prove independence theorems for zeta values at a fixed infinite argument, for different families over different base fields. 02 treats dilations $\zeta_D^{(j)}(kx)$ at a positive infinite real x , together with x , $L_{\text{St}}(x)$ and $\Gamma_{\text{St}}(x)$. 03 treats real shifts $\zeta_D^{(k)}(X + a)$ at a positive infinite $X \in \Pi$, over fields containing a slow Hahn field and Gamma values. Both keep the standard literature in view. Arithmetic-function independence has a classical theory, with Pong [9] an explicit antecedent. No novelty is claimed for the functional-independence principle [02]. What is new is the exact evaluation at one surreal argument, over the stated base fields. Standard parts destroy the information, since $\operatorname{st} \zeta_D(kx) = 1$, and nothing here concerns the ordinary values $\zeta(2), \zeta(3), \dots$ [02].

10.1 Dilations

For distinct primes $p_1, \dots, p_M$, the prime projection $\operatorname{pr}(a) = \sum_{\alpha \in \mathbb{N}^M} a(p_1^{\alpha_1} \dots p_M^{\alpha_M}) Z^\alpha \in \mathbb{C}[[Z_1, \dots, Z_M]]$ preserves convolution, since every divisor of an integer supported on these primes is supported on them too. It turns δ into $\delta_Z = -\sum_i (\log p_i) Z_i \partial_{Z_i}$. Let $a_{k,0}$ be the indicator of perfect k th powers and $a_{k,j} = \delta^j a_{k,0}$. Then

$$A_{k,0} = \operatorname{pr}(a_{k,0}) = \prod_{i=1}^M (1 - Z_i^k)^{-1}, \quad A_{k,j} = \delta_Z^j A_{k,0}, \quad \mathcal{D}_s(a_{k,j}) = k^j \zeta_D^{(j)}(ks).$$

The factor k^j appears because δ differentiates in s , whereas $\zeta_D^{(j)}(ks)$ differentiates before dilating.

Lemma 10.1 (A nonzero Jacobian [02]). *Let $\mathcal{K} \subset \mathbb{N}_{\geq 1}$ be finite, $J_k \geq 0$ for $k \in \mathcal{K}$, and $M = \sum_k (J_k + 1)$. For M distinct primes, the Jacobian of $\{A_{k,j} : k \in \mathcal{K}, 0 \leq j \leq J_k\}$ with respect to $Z_1, \dots, Z_M$ is nonzero.*

Proof. The lowest nonconstant homogeneous part of $A_{k,j}$ is $\sum_i (-k \log p_i)^j Z_i^k$. The lowest-degree part of the Jacobian is therefore $J_0 = \det[k(-k \log p_i)^j Z_i^{k-1}]$, and higher parts cannot cancel it. Partition the columns into groups I_k of sizes $J_k + 1$ and look at the coefficient

of $\prod_k \prod_{i \in I_k} Z_i^{k-1}$. Distinct k give distinct exponents, including $k = 1$ with exponent 0, so the monomial forces the rows of k onto the columns of I_k . Up to a sign and powers of k , the coefficient is $\prod_k \det[(\log p_i)^j]_{j \leq J_k, i \in I_k}$, a product of Vandermonde determinants in distinct reals. $\square$

Lemma 10.2 (Formal Jacobian criterion [02]). *Power series $F_1, \dots, F_M \in \mathbb{C}[[Z_1, \dots, Z_M]]$ with nonzero Jacobian are algebraically independent over $\mathbb{C}$.*

Proof. Let $P \neq 0$ be a relation of least degree. Differentiating $P(F) = 0$ gives a linear system with the Jacobian as matrix, invertible over the fraction field. So every $\partial_j P(F) = 0$, and by minimality every $\partial_j P$ is zero. In characteristic zero P is then constant, a contradiction. $\square$

Theorem 10.3 (Dilation jets [02]). *For every $s \in \mathcal{T}_+$, complex or real, $\{\zeta_D^{(j)}(ks) : k \geq 1, j \geq 0\}$ is algebraically independent over $\mathbb{C}$. So is $\{\zeta_D^{(j)}(qs) : q \in \mathbb{Q}_{>0}, j \geq 0\}$.*

Proof. A relation uses finitely many pairs; enlarge them to initial segments $j \leq J_k$. By the two lemmas the projected $A_{k,j}$ are independent, hence so are the $a_{k,j}$, and injectivity of $\mathcal{D}_s$ with the nonzero scalars k^j transfers this to the values. For rational dilations use a common denominator M and work at $s/M \in \mathcal{T}_+$. $\square$

The only arithmetic input is that distinct primes have distinct logarithms. No conjecture on logarithms of primes (such as Schanuel's) is used.

10.2 Leading forms and the Gamma scale

Fix a positive infinite real x . Let $\mathcal{B}_x = \mathcal{D}_x(\mathcal{A})$ and $\mathcal{F}_x^{\text{Dir}} = \text{Frac}(\mathcal{B}_x[x])$. The needed scale inequalities, for ordinary $c > 0$ and $d, h \geq 0$, are $(\log x)^h \prec x$ ($h \geq 1$) and $x^d(\log x)^h \prec e^{cx}$. They follow from the elementary theory of $(\mathbf{No}, \exp)$ [4].

Lemma 10.4 (Polynomial coefficients [02]). *Let $a_0, \dots, a_D \in \mathcal{A}$, not all zero, and $P_n(T) = \sum_d a_d(n)T^d$. Let n_0 be the least index with $P_{n_0} \neq 0$, with highest term cT^{d_0} . Then $\sum_d x^d \mathcal{D}_x(a_d) = cx^{d_0}n_0^{-x}(1 + \eta)$ with $\eta \in \mathfrak{m}$. So $\mathcal{A}[T] \rightarrow \mathbf{SC}$, $T \mapsto x$, is injective.*

Proof. The left side is $\sum_n^H P_n(x)n^{-x}$, and $P_{n_0}(x) = cx^{d_0}(1 + \eta')$. After division by the n_0 term, the term at $n > n_0$ has monomial part $(q_n/q_{n_0})x^e$ with $|e| \leq D$. Since $q_n/q_{n_0} = \exp(-x \log(n/n_0))$ and $x^D \prec \exp(x \log(n/n_0))$, we get $q_n/q_{n_0} \prec x^{-D}$, and every such monomial is infinitesimal. The normalized tail is a finite sum of strong arithmetic families times fixed factors, with support of positive valuation, so its sum is infinitesimal. $\square$

The step $q_n/q_{n_0} \prec x^{-D}$ is the one line that 02's proof leaves implicit (error 10).

Theorem 10.5 (Leading forms [02]). *Every nonzero $f \in \mathcal{F}_x^{\text{Dir}}$ is uniquely $cx^d q^{-x}(1 + \eta)$ with $c \in \mathbb{C}^\times$, $d \in \mathbb{Z}$, $q \in \mathbb{Q}_{>0}$ and $\eta \in \mathfrak{m}$. Hence the family $\{\zeta_D^{(j)}(kx)\}$ is algebraically independent over $\mathbb{C}(x)$.*

Proof. Apply Lemma 10.4 to a numerator and a denominator. For uniqueness, $c'x^{d'}q'^{-x} \sim 1$ forces $q' = 1$ by exponential separation, then $d' = 0$, then $c' = 1$. For the second claim, clear denominators; the injectivity in Lemma 10.4 makes every coefficient vanish as an arithmetic function, and Theorem 10.3 finishes. $\square$

Lemma 10.6 (A logarithmic scale [02]). *$L_{\text{St}}(x) = x \log x (1 + \varepsilon)$ with $\varepsilon \in \mathfrak{m}_{\mathbb{R}}$. $L_{\text{St}}(x)$ is transcendental over $\mathcal{F}_x^{\text{Dir}}$, and every nonzero element of $\mathcal{F}_x^{\text{Dir}}(L_{\text{St}}(x))$ is uniquely $cx^d(\log x)^h q^{-x}(1 + \eta)$ with $c \in \mathbb{C}^\times$, $d, h \in \mathbb{Z}$, $q \in \mathbb{Q}_{>0}$ and $\eta \in \mathfrak{m}$.*

Proof. In $P(x, L_{\text{St}}(x))$ the monomial $U^d V^h$ has leading form $x^{d+h}(\log x)^h$. The lexicographically largest pair $(d+h, h)$ determines a unique original (d, h) , whose term dominates the others. With arithmetic coefficients, write the expression as $\sum_n^H P_n(x, L_{\text{St}}(x))n^{-x}$ with uniformly bounded degrees. Choose the least n_0 , and then the largest pair. Later indices have an exponential disadvantage that dominates fixed powers of x and $\log x$. So $\mathcal{A}[U, V] \rightarrow \mathbf{SC}$, $(U, V) \mapsto (x, L_{\text{St}}(x))$, is injective, and clearing denominators gives transcendence over $\mathcal{F}_x^{\text{Dir}}$. Applying the leading term to a numerator and a denominator gives the leading form. Uniqueness follows by comparing, in turn, the exponential scale, the power of x , the power of $\log x$ and the coefficient. $\square$

Theorem 10.7 (Joint scale separation [02]). *$L_{\text{St}}(x)$ and $\Gamma_{\text{St}}(x)$ are algebraically independent over $\mathcal{F}_x^{\text{Dir}}$. In fact $\Gamma_{\text{St}}(x)$ is transcendental over $\mathcal{F}_x^{\text{Dir}}(L_{\text{St}}(x))$. More generally, every positive real G with $\log G/x > \mathbb{R}$ is transcendental over $\mathcal{F}_x^{\text{Dir}}(L_{\text{St}}(x))$.*

Proof. A nonzero coefficient b has $\log |b|/x = -\log q + \text{infinitesimal}$, a finite slope, whereas $\log \Gamma_{\text{St}}(x)/x = L_{\text{St}}(x)/x \sim \log x$ is infinite. In $\sum_{r \leq R} b_r \Gamma_{\text{St}}(x)^r = 0$ with $b_R \neq 0$, each lower term relative to the top term is $\exp(\log |b_r/b_R| - (R-r)L_{\text{St}}(x)) \in \mathfrak{m}$. The same argument applies to any G with infinite slope. $\square$

Theorem 10.8 (Principal independence [02]). *For every positive infinite real x , the family $\{x, L_{\text{St}}(x), \Gamma_{\text{St}}(x)\} \cup \{\zeta_D^{(j)}(kx) : k \geq 1, j \geq 0\}$ is algebraically independent over $\mathbb{C}$. The derivatives are coordinate derivatives.*

Proof. [Theorem 10.3](#) gives the jets. [Lemma 10.4](#) adjoins x , [Lemma 10.6](#) adjoins $L_{\text{St}}(x)$ even over $\mathcal{F}_x^{\text{Dir}}$, and [Theorem 10.7](#) adjoins $\Gamma_{\text{St}}(x)$. $\square$

The Gamma step is robust: an infinitesimal relative perturbation of the baseline changes nothing. The log-Gamma step uses the scale $x \log x$ of L_{St} , not a uniqueness property of surreal Gamma. The theorem is compatible with the gauge nonuniqueness of the gamma-functions report, whose members all have the same growth scales.

10.3 Shifts

Lemma 10.9 (Chebyshev determinant [03]). *For distinct reals $a_1, \dots, a_d$, integers $m_i \geq 1$, $r = \sum m_i$, and distinct reals $t_1, \dots, t_r$, the matrix $[e^{-a_i t_j} (-t_j)^k]_{(i,k),j}$ ($0 \leq k < m_i$) is nonsingular over $\mathbb{C}$.*

Proof. A nonzero real exponential polynomial $\sum_i e^{-a_i t} P_i(t)$ with $\deg P_i < m_i$ has at most $r-1$ real zeros. For one exponent this is the polynomial bound. Otherwise multiply by $e^{a_1 t}$ and differentiate m_1 times. The first polynomial disappears, and each other term becomes $e^{(a_1 - a_i)t} (D + a_1 - a_i)^{m_1} P_i$, which is nonzero of the same degree. Rolle's theorem and induction finish. A singular real matrix would give such a function vanishing at r points, and a complex null vector gives a real one. $\square$

This is the only place where an ordinary mean-value argument is used. No surreal Rolle theorem is invoked.

Lemma 10.10 (Finite-prime independence [03]; proof completed [merge]). *Let $F \supseteq \mathbb{C}$ be a field, S a finite set of primes, and x_p the arithmetic function supported at p with value 1. For distinct reals a_i and integers $m_i \geq 1$, the functions $A_{a_i, k}(n) = n^{-a_i} (-\log n)^k$ ($0 \leq k < m_i$, value at $n=1$ as in [Definition 8.4](#)) satisfy no nonzero polynomial relation with coefficients in $F[x_p : p \in S]$, or, after clearing denominators, in its fraction field.*

Proof. Let $r = \sum m_i$ and choose r distinct primes $q_j \notin S$. Projecting onto integers supported on $S \cup \{q_j\}$ is an algebra map onto $F[[x_p, t_j]]$. With $E_i = \prod_{p \in S} (1 - p^{-a_i} x_p)^{-1}$ and $D = -\sum_p (\log p) x_p \partial_{x_p} - \sum_j (\log q_j) t_j \partial_{t_j}$, the projection of $A_{a_i, k}$ is $Y_{i, k} = D^k (E_i \prod_j (1 - q_j^{-a_i} t_j)^{-1})$. Every t -coefficient of $Y_{i, k}$ is a rational function in the x_p that is regular at $x = 0$: it is obtained from products of the factors $(1 - p^{-a_i} x_p)^{-1}$ by $D_S = -\sum_p (\log p) x_p \partial_{x_p}$. Hence $Y_{i, k} \in R[[t]]$, where $R = F(x) \cap F[[x]]$ is the local ring at 0, and $R[[t]]$ embeds both in $F(x)[[t]]$ and in $F[[x, t]]$. At $t = 0$ the Jacobian has entries $q_j^{-a_i} \sum_{b \leq k} \binom{k}{b} D_S^{k-b} E_i (-\log q_j)^b$. For each i its row block is a lower-triangular transform, with diagonal $E_i \neq 0$, of the rows $q_j^{-a_i} (-\log q_j)^b$. So the Jacobian is invertible over $F(x)$ by [Lemma 10.9](#) with $t_j = \log q_j$. The formal inverse function theorem (solve degree by degree, one finite linear system over $F(x)$ at each degree) shows that the $Y_{i, k}$ are algebraically independent over $F(x)$. A relation P with coefficients in $F[x_p] \subseteq R$ satisfies $P(Y) = 0$ in $F[[x, t]]$, hence in $R[[t]]$ and in $F(x)[[t]]$, so $P = 0$. $\square$

03's proof works in $F(x)[[t]]$ although the projected objects live in $F[[x, t]]$, and $F[[x]]$ is not a subring of $F(x)$. The passage through $R[[t]]$ supplies the missing step (error 17). *Antecedent.* For $F = \mathbb{C}$ the lemma is a special case of Pong's Theorem 5.5 [9]: take the function $f = \mathbf{1}$ and $\alpha_i = -a_i$, independence over the algebra $\mathcal{S} \supseteq \mathbb{C}[x_p]$ of arithmetic functions with finitely generated support. Pong's proof also chooses fresh primes and shows that a determinant is nonzero. The F -general form follows from the case $F = \mathbb{C}$ by linear disjointness. The 2020 Addendum [10] was not checked. 03 cites Pong as an antecedent but presents its proof as self-contained. The new content of this subsection is the Hahn-support separation in [Lemma 10.12](#) and [Theorems 10.13](#) and [10.14](#) (error 18). Pong's Corollary 5.6 and Example 5.5 (Ostrowski) are the classical-analytic shadow of [Theorem 10.11](#).

Fix $X \in \Pi$, $X > 0$, and write $q_n = e^{-X \log n}$ as in [Section 8](#). Then $\text{ev}_X(a) = \sum_n^H a(n) q_n$ is the $c = \varepsilon = 0$ case of [Theorem 8.1](#), and it is injective [03].

Theorem 10.11 (Shift jets over a monomial field [03]). $\{\zeta_D^{(k)}(X + a) : a \in \mathbb{R}, k \in \mathbb{N}\}$ is algebraically independent over $M_X = \mathbb{C}(e^{-\lambda X} : \lambda \in \mathbb{R})$.

Proof. Take finitely many shifts a_i , all derivatives $k < m_i$, and a relation cleared of denominators, $\sum_{j \leq N} e^{-\lambda_j X} P_j(Y) = 0$, with distinct λ_j and complex polynomials P_j , not all zero. Each $P_j(Y)$ is the evaluation of an arithmetic function with exponents in $\log \mathbb{N}_{\geq 1}$. So terms from different summands can share a monomial only if $\lambda_j - \lambda_\ell \in \log \mathbb{Q}_{>0}$. Distinct cosets modulo $\log \mathbb{Q}_{>0}$ have disjoint supports and vanish separately. In one coset, factor out one $e^{-\lambda_0 X}$; the other factors are r^{-X} with $r \in \mathbb{Q}_{>0}$, involving a finite set S of primes. Clearing negative prime exponents gives a nonzero relation $Q((p^{-X})_{p \in S}, Y) = 0$. By injectivity of ev_X this is a relation among arithmetic functions, contradicting [Lemma 10.10](#) with $F = \mathbb{C}$. $\square$

The base field must be a *generated* field. Over a Hahn completion containing the zeta series, transcendence would fail trivially [03].

Assume now that X is a positive infinite Conway monomial, and let $F_X^{\text{Hahn}} = \{\sum_{r \in S}^H c_r X^{-r} : S \subseteq \mathbb{R} \text{ well ordered}\} \cong \mathbb{C}((t^{\mathbb{R}}))$ with $t = X^{-1}$. Since $X/\log X$ is infinite, the monomials $e^{-\lambda X} X^{-r} = \exp(-\lambda X - r \log X)$ are exact Conway monomials, ordered lexicographically in (λ, r) .

Lemma 10.12 (Slow coefficients [03]). *For arbitrary $f_n \in F_X^{\text{Hahn}}$, $\sum_n^H f_n e^{-X \log n}$ is strongly summable, and it realizes an injective homomorphism from the arithmetic algebra over F_X^{Hahn} . Its block at first-scale exponent $\log n$ is exactly f_n .*

Proof. The support is $\bigcup_n \{\log n\} \times \text{supp } f_n$ in the lexicographic order. The first coordinates increase and each fibre is well ordered, so the union is well ordered, and different fibres are disjoint. Products use the finitely many factorizations of n . $\square$

Theorem 10.13 (Over the slow Hahn field [03]). *For a positive infinite monomial X , the shift family of Theorem 10.11 is independent over $E_X = F_X^{\text{Hahn}}(e^{-\lambda X} : \lambda \in \mathbb{R})$, the generated field (not a further Hahn completion).*

Proof. Repeat the previous proof with F_X^{Hahn} in place of $\mathbb{C}$. Distinct λ give monomials that are linearly independent over F_X^{Hahn} . Lemma 10.12 gives injectivity and separates cosets by their first coordinate. Finish with Lemma 10.10 for $F = F_X^{\text{Hahn}}$. $\square$

Theorem 10.14 (Over a field containing the Gamma ray [03]). *Let $T_X = e^{X \log X}$ and $E_X^\Gamma = E_X(T_X)$. The shift family is algebraically independent over E_X^Γ . This field contains $\Gamma_{\text{St}}(X + b) = \Gamma_\chi(X + b)$ for every ordinary real b and every phase χ .*

Proof. The field generated by E_X and the jets lies in the Hahn field on the two-scale group, and $X \log X \gg X \gg \log X$. So integer powers of T_X lie in distinct first-coordinate fibres of a three-scale workspace. Hence T_X is transcendental over that field, and a relation $\sum_j T_X^j P_j(Y) = 0$ splits into $P_j(Y) = 0$, which contradicts Theorem 10.13. By Theorem 11.2, $\Gamma_{\text{St}}(X + b) = \sqrt{2\pi} T_X e^{-X} X^{b-1/2} \text{Exp}_m(S_{\text{St}}(X) + Q_b(X))$. The last factor lies in $\mathbb{C}[[X^{-1}]] \subset F_X^{\text{Hahn}}$, and the value is real and phase-free. $\square$

The Gamma values are not claimed to be mutually independent: the recurrence relates them. Coefficients from all of **SC** are not allowed, since every element is algebraic over a field that contains it [03]. Example: no algebraic equation for $\zeta_D(\omega)$ has nonzero coefficients in the field generated by $\mathbb{C}((\omega^{-\mathbb{R}}))$, all $e^{-\lambda\omega}$, and all $\Gamma_{\text{St}}(\omega + b)$ with $b \in \mathbb{R}$.

10.4 How the two independence theorems compare

The following table compares the two theorems.

	02 (Sections 10.1 and 10.2)	03 (Section 10.3)
Point	any positive infinite real x ; complex $s \in \mathcal{T}_+$ for the jets alone	positive $X \in \Pi$; a monomial for the Hahn fields
Family	dilations $\zeta_D^{(j)}(kx)$, $\zeta_D^{(j)}(qx)$, with $x, L_{\text{St}}(x), \Gamma_{\text{St}}(x)$	real shifts $\zeta_D^{(k)}(X + a)$
Base field	$\mathbb{C}; \mathbb{C}(x); \mathcal{F}_x^{\text{Dir}}$ for $L_{\text{St}}, \Gamma_{\text{St}}$	$M_X; E_X; E_X^\Gamma \ni \Gamma_{\text{St}}(X + b)$
Gamma	$L_{\text{St}}(x), \Gamma_{\text{St}}(x)$ independent over $\mathcal{F}_x^{\text{Dir}}$ (all Dirichlet values and x)	Gamma values in the base field; no mutual independence

Neither theorem implies the other. For a positive infinite monomial X , the overlap $a = 0$ (03, over $E_X \supseteq \mathbb{C}(X)$) and $k = 1$ (02) gives independence of $\{\zeta_D^{(j)}(X)\}$ over $\mathbb{C}(X)$; 03's field M_X does not contain X . 03's base field E_X^Γ contains X , all real powers of X and $\Gamma_{\text{St}}(X)$, but not $\log X$. Indeed E_X^Γ lies in the Hahn field whose monomials are $\exp(jX \log X - \lambda X - r \log X)$ ($j \in \mathbb{Z}, \lambda, r \in \mathbb{R}$), and the leading monomial $\exp(\text{pp}(\log \log X))$ of $\log X$ is not of this form, since $0 < \text{pp}(\log \log X) \prec \log X$. So E_X^Γ does not contain $L_{\text{St}}(X)$ either: that would give $\log X = (L_{\text{St}}(X) + X - \frac{1}{2} \log 2\pi - S_{\text{St}}(X))/(X - \frac{1}{2}) \in E_X^\Gamma$. 02 covers dilations and nonmonomial x , but has no Hahn coefficients and no shifts. 02's dilation family, indicators of k th powers times log powers, is not of Pong's form $T^\alpha \partial^j f$, and whether another classical result covers dilations was not searched. A combined dilation-and-shift statement over 03's field is proved by neither source and is open (Question 23.2). 01's real inverse $s_0(u)$ involves $\log u$ and $u^{\log_2(3/2)}$, which is consistent with the independence of x and $\zeta_D(x)$.

11 The Stirling logarithm at infinity

11.1 Definition

On the Stirling half-plane $\mathcal{S}_\infty = \{\operatorname{Re} z > 0, |z| \text{ infinite}\}$ put

$$L_{\text{St}}(z) = \left(z - \frac{1}{2}\right) \operatorname{Log} z - z + \frac{1}{2} \log 2\pi + S_{\text{St}}(z), \quad S_{\text{St}}(z) = \sum_{k \geq 1}^H \frac{B_{2k}}{2k(2k-1)} z^{1-2k}, \quad (11.1)$$

with the principal finite angle. The series is strongly summable because $z^{-1} \in \mathfrak{m}$, whatever the factorial growth of the Bernoulli numbers [01–06]. At real $x > \mathbb{R}$ put $\Gamma_{\text{St}}(x) = \exp_{\mathbf{No}} L_{\text{St}}(x) > 0$. With $\Gamma^\#$ on positive finite arguments this is the positive-real baseline on $\mathbf{No}_{>0}$. It satisfies $\Gamma(x+1) = x\Gamma(x)$ at every $x \in \mathbf{No}_{>0}$, by (6.6) at finite x and by applying $\exp_{\mathbf{No}}$ to Proposition 11.1 at infinite x [05]. On real $x > \mathbb{R}$, L_{St} is the gamma-functions report's L_0 (eq:L0-infinite); on $\{x > \mathbb{R}, y \in \mathcal{O}_{\mathbb{R}}\}$ it is its $\tilde{L}_0$ (eq:complex-infinite, eq:log-infinite-complex). The strong interpretation of Stirling's series at infinite arguments is part of the Costin–Ehrlich extension framework [5], which treats log-Gamma explicitly, and Berarducci and Mantova treat transseries as surreal functions [7]. The classical series is an asymptotic expansion, valid in sectors $|\arg z| \leq \pi - \delta$ [14]. On $\mathcal{S}_\infty$, (11.1) is a *prescription*. It also covers points such as $\varepsilon + iT$ with $\varepsilon \in \mathfrak{m}_{>0}$ and $T > \mathbb{R}$ (a doubt raised in the review of 06). Nothing ties its values there to the analytic behaviour of a Gamma function; 05 calls it *bare Stirling* for this reason.

Proposition 11.1 (Recurrence, Gauss and formal uniqueness [01, 02, 03, 05, 06]). *On $\mathcal{S}_\infty$, for ordinary $m \geq 1$,*

$$L_{\text{St}}(z+1) - L_{\text{St}}(z) = \operatorname{Log} z, \quad \sum_{r=0}^{m-1} L_{\text{St}}\left(z + \frac{r}{m}\right) = L_{\text{St}}(mz) + \frac{m-1}{2} \log 2\pi + \left(\frac{1}{2} - mz\right) \log m.$$

Among expressions $(z - \frac{1}{2}) \operatorname{Log} z - z + c + \sum_{k \geq 1} a_k z^{-k}$ with $a_k \in \mathbb{C}$, the recurrence determines every a_k once c is fixed.

Proof. Work formally with $D = d/dZ$ and $u/(e^u - 1) = \sum B_n u^n/n!$. Then $\frac{D}{e^D - 1} \log Z = \log Z - \frac{1}{2Z} - \sum_{k \geq 1} \frac{B_{2k}}{2k} Z^{-2k} = L'_{\text{St}}(Z)$, and multiplying by $e^D - 1$, the formal shift, gives $L'_{\text{St}}(Z+1) - L'_{\text{St}}(Z) = 1/Z$. So $L_{\text{St}}(Z+1) - L_{\text{St}}(Z) - \log Z$ has zero derivative. Its expansion has zero constant term, so it vanishes. For Gauss, the difference R_m of the two sides satisfies $R_m(Z+1/m) = R_m(Z)$ by the recurrence. Expanding $\operatorname{Log}(Z + \frac{r}{m}) = \operatorname{Log} Z + \operatorname{Log}_{\mathfrak{m}}(1 + \frac{r}{mZ})$, the $\operatorname{Log} Z$ terms agree because $\sum_{r < m} (Z + \frac{r}{m} - \frac{1}{2}) = mZ - \frac{1}{2}$, the $\log m$ terms cancel, and the constant terms are $\frac{m}{2} \log 2\pi$ on both sides, since the constant $\frac{r}{m}$ of $(Z + \frac{r}{m} - \frac{1}{2}) \operatorname{Log}_{\mathfrak{m}}(1 + \frac{r}{mZ})$ cancels the $-\frac{r}{m}$ of $-(Z + \frac{r}{m})$. So R_m is a series in negative powers of Z [merge]. A nonzero such series with leading term cZ^{-d} changes by $-(cd/m)Z^{-d-1} \neq 0$ under the shift, so $R_m = 0$. Uniqueness is the same argument: a nonzero difference $\sum a_k z^{-k}$ with first term $a_k z^{-k}$ has shift difference $-ka_k z^{-k-1} + \dots \neq 0$. Substituting $Z = z$ is legitimate by Lemma 4.3, using $\operatorname{Log}(z+a) = \operatorname{Log} z + \operatorname{Log}_{\mathfrak{m}}(1+a/z)$ on $\mathcal{S}_\infty$ for finite a with $z+a \in \mathcal{S}_\infty$, and $\operatorname{Log}(mz) = \operatorname{Log} z + \log m$. $\square$

These identities duplicate the gamma-functions report's `lem:formal-shift`, `eq:baseline-recurrence`, `lem:periodic` and `prop:gauss0`, which prove them on real arguments by coefficient matching and a periodicity lemma. The proof above is the Bernoulli-operator route of 02, 03 and 05. 01, 04 and 06 instead derive the recurrence (01 also Gauss) from the Hurwitz shift and distribution identities; that route is Remark 12.7 [01, 04, 06]. The uniqueness holds only among normalized formal series. It does not exclude corrections smaller than every inverse power (Section 11.5) [01, 05].

The formal logarithmic derivative is

$$\psi_{\text{St}}(z) = \text{Log } z - \frac{1}{2z} - \sum_{k \geq 1}^{\text{H}} \frac{B_{2k}}{2k} z^{-2k}. \quad (11.2)$$

At positive infinite real A , the raw classical formula $\psi'(A) = \sum_{n \geq 0} (A+n)^{-2}$ must not be used. Every summand has the leading monomial of A^{-2} with the same coefficient, so the family is not strongly summable. The derivative of (11.2) is the correct object [01, 02, 03, 04]. The gamma-functions report records the same guardrail in its unlabelled warning “An illegitimate substitution into an ordinary sum”.

11.2 Translation and the ratio cocycle

Theorem 11.2 (Bernoulli translation formula [02, 03, 05]). *Let $Z \in \mathbf{SC}$ with $Z^{-1} \in \mathbf{m}$, let $a \in \mathcal{O}$, and let $\text{Log } Z$ be a logarithm of Z with finite imaginary part. Put $\mathcal{L}(Z+a) = (Z+a - \frac{1}{2})(\text{Log } Z + \text{Log}_{\mathbf{m}}(1+a/Z)) - (Z+a) + \frac{1}{2} \log 2\pi + S_{\text{St}}(Z+a)$. Then*

$$\mathcal{L}(Z+a) = \left(Z+a - \frac{1}{2}\right) \text{Log } Z - Z + \frac{1}{2} \log 2\pi + \sum_{n \geq 1}^{\text{H}} \frac{(-1)^{n+1} B_{n+1}(a)}{n(n+1)} Z^{-n}. \quad (11.3)$$

If $Z, Z+a \in \mathcal{S}_{\infty}$ with the principal branches, then $\mathcal{L} = L_{\text{St}}$, and

$$\begin{aligned} L_{\text{St}}(Z+a) - L_{\text{St}}(Z) - a \text{Log } Z &= Q_a(Z) := \sum_{n \geq 1}^{\text{H}} \frac{(-1)^{n+1} (B_{n+1}(a) - B_{n+1})}{n(n+1)} Z^{-n} \\ &= \frac{a(a-1)}{2Z} - \frac{a(a-1)(2a-1)}{12Z^2} + \cdots \end{aligned}$$

Hence, for every phase, $\Gamma_{\chi}(Z+a)/(\Gamma_{\chi}(Z)E_{\chi}(a \text{Log } Z)) = \text{Exp}_{\mathbf{m}}(Q_a(Z))$, independent of χ . At $a=1$, $Q_1=0$ gives the recurrence again.

Proof. Work formally in $u = Z^{-1}$, with a an indeterminate and $\text{Log } Z$ a formal symbol of derivative $1/Z$. Using $B_1(a) = a - \frac{1}{2}$, the Z -derivative of the right side is $\text{Log } Z + \sum_{n \geq 1} (-1)^{n+1} B_n(a) Z^{-n}/n$. The Z -derivative of the left side is $\psi_{\text{St}}(Z+a)$ expanded in a/Z . By the binomial theorem and $B_n(a) = \sum_j \binom{n}{j} B_j a^{n-j}$ its coefficient of Z^{-n} is the same, which is 03's computation of $L'(z+a)$. So the two sides differ by a constant in Z . Both have constant term $\frac{1}{2} \log 2\pi$: on the left, the term $Z \cdot a/Z$ from $(Z+a - \frac{1}{2}) \text{Log}_{\mathbf{m}}(1+a/Z)$ cancels $-a$. So (11.3) holds in $\mathbb{Q}[a]((u))$ together with the log terms. To evaluate, write $a = c + \delta$ with $c \in \mathbb{C}$ and $\delta \in \mathbf{m}$. Every coefficient is a polynomial in δ , so the family is jointly strongly summable in u and δ (Lemma 4.3). This joint evaluation is the step that 02's (8.6) asserts without detail (error 11). On $\mathcal{S}_{\infty}$ the principal branches satisfy $\text{Log}(Z+a) = \text{Log } Z + \text{Log}_{\mathbf{m}}(1+a/Z)$, since both sides are logarithms of $Z+a$ with imaginary parts differing by an infinitesimal. Apply E_{χ} and its local law on the infinitesimal $Q_a(Z)$. Finally $B_{n+1}(1) = B_{n+1}$ for $n \geq 1$. $\square$

03 proves this on $\mathcal{T}_+$, 02 for real $x > \mathbb{R}$ and $h \in \mathcal{O}$, and 05 as a formal identity at infinite $|Z|$ (its eq. 6.11, the surreal form of DLMF 5.11.8). The evaluation at $Z = \pm iT$, used in Theorem 17.1, needs only the first statement.

Theorem 11.3 (Ratio cocycle [05]). *For $|Z|$ infinite and finite a, b, c , put $Q_{a,b}(Z) = \sum_{n \geq 1}^{\text{H}} \frac{(-1)^{n+1} (B_{n+1}(a) - B_{n+1}(b))}{n(n+1)} Z^{-n}$ and $R_{a,b} = \text{Exp}_{\mathbf{m}}(Q_{a,b})$. Then $R_{a,b}$ is well defined and nonzero, and*

$$\begin{aligned} R_{a,b} R_{b,c} &= R_{a,c}, \quad R_{a,a} = 1, \quad R_{1,0} = 1, \quad \overline{R_{a,b}(Z)} = R_{\bar{a},\bar{b}}(\bar{Z}), \\ \frac{R_{a,b}(Z+1)}{R_{a,b}(Z)} &= \frac{1+a/Z}{1+b/Z} (1+1/Z)^{-(a-b)}, \quad R_{a,b}(Z) = 1 + \frac{(a-b)(a+b-1)}{2Z} + \mathcal{O}(Z^{-2}). \end{aligned}$$

The powers are near-one powers. For example $R_{1/2,0}(Z) = 1 - \frac{1}{8Z} + \frac{1}{128Z^2} + \frac{5}{1024Z^3} - \frac{21}{32768Z^4} + O(Z^{-5})$.

Proof. Expand the Bernoulli polynomials at the standard parts of a, b and evaluate jointly in Z^{-1} , $a - \text{st } a$, $b - \text{st } b$. The cocycle follows by cancelling the middle polynomial, $R_{1,0} = 1$ from $B_k(1) = B_k$ ($k \geq 2$), and conjugation from the real coefficients. For the shift, subtract the two instances of (11.3) and use the recurrence: $Q_{a,b}(Z+1) - Q_{a,b}(Z) = \text{Log}_m(1+a/Z) - \text{Log}_m(1+b/Z) - (a-b)\text{Log}_m(1+1/Z)$. The leading term uses $B_2(a) = a^2 - a + \frac{1}{6}$. $\square$

$Q_{a,b} = Q_a - Q_b$, where Q_a is 03's R_a , and at $b = 0$ with $a = h$ it is the correction in 02's (8.6). The leading term $(a-b)(a+b-1)/(2Z)$ matches 03's $a(a-1)/(2z)$. The coefficient $-21/32768$ is an addition to 05's expansion, confirmed numerically in the review of 05.

Remark 11.4 (Reading $R_{a,b}$ as a Gamma ratio [05]). 05 glosses $R_{1/2,0}$ as the formal normalization of $\Gamma(Z + \frac{1}{2})/(Z^{1/2}\Gamma(Z))$, and the same reading makes $R_{a,b}$ the formal normalization of $\Gamma(Z + a)/(Z^{a-b}\Gamma(Z + b))$. The identities hold for every infinite Z , including negative real Z , but the gloss is meaningful only away from the negative real direction, that is, in sectors $|\arg Z| \leq \pi - \delta$. On the negative real axis the classical ratio carries a non-flat quotient of sines coming from reflection (error 34). $R_{a,b}$ is a formal, phase-free object, and no claim is made that exponentially small corrections vanish.

11.3 The phase estimate and the finite-phase locus

Proposition 11.5 (Phase estimate [01, 02, 03, 04, 05, 06]). *Let $x > \mathbb{R}$ and let y be finite. Then*

$$\text{Im } L_{\text{St}}(x + iy) = y \log x - \frac{y}{2x} + \left(\frac{y^2}{6} - \frac{1}{12}\right) \frac{y}{x^2} + O\left(\frac{y}{x^3}\right),$$

so $\text{Im } L_{\text{St}}(x + iy) - y \log x \in \mathfrak{m}_{\mathbb{R}}$. Consequently $E_{\text{fp}}(L_{\text{St}}(z))$ is defined exactly on Ω_{∞} . There $\Gamma_{\text{fp}} = E_{\text{fp}} \circ L_{\text{St}}$ is zero-free and satisfies the recurrence and every finite Gauss formula. A nonzero ordinary imaginary displacement $x + iy$ ($y \in \mathbb{R}^{\times}$) always leaves Ω_{∞} .

Proof. Expand $\text{Log}(x + iy) = \log x + \text{Log}_m(1 + iy/x)$ and $(x + iy)^{-1}$ in y/x . The term y from $(x - \frac{1}{2}) \text{Im } \text{Log}_m(1 + iy/x)$ cancels $-\text{Im } z = -y$, leaving $y \log x - y/(2x) + y^3/(6x^2) + \dots$, and $\text{Im } S_{\text{St}}(z) = -y/(12x^2) + \dots$. Adding an infinitesimal does not change finiteness. Ω_{∞} is preserved by $z \mapsto z + 1$, $z \mapsto z + j/q$ and $z \mapsto qz$, since $\log(x + j/q) - \log x \in \mathfrak{m}$ and $\log(qx) = \log x + \log q$. Apply E_{fp} to Proposition 11.1. For ordinary $y \neq 0$, $y \log x$ is infinite. $\square$

This is the gamma-functions report's `lem:phase` and `thm:phase-domain`. Those results are stronger: they cover every gauge a , the finite-real-part part of the tube, and fine holomorphy and fine-openness. They are cited here, not claimed. The expansion is printed with the corrected remainder: 01's proof of its Theorem 6.2 omits the terms of order y^3/x^2 and y/x^2 (error 4). 05 writes $\text{Im } L_{\text{St}} = y \log x(1 + \eta)$, a weaker consistent form, and “usually” where “always” is correct (error 30). Examples [01, 02], as in the gamma-functions report's example after `thm:phase-domain`: $x + i/\log x \in \Omega_{\infty}$ with $\Gamma_{\text{fp}}(x + i/\log x)/\Gamma_{\text{St}}(x) = e^i(1 + \eta)$, while $x + i \notin \Omega_{\infty}$. Purely real positive infinite arguments always lie in Ω_{∞} . The locus is not a maximal-domain or impossibility statement [01, 02]. The next proposition makes that precise.

Proposition 11.6 (Ω_{∞} is the locus of phase independence [merge]). *Let $z = x + iy$ with $x > \mathbb{R}$ and y arbitrary. All phase exponentials E_{χ} give the same value $E_{\chi}(L_{\text{St}}(z))$ if and only if $\text{Im } L_{\text{St}}(z)$ is finite, if and only if $z \in \Omega_{\infty}$. If $p = \text{pp}(\text{Im } L_{\text{St}}(z)) \neq 0$ and M is the leading monomial of p , the phases $\chi_{\lambda,M}$ give $E_{\chi_{\lambda,M}}(L_{\text{St}}(z)) = e^{i\lambda c_M(p)} \Gamma_{\text{Exp}}(z)$ for every $\lambda \in \mathbb{R}$, so z carries a whole circle of values.*

Proof. If $\text{Im } L_{\text{St}}(z)$ is finite, every E_χ agrees with E_{fp} there. If y is finite, $\text{Im } L_{\text{St}}(z)$ is finite exactly on Ω_∞ by [Proposition 11.5](#). Suppose y is infinite, say positive. Then $\text{Im } L_{\text{St}}(z) = (x - \frac{1}{2}) \text{Arg } z + y(\log |z| - 1) + \text{Im } S_{\text{St}}(z)$. Here $\text{Arg } z > 0$ is finite, $\log |z| - 1 \geq \log y - 1$ is positive infinite, and $\text{Im } S_{\text{St}}(z)$ is infinitesimal. So $\text{Im } L_{\text{St}}(z)$ is positive infinite, and similarly for negative y . Hence $\text{Im } L_{\text{St}}(z)$ is finite exactly on Ω_∞ . Finally, if $p \neq 0$ then $c_M(p) \neq 0$, and $\Psi_{\chi, \lambda, M}(\text{Im } L_{\text{St}}) = e^{i\lambda c_M(p)} \text{cis}(\text{fin Im } L_{\text{St}})$. $\square$

This turns “ $E_{\text{fp}} \circ L_{\text{St}}$ is defined exactly on Ω_∞ ”, which is 01’s Theorem 6.2 and the gamma-functions report’s `thm:phase-domain`, into a statement that stays true when `Exp` is available everywhere. 01’s “defined if and only if” is correct for E_{fp} . It does not mean that Gamma is undefined elsewhere (error 5).

11.4 Gamma with a named phase

Proposition 11.7 (Phase Gamma on $\mathcal{S}_\infty$ [01, 02, 03, 06]). *Let $E : \mathbf{SC} \rightarrow \mathbf{SC}^\times$ be any additive-to-multiplicative homomorphism extending E_{fp} . Then $\Gamma_{E, \text{St}} = E \circ L_{\text{St}}$ is nonzero on $\mathcal{S}_\infty$ and satisfies $\Gamma_{E, \text{St}}(z + 1) = z \Gamma_{E, \text{St}}(z)$ [06]. For a phase χ , $\Gamma_\chi = E_\chi \circ L_{\text{St}}$ moreover commutes with conjugation and satisfies every finite Gauss formula on $\mathcal{S}_\infty$, with $m^{1/2-mz} := E_\chi((\frac{1}{2} - mz) \log m)$. It agrees with Γ_{St} on real $x > \mathbb{R}$ and with Γ_{fp} on Ω_∞ . It has the local law $\Gamma_\chi(z + h) = \Gamma_\chi(z) \text{Exp}_m(L_{\text{St}}(z + h) - L_{\text{St}}(z))$ for increments with h/z and $h(1 + |\text{Log } z|)$ infinitesimal.*

Proof. $E(\text{Log } z) = z$ because $\text{Arg } z$ is finite. Apply E to [Proposition 11.1](#). A homomorphism into $\mathbf{SC}^\times$ does not vanish. The local law expands $L_{\text{St}}(z + h)$ jointly in z^{-1} and h/z ; the linear coefficient contains $\text{Log } z$. $\square$

06 does not assume the modulus law $|E| = \exp \text{Re}$; the other clauses use it through [Proposition 5.3](#). 02 states the theorem on $\mathcal{T}_+$ with E_θ , 01 on the wing with E_λ , and 03 on $\mathcal{T}_+$ with every phase. Γ_χ , and $\Gamma_{E, \text{St}}$ for every E , satisfies the recurrence on $\mathcal{S}_\infty$, but *not* reflection at any point of the vertical strip ([Theorem 17.1\(e\)](#)). 06 does not claim reflection there, so this is a missing remark in 06, not an error (error 48).

Each Γ_χ can be joined with $\Gamma^\#$ on $\mathcal{O}$, which is disjoint from $\mathcal{S}_\infty$. The result is a union of prescriptions on disjoint domains: it agrees with the ordinary Γ on $\mathbb{C}$ because $\Gamma^\#$ does, and no identity linking the two pieces is asserted [01].

Example 11.8 (Phase dependence [01, 02, 03]).

- (a) [01] At $x = \exp_{\mathbf{No}}(\omega)$, $\text{Im } L_{\text{St}}(x + i) = \omega + \delta$ with $\delta \in \mathbf{m}$. So $\Gamma_{\chi\lambda}(x + i)/\Gamma_{\text{Exp}}(x + i) = e^{i\lambda}$, and $\lambda = 0, \pi$ give opposite values at one point while agreeing on the real axis.
- (b) [02] For every positive infinite real x and every ordinary ϑ , there is an explicit $\theta = \theta_x$ (a multiple of the coefficient of one monomial of $\text{pp}(\log x)$) with $\Gamma_{\chi_\theta}(x + i)/\Gamma_{\text{Exp}}(x + i) = e^{i\vartheta}$. The functional depends on x (error 12).
- (c) [03] For a positive infinite monomial X , $\log X \in \Pi$, so $\text{Exp}(i \log X) = 1$ and $\Gamma_{\text{Exp}}(X + i)/\Gamma_{\text{Exp}}(X) = \text{Exp}_m(Q_i(X)) = 1 - \frac{1+i}{2X} + O(X^{-2})$. At the smaller displacement $i\vartheta/\log X$ the phase is finite, and $\Gamma_{\text{Exp}}(X + i\vartheta/\log X)/\Gamma_{\text{St}}(X) = e^{i\vartheta}(1 + \varepsilon)$.

For $z \in \mathcal{S}_\infty$ the phase-free factor is

$$\text{Exp}_m(S_{\text{St}}(z)) = 1 + \frac{1}{12z} + \frac{1}{288z^2} - \frac{139}{51840z^3} - \frac{571}{2488320z^4} + \frac{163879}{209018880z^5} + O(z^{-6}),$$

and $\Gamma_{\text{Exp}}(\omega) = \Gamma_{\text{St}}(\omega) = \sqrt{2\pi} \omega^{\omega-1/2} e^{-\omega} \text{Exp}_m(\frac{1}{12\omega} - \frac{1}{360\omega^3} + \frac{1}{1260\omega^5} - \frac{1}{1680\omega^7} + \dots)$, where $\omega^{\omega-1/2} = \exp_{\mathbf{No}}((\omega - \frac{1}{2}) \log \omega)$ involves no phase [01, 02, 03, 04].

11.5 Three mechanisms of non-uniqueness

- (1) *Phase choice off* Ω_∞ [01, 02, 03, 06]: the functions $E_\chi \circ L_{\text{St}}$ for different χ (Proposition 11.6 and Example 11.8). No real value changes.
- (2) *Flat periodic gauges on the positive reals* [04, 05]: 05's $\Gamma_{\text{St}} \cdot e^{h_\lambda^{05}}$ with $h_\lambda^{05}(x) = \lambda e^{-\text{pp}(x)}$ at infinite x and 0 at finite x , and 04's $\Gamma_{\text{St}} \cdot \exp(\pi e^{-\text{pp}(A)})$ (Proposition 12.8). Both keep the recurrence and every finite value, and the corrections are smaller than every inverse power, since $x = \text{pp}(x) + O(1)$ and $e^{-P} P^N \in \mathfrak{m}$. Both break every Gauss formula at infinite arguments [merge]: for $h = \lambda e^{-\text{pp}}$ with $\lambda \neq 0$ and ordinary $q \geq 2$, $\sum_{r < q} h(x + r/q) = q\lambda e^{-\text{pp}(x)} \neq \lambda e^{-q\text{pp}(x)} = h(qx)$, because $e^{(q-1)\text{pp}(x)} \neq 1/q$. Neither source claims Gauss, so this is a missing statement, not an error (error 23).
- (3) *The convex gauge family* of the gamma-functions report (`cor:headline`, `thm:main`): $h_a(x) = a(M(x))\text{pp}(x)$, for example $a_\lambda(m) = \lambda e^{-m}$, which is 06's gauge. It keeps the recurrence, every Gauss formula, strict log-convexity and all Stirling estimates. 06 checks directly that it keeps the recurrence and every finite Gauss formula, since $M(x+1) = M(x)$, $h(qx) = qh(x)$ and $h(x + j/q) = h(x)$ [06]. Strict log-convexity, the Stirling estimates and the sharp classification belong to that report and are not reproved.

Reconciliation. Gamma at infinity is non-unique in three independent ways. The first changes no real value. The last two change no finite value and no asymptotic expansion, and only the third keeps Gauss multiplication. A small logarithmic difference need not be a small absolute difference of Gamma values (`warn:absolute-difference`). The member $a = 0$, the baseline of every source, is the one singled out by the scalar chain rule for the Berarducci–Mantova derivation (`thm:BM-all`). The sources fix it by prescription, not by that derivation [03].

12 Hurwitz zeta at an infinite parameter

12.1 Definition and domain

Let $A > \mathbb{R}$ be real and $s \in \mathcal{O}$. With $(s)_j = s(s+1) \cdots (s+j-1)$ and a named phase for $A^w = E_\chi(w \log A)$, put

$$\zeta_{\text{H}}(s, A) = \frac{A^{1-s}}{s-1} + \frac{A^{-s}}{2} + \sum_{k \geq 1}^{\text{H}} \frac{B_{2k}}{(2k)!} (s)_{2k-1} A^{1-s-2k}, \quad \widehat{\zeta}_{\text{H}}(s, A) = A^{s-1} \zeta_{\text{H}}(s, A). \quad (12.1)$$

This takes the classical large-parameter expansion (DLMF 25.11.43 [17]) as an exact definition [01, 02, 03, 04, 06]. Since A^{1-s} factors out, $\widehat{\zeta}_{\text{H}}$ is phase-free, and $\zeta_{\text{H}}(s, A)$ is phase-free exactly when $(\text{Im } s) \log A$ is finite, that is, on $\mathcal{S}_A$ [01, 02]. For finite s with $(\text{Im } s) \log A$ infinite its value depends on the phase [03], and every identity below holds for every phase.

Proposition 12.1 (Existence [01, 02, 03]). *For every finite s the family in (12.1) is strongly summable.*

Proof. Write $s = c + \varepsilon$. Each $(s)_{2k-1}$ is a polynomial in ε with complex coefficients, and the k th term carries A^{-2k} after A^{1-s} is factored out. Apply Lemma 4.3 to A^{-1} and ε jointly. $\square$

At $s \neq 1$ the factor $1/(s-1)$ is an ordinary field inverse, even when s is infinitely close to 1; no separate interpretation is needed (01's wording, error 6). *Parameter germs.* At a finite centre s_0 , write $A^{-s_0-h} = A^{-s_0} \text{Exp}_{\mathfrak{m}}(-h \log A)$. This is a strongly evaluable germ when $h \log A \in \mathfrak{m}$, but not for every infinitesimal h [02, 03, 04, 06]. 04 uses the germ at 0 ($s \in \mathfrak{m}$, $s \log A \in \mathfrak{m}$) and separate prescriptions at $s = -m$. 06 uses ordinary real s with germs. 01 uses the formal germ $\zeta_{\text{H}}(S, A) \in \mathbf{SC}[[S]]$, evaluable when $S \log A \in \mathfrak{m}$.

Warning 12.2 (The raw shifted sum [01, 02, 03, 04, 06]). For real $A > \mathbb{R}$ and every $s \in \mathcal{S}_A$, in particular every ordinary real σ including $\sigma = 0$ and the negative integers [02], the family $((A + n)^{-s})_{n \geq 0}$ is not strongly summable. Every member is $A^{-s}(1 + n/A)^{-s} = A^{-s}(1 + \varepsilon_n)$ with $\varepsilon_n \in \mathfrak{m}$, so all members have the same leading monomial and coefficient. So (12.1) is not a rearrangement of a strong sum of $(A + n)^{-s}$. Substituting $A = 1$ into an infinite-parameter definition does not give a global Riemann zeta [06]. No relation $\zeta(s) = \sum_{n < A} n^{-s} + \zeta_H(s, A)$ with omnific A is used anywhere [03].

Theorem 12.3 (Normalized unit [03]). *For every finite s , $(s - 1)\hat{\zeta}_H(s, A) = 1 + \varepsilon(s, A)$ with $\varepsilon(s, A) \in \mathfrak{m}$. So $\zeta_H(\cdot, A)$ is locally meromorphic in finite s , has a simple pole with residue 1 at $s = 1$ and no other poles, and has no zeros at finite s . The normalized factor and the zero-freeness are independent of the phase.*

Proof. $(s - 1)\hat{\zeta}_H$ has constant term 1, and every other term carries a positive power of A^{-1} with a coefficient polynomial in s , so their joint sum is infinitesimal. For $s \neq 1$ both $(s - 1)^{-1}$ and A^{1-s} are nonzero. At $s_0 = 1$, $A^{1-s} = 1 + O(h \log A)$ and the principal part of $\hat{\zeta}_H$ is $1/h$. $\square$

Consistently, $\zeta_H(-m, A) = -B_{m+1}(A)/(m + 1) \neq 0$ (Theorem 12.6).

12.2 Identities

Theorem 12.4 (Hurwitz identities [01, 02, 03, 04, 06]). *For real $A > \mathbb{R}$, finite s , ordinary $q, m \geq 1$, and every phase, meromorphically in s :*

$$\begin{aligned} \zeta_H(s, A) - \zeta_H(s, A + 1) &= A^{-s}, & \partial_A \zeta_H(s, A) &= -s \zeta_H(s + 1, A), \\ \sum_{r=0}^{m-1} \zeta_H\left(s, \frac{A+r}{m}\right) &= m^s \zeta_H(s, A), & \sum_{j=0}^{q-1} \zeta_H\left(s, A + \frac{j}{q}\right) &= q^s \zeta_H(s, qA), \end{aligned}$$

and $\text{Res}_{s=1} \zeta_H(s, A) = 1$. The two distribution identities are the same after $A \mapsto A/q$ [01, 03].

Proof. Work over $\mathbb{C}(s)$ in $A^{1-s}\mathbb{C}(s)[[A^{-1}]]$ with $D = d/dA$ and $D^{-1}A^{-s} = A^{1-s}/(1-s)$, with no integration constant. The Bernoulli identity $(1 - e^D)^{-1} = -D^{-1} + \frac{1}{2} - \sum_{k \geq 1} \frac{B_{2k}}{(2k)!} D^{2k-1}$ applied to A^{-s} gives exactly (12.1), since $D^{2k-1}A^{-s} = -(s)_{2k-1}A^{-s-2k+1}$. Multiplying by $1 - e^D$ gives the shift. The operator e^D acts as $A \mapsto A + 1$ through the binomial series $(A + 1)^w = A^w \sum_i \binom{w}{i} A^{-i}$. So the coefficient of A^{1-s-n} in $e^D \zeta_H$ is the finite sum $\sum_{j+i=n} c_j \binom{1-s-j}{i}$, where c_j is the coefficient of A^{1-s-j} in ζ_H . The regrouping is therefore coefficientwise finite. This is the detail that 04's proof of its Theorem 8.1 sketches (error 25). Since $\zeta_H(s, A) = (1 - e^D)^{-1}A^{-s}$ and the operator commutes with D , $\partial_A \zeta_H(s, A) = (1 - e^D)^{-1}(-sA^{-s-1}) = -s \zeta_H(s + 1, A)$; termwise this is $(s)_{2k-1}(s + 2k - 1) = s(s + 1)_{2k-1}$. For distribution, the difference $W(A) = \sum_{r < m} \zeta_H(s, (A + r)/m) - m^s \zeta_H(s, A)$ is 1-periodic by the shift identity, and $m^{1-s}W \in A^{1-s}\mathbb{C}(s)[[A^{-1}]]$. A nonzero element of this space with first term $c_j(s)A^{1-s-j}$ has forward difference beginning $c_j(s)(1 - s - j)A^{-s-j} \neq 0$ over $\mathbb{C}(s)$, so $W = 0$ [03]. To specialize, write $s = c + \varepsilon$. Affine changes of A use $\log(A + r) = \log A + \text{Log}_{\mathfrak{m}}(1 + r/A)$, and the leading factors are exponentiated by the homomorphism E_χ . Every coefficient identity is polynomial in ε , so strong evaluation in A^{-1} and ε proves the displayed identities. Poles and removable points are handled as germ identities. The residue comes from the first term alone. $\square$

Second route [01]: the geometric operator identity $\sum_{j < q} e^{jD/q} = (e^D - 1)/(e^{D/q} - 1)$ gives the distribution formula directly, because the derivative with respect to qA is D/q .

Corollary 12.5 (Formal uniqueness [03]). *Over $\mathbb{C}(s)$, the shift equation $Z(s, A + 1) - Z(s, A) = -A^{-s}$ has exactly one solution in $A^{1-s}\mathbb{C}(s)[[A^{-1}]]$, namely (12.1).*

The uniqueness is formal, before specialization. At $s = -m$, adding a constant in A preserves the specialized difference equation, and it is coherence in s that fixes these constants. The corollary does not classify every function satisfying the difference equation, and flat Hurwitz sectors are not excluded ([Proposition 12.8](#)).

Theorem 12.6 (Bernoulli values and the Lerch identity [01, 02, 03, 04, 06]). *For real $A > \mathbb{R}$ and ordinary $m \geq 0$, and independently of the phase:*

$$\begin{aligned}\zeta_{\mathrm{H}}(-m, A) &= -\frac{B_{m+1}(A)}{m+1}, & \zeta_{\mathrm{H}}(0, A) &= \frac{1}{2} - A, & \zeta_{\mathrm{H}}(-1, A) &= -\frac{A^2}{2} + \frac{A}{2} - \frac{1}{12}, \\ \zeta_{\mathrm{H}}(-2, A) &= -\frac{A^3}{3} + \frac{A^2}{2} - \frac{A}{6}, & \zeta_{\mathrm{H}}(-3, A) &= -\frac{A^4}{4} + \frac{A^3}{2} - \frac{A^2}{4} + \frac{1}{120}, \\ \partial_s \zeta_{\mathrm{H}}(s, A)|_{s=0} &= L_{\mathrm{St}}(A) - \frac{1}{2} \log 2\pi, & \Gamma_{\mathrm{St}}(A) &= \sqrt{2\pi} \exp(\partial_s \zeta_{\mathrm{H}}(s, A)|_{s=0}), \\ \zeta_{\mathrm{H}}(1+h, A) &= h^{-1} - \psi_{\mathrm{St}}(A) + \mathrm{O}(h) \quad (h \log A \in \mathfrak{m}), \\ \psi_{\mathrm{St}}^{(m)}(A) &= (-1)^{m+1} m! \zeta_{\mathrm{H}}(m+1, A) \quad (m \geq 1).\end{aligned}$$

For example $\psi'_{\mathrm{St}}(A) = A^{-1} + \frac{1}{2}A^{-2} + \sum_{j \geq 1}^{\mathrm{H}} B_{2j} A^{-2j-1}$.

Proof. At $s = -m$ the rising factorial $(s)_{2k-1}$ vanishes once $2k-1 > m$, and $(-m)_{2k-1} = -m!/(m-2k+1)!$ for $2k-1 \leq m$. With $B_{m+1}(A) = \sum_j \binom{m+1}{j} B_j A^{m+1-j}$ and the vanishing of odd Bernoulli numbers beyond B_1 , this gives $-B_{m+1}(A)/(m+1)$. At $s = 0$ the derivatives of the first two terms are $A \log A - A$ and $-\frac{1}{2} \log A$. Since $(s)_{2k-1}$ vanishes at 0 with derivative $(2k-2)!$, the k th term contributes $\frac{B_{2k}}{2k(2k-1)} A^{1-2k}$. At $s = 1+h$, $A^{-h} = 1 - h \log A + \mathrm{O}(h^2)$ and $(1)_{2k-1} = (2k-1)!$, which give the constant $-\log A + \frac{1}{2A} + \sum^{\mathrm{H}} \frac{B_{2k}}{2k} A^{-2k} = -\psi_{\mathrm{St}}(A)$. Differentiating (11.2) m times and comparing with $\zeta_{\mathrm{H}}(m+1, A)$ through $(m+1)_{2k-1} = (2k+m-1)!/m!$ gives the polygamma formula. All centres are real, so only real exponentiation is used. $\square$

Classically the Lerch identity is $\zeta'(0, A) = \log \Gamma(A) - \frac{1}{2} \log 2\pi$ (DLMF 25.11.18). Here it is verified for the normal forms, with no strong sum over $(A+n)^{-s}$ (01). The Lerch identity is a normalization principle: the baseline's logarithm comes from a coherent spectral family. It does not remove flat deformations if unprescribed flat Hurwitz sectors are allowed [02, 03]. The value $-1/12$ in $\zeta_{\mathrm{H}}(-1, A)$ is the coherence-selected constant of a polynomial tail. It does not make $1+2+3+\dots$ a strong sum, since that family is not strongly summable [02, 03, 06].

Remark 12.7 (Second route to the Stirling identities [01, 04, 06]). Differentiating the shift identity at $s = 0$ and using the Lerch identity gives $L_{\mathrm{St}}(A+1) - L_{\mathrm{St}}(A) = \log A$ (01 Theorem 6.1; 04 Theorem 8.1; 06 Proposition 6.1). Differentiating the distribution identity gives Gauss (01 Theorem 6.1). The identities are formal in inverse powers, so they extend to complex $z \in \mathcal{S}_{\infty}$ on the principal branch. This route is independent of [Proposition 11.1](#)'s proof.

Proposition 12.8 (A flat Hurwitz modification [04]). *Let $h(A) = \exp(-\mathrm{pp}(A))$ for positive infinite A and $h(A) = 0$ for finite A . On the local s -germs at the ordinary integers, where values are prescribed, put $\tilde{\zeta}_{\mathrm{H}}(s, A) = \zeta_{\mathrm{H}}(s, A) + h(A) \sin^{\#}(\pi s)$. Then $\tilde{\zeta}_{\mathrm{H}}$ satisfies the same shift identity, and the correction $h(A) \sin^{\#}(\pi s)$ vanishes at every ordinary integer s and is smaller than every ordinary inverse power of A . But $\partial_s \tilde{\zeta}_{\mathrm{H}}|_{s=0} = \partial_s \zeta_{\mathrm{H}}|_{s=0} + \pi h(A)$, so the corresponding Gamma is $\Gamma_{\mathrm{St}} \cdot \exp(\pi h(A))$, with the same recurrence and the same finite values.*

Proof. $\mathrm{pp}(A+1) = \mathrm{pp}(A)$, so h is 1-periodic. $\mathrm{pp}(A)$ differs from A by a finite amount and dominates every multiple of $\log A$, so $A^N h(A) \in \mathfrak{m}$. The sine vanishes at integers and has derivative π at 0. $\square$

The modification uses only the integer germs and does not extend 04's germ domain to the whole finite s -plane. By [Section 11.5\(2\)](#) it breaks every Gauss formula at infinite A . The

Stirling–Hurwitz values are thus a *selected* normalization, and no uniqueness, convexity or Gauss claim is made for the correction [04].

13 The right wing

$\mathcal{T}_+$ has finite imaginary part. On the wing $\mathcal{W} = \{\operatorname{Re} s > \mathbb{R}\}$ the terms $E_\chi(-s \log n)$ need a phase. *Reconciliation.* ζ_D is zero-free on $\mathcal{T}_+$ in all six sources. 02’s Warning 6.4, which confines zero-freeness to $\mathcal{T}_+$, concerns the raw ζ_D (error 13). Its own Proposition 11.3, 03’s Proposition 13.1, and the next proposition concern wing sums with a named phase. 04 states the Dirichlet chart for finite imaginary part only and omits the extension under Exp (error 24). 05 defines nothing at infinite imaginary part, so no source asserts zero-freeness there without naming a phase.

Proposition 13.1 (Wing sums for ordinary-circle phases [02, 03]; general form [merge]). *Let $\chi(\Pi) \subseteq S^1(\mathbb{C})$, and let $s = x + iy \in \mathcal{W}$ with $Y = \operatorname{pp}(y)$ and $f = \operatorname{fin}(y)$. Then $(E_\chi(-s \log n))_{n \geq 1}$ is strongly summable, and*

$$\zeta_\chi^\mathcal{W}(s) = \sum_n^H E_\chi(-s \log n) = \mathcal{D}_{x+if}(a), \quad a(n) = \chi(-Y \log n) \in S^1(\mathbb{C}),$$

with a completely multiplicative. So $\zeta_\chi^\mathcal{W}$ is a zero-free Euler product, locally analytic, and agrees with ζ_D on $\mathcal{T}_+$. For E_θ , $\zeta_{\chi_\theta}^\mathcal{W}(s) = \zeta_D(\pi_\theta(s))$ with $\pi_\theta(x + iy) = x + i(\operatorname{fin} y + \theta(\operatorname{pp} y))$ [02]. For Exp, $\zeta_{\operatorname{Exp}}^\mathcal{W}(s) = \zeta_D(x + i \operatorname{fin} y)$, and this is invariant under $s \mapsto s + iY'$ for every $Y' \in \Pi$ [03].

Proof. $-y \log n = -Y \log n - f \log n$ with $Y \log n \in \Pi$ and $f \log n$ finite. So $E_\chi(-s \log n) = E_{\operatorname{fp}}(-(x + if) \log n) \chi(-Y \log n)$, and $x + if \in \mathcal{T}_+$. Theorem 8.1 allows arbitrary complex coefficients. Complete multiplicativity comes from the character property. Zero-freeness and the Euler product are Theorem 8.5(d). An infinitesimal increment changes f and x but not Y , so the local germ is that of Proposition 8.3. For χ_θ , $a(n) = n^{-i\theta(Y)}$. $\square$

Corollary 13.2 (A continuum of wing values [02]). *For real $x > \mathbb{R}$ and nonzero $Y \in \Pi$, the values $\zeta_{\chi_\theta}^\mathcal{W}(x + iY)$, as θ ranges over explicit coefficient functionals, include the continuum of pairwise distinct values $\zeta_D(x + it)$, $t \in \mathbb{R}$.*

Proof. Scale a coefficient functional so that $\theta(Y) = t$. If $\zeta_D(x + it) = \zeta_D(x + iu)$, injectivity of $\mathcal{D}_x$ gives $e^{-i(t-u) \log n} = 1$ for all n . At $n = 2, 3$ this makes $\log 2 / \log 3$ rational unless $t = u$. $\square$

Proposition 13.3 (Wing sums can fail [merge]). *Assume choice for a Hamel basis of $\mathbb{R}$. There is a phase χ , whose character values are $\operatorname{cis} \delta$ with infinitesimal δ (values in $1 + \mathfrak{m}$, off $S^1(\mathbb{C})$), for which $(E_\chi(-s \log n))_n$ is not strongly summable at $s = \omega + i\omega \in \mathcal{W}$.*

Proof. The $\log p$ are $\mathbb{Q}$ -independent, so there is a $\mathbb{Q}$ -linear $\mathfrak{s} : \mathbb{R} \rightarrow \mathfrak{m}_\mathbb{R}$ with $\mathfrak{s}(\log p) = \omega^{-\omega^2/p}$ for every prime p . Let $\chi(P) = \operatorname{cis}(\mathfrak{s}(c_\omega(P)))$. Then $E_\chi(-s \log p) = \exp_{\mathbf{No}}(-\omega \log p) \operatorname{cis}(-\omega^{-\omega^2/p})$, and $\exp_{\mathbf{No}}(-\omega \log p) = \omega^{-\omega \log p}$ is a monomial. So the p th member contains the exponent $\omega \log p + \omega^2/p$ (in the valuation convention) with coefficient $-i$, and no other part of that member cancels it. For $p < p'$, $(\omega \log p + \omega^2/p) - (\omega \log p' + \omega^2/p') = \omega^2 \frac{p' - p}{pp'} - \omega \log \frac{p'}{p} > 0$. So the union of supports contains an infinite strictly decreasing sequence and is not well ordered. $\square$

This makes precise 03’s caution: “For general admissible phases, the termwise reduction used in its proof need not hold.” No source asserts anything on $\mathcal{W}$ without naming a phase.

Corollary 13.4 (Wing fibres [merge]; from [01, 03]). *For every nonzero $u \in \mathfrak{m}$, $\{s \in \mathcal{W} : \zeta_{\operatorname{Exp}}^\mathcal{W}(s) = 1 + u\} = \{s_k(u) + iY : k \in \mathbb{Z}, Y \in \Pi\}$.*

Proof. $\zeta_{\text{Exp}}^{\mathcal{W}}(x + iy) = 1 + u$ if and only if $x + i \text{fin } y$ lies in the $\mathcal{T}_+$ fibre, $\{s_k(u)\}$ by [Theorem 9.4](#) and [Corollary 9.6](#). Since $\text{Im } s_k(u)$ is finite, $x + iy = s_k(u) + i \text{pp}(y)$, and conversely $\text{fin}(\text{Im } s_k + Y) = \text{Im } s_k$. $\square$

None of this extends the completed Gamma–zeta package to $\mathcal{V}$, and there is no value at $\frac{1}{2} + i\omega$ [02, 03]. A formal Stirling series can be summable on large complex sectors while its branch, Stokes data and compatibility across sectors remain unspecified [03].

Part IV

Global objects on the horizontal tube

This part is source 03’s construction, stated for an arbitrary phase Ψ_χ and renamed as in [Table 1](#). The domain is the horizontal tube $\mathcal{T}_h = \mathcal{T}_- \sqcup \mathcal{O} \sqcup \mathcal{T}_+ = \{\text{Im } s \in \mathcal{O}_{\mathbb{R}}\}$. It contains every ordinary complex number and points such as $\omega + i$ and $-e^\omega + 3i$, but not $\frac{1}{2} + i\omega$. A function is *locally Hahn-holomorphic* if near each point it is given by a specified strongly evaluable Taylor germ, and *locally Hahn-meromorphic* if it is locally a quotient of such germs. A pole is a Laurent germ, not a field value. Every point has a fine neighbourhood inside its own region, so the definitions below give local analytic data without asserting a connected continuation between $\mathcal{T}_-$, $\mathcal{O}$ and $\mathcal{T}_+$. On $\mathcal{T}_-$, reflection and the functional equation are *definitions*. The content is that these definitions are consistent with the independently constructed pieces, have the stated local structure, and obey exact divisor and cancellation theorems [03].

14 Reflection forces infinite Gamma poles

Theorem 14.1 (Forced poles [03, 06]). *Let Ψ_χ be any phase.*

- (a) [06] *Let G be a function with values in $\mathbf{SC}$ on a set of real surreals, satisfying $G(x)G(1-x)\text{Sin}_\chi(\pi x) = \pi$ whenever $G(x)$ and $G(1-x)$ are both defined. Then for every $x \in \mathcal{I}(\Psi_\chi)$, G is undefined at x or at $1-x$. For Exp, this holds for every omnific integer x . In particular no G defined at every real surreal except possibly the ordinary nonpositive integers satisfies the relation.*
- (b) [03] *If g is locally Hahn-holomorphic on $\mathcal{T}_h$, nonzero at every positive infinite real point, and satisfies $g(z)g(1-z) = \text{Sin}_\chi(\pi z)/\pi$, then $g(x) = 0$ at every $x \in \mathcal{I}(\Psi_\chi)$ with $x < -\mathbb{R}$. So a zero-free meromorphic Gamma $1/g$ with reflection has poles there.*

Proof. By [Theorem 5.7](#), $\text{Sin}_\chi(\pi x) = 0$ for every $x \in \mathcal{I}(\Psi_\chi)$, so the relation cannot hold at x with both values defined. $\mathcal{I}$ has positive infinite elements x , and then x and $1-x$ are both infinite. For (b), $1-x > \mathbb{R}$, so $g(1-x) \neq 0$ and $g(x) = 0$. $\square$

Both parts are one-line consequences of `trigonometry:thm:infiniteperiods` and `trigonometry:cor:globalzeros`, so forced poles are immediate once infinite periods exist. 06 presents the existence of infinite periods as known, citing the trigonometry report but not this label, and only its application to Gamma as new (error 41). The pole set of [Theorem 15.1](#) satisfies (a): for $x \in \mathcal{I}$ with $x > \mathbb{R}$, the element $1-x \in \mathcal{I}$ is a pole below $-\mathbb{R}$; for finite $x \in \mathcal{I} = \mathbb{Z}$, either $x \leq 0$ or $1-x \leq 0$. *Reconciliation:* 03 builds Gamma with reflection on $\mathcal{T}_h$ while 06 proves that reflection with only ordinary poles is impossible. These are the same fact. 06’s theorem concerns arbitrary functions and is part (a); 03’s is part (b), the form for the local germ class, which also says at which of x , $1-x$ the pole lies. The conclusion is not that global

Gamma functions are impossible. A global reflection formula must allow extra infinite poles, restrict its domain, or change another part of the prescription [06]. The ordinary nonpositive integers are the complete pole set only for the finite lift. Transported models choose the first alternative, with poles at the internal integers (Section 20.2). The obstruction uses no contour integral, no Stirling coefficient and no hypothesis on primes [03].

15 Gamma on the horizontal tube

Define the reciprocal

$$g_\chi(z) = \begin{cases} (1/\Gamma)^\#(z), & z \in \mathcal{O}, \\ E_\chi(-L_{\text{St}}(z)), & z \in \mathcal{T}_+, \\ \frac{\text{Sin}_\chi(\pi z)}{\pi} \Gamma_\chi(1-z), & z \in \mathcal{T}_-, \end{cases}$$

where $\Gamma_\chi(1-z) = E_\chi(L_{\text{St}}(1-z))$ on $\mathcal{T}_+$. Put $\Gamma_\chi = 1/g_\chi$, read meromorphically at zeros of g_χ . On $\mathcal{T}_+ \subset \mathcal{S}_\infty$ this is the Γ_χ of Proposition 11.7, so Γ_χ is one function on $\mathcal{S}_\infty \cup \mathcal{T}_h$.

Theorem 15.1 (Gamma on $\mathcal{T}_h$ [03]). *g_χ is locally Hahn-holomorphic on $\mathcal{T}_h$. Γ_χ has no zeros, and its poles are simple and form exactly*

$$\mathbb{Z}_{\leq 0} \cup (\mathcal{I}(\Psi_\chi) \cap \{x < -\mathbb{R}\}).$$

As identities of meromorphic germs on $\mathcal{T}_h$,

$$\Gamma_\chi(z+1) = z\Gamma_\chi(z), \quad g_\chi(z)g_\chi(1-z) = \frac{\text{Sin}_\chi(\pi z)}{\pi}, \quad \Gamma_\chi(z)\Gamma_\chi(1-z) = \frac{\pi}{\text{Sin}_\chi(\pi z)},$$

and at a real pole x , $\text{Res}_{z=x} \Gamma_\chi(z) = \text{Cos}_\chi(\pi x)/\Gamma_\chi(1-x)$. For Exp the pole set is exactly $\mathbf{Oz}_{\leq 0}$.

Proof. The first two cases are holomorphic by Theorem 6.3 and Proposition 11.7. On $\mathcal{T}_-$ both factors are holomorphic and the second is nonzero. So the zeros of g_χ on $\mathcal{T}_-$ are those of $\text{Sin}_\chi(\pi z)$, which are real, simple and equal to $\mathcal{I} \cap \{x < -\mathbb{R}\}$ by Theorem 5.7. The finite zeros are the ordinary nonpositive integers, and there are none on $\mathcal{T}_+$. For left-infinite z , the recurrence on $\mathcal{T}_+$ gives $\Gamma_\chi(1-z) = (-z)\Gamma_\chi(-z)$, and Sin_χ changes sign under $z \mapsto z+1$ (since $\Psi_\chi(\pi) = -1$), so $g_\chi(z) = zg_\chi(z+1)$. The other regions use the finite germ identity and Proposition 11.1. Reflection on $\mathcal{T}_-$ is the definition; on $\mathcal{T}_+$ apply it to $1-z$ and use $\text{Sin}_\chi(\pi(1-z)) = \text{Sin}_\chi(\pi z)$; on $\mathcal{O}$ it is the lifted classical identity. At a pole x write $z = x+h$: then $g_\chi(x+h) = h \text{Cos}_\chi(\pi x)\Gamma_\chi(1-x) + \mathcal{O}(h^2)$, and $\text{Cos}_\chi(\pi x) = \pm 1$, which gives the residue. It reduces to $(-1)^n/n!$ at $x = -n$. Finally $\mathcal{I}(\Phi) = \mathbf{Oz}$ and $\mathbf{Oz} \cap \mathcal{O}_\mathbb{R} = \mathbb{Z}$. $\square$

Theorem 15.2 (Gauss on $\mathcal{T}_h$ [03]). *For every ordinary $m \geq 1$ and $z \in \mathcal{T}_h$, $\prod_{r < m} \Gamma_\chi(z+r/m) = (2\pi)^{(m-1)/2} m^{1/2-mz} \Gamma_\chi(mz)$ meromorphically, with $m^{1/2-mz} = E_\chi((\frac{1}{2} - mz) \log m)$.*

Proof. The finite case is classical and the case $\mathcal{T}_+$ is Proposition 11.7. For $z \in \mathcal{T}_-$ use $\prod_{r < m} \text{Sin}_\chi(\pi(z+r/m)) = 2^{1-m} \text{Sin}_\chi(\pi mz)$. To prove it, replace each sine by exponentials and factor $U^m - 1$ over the ordinary m th roots of unity; only the group law and finite phases enter. The complementary arguments $1-z-r/m$ permute the $w+r/m$ with $w = 1/m - z \in \mathcal{T}_+$, so their Gamma product is $(2\pi)^{(m-1)/2} m^{mz-1/2} \Gamma_\chi(1-mz)$. Multiply the reciprocal definitions. The computation is an identity of holomorphic reciprocal germs, so no division by a vanishing sine occurs. $\square$

Proposition 15.3 (Uniqueness for fixed data [03]). *Once the phase, the finite lift and the exact Stirling branch on $\mathcal{T}_+$ are fixed, there is exactly one zero-free locally meromorphic Gamma on $\mathcal{T}_h$ satisfying reflection.*

Proof. Reflection determines the reciprocal, with its germs at the zeros, on $\mathcal{T}_-$ from the already specified values on $\mathcal{T}_+$. Existence is [Theorem 15.1](#). $\square$

This is uniqueness for fixed data, and it is immediate: reflection is the definition of g_χ on $\mathcal{T}_-$. It is not a surreal Bohr–Mollerup theorem, and the gauges of [Section 11.5](#) show that the positive branch is itself a selection.

Example 15.4 (The canonical case [03]). $\text{Res}_{z=-\omega} \Gamma_{\text{Exp}}(z) = 1/\Gamma_{\text{St}}(1+\omega)$, because $\text{Cos}(\pi\omega) = \text{cos}(\text{fin } \pi\omega) = 1$. By reflection (a direct consequence of [Theorem 15.1](#), not printed in 03), $\Gamma_{\text{Exp}}(-\omega - \frac{1}{2}) = -\pi/\Gamma_{\text{St}}(\omega + \frac{3}{2})$, because $\text{Sin}(-\pi\omega - \frac{\pi}{2}) = \text{sin}(-\frac{\pi}{2}) = -1$. On real $x < -\mathbb{R}$, $\Gamma_{\text{Exp}}(x) = \pi/(\text{sin}(\pi \text{fin } x)\Gamma_{\text{St}}(1-x))$ away from $\mathbf{Oz}$.

16 Zeta and ξ on the horizontal tube

With the same phase everywhere, put

$$\mathcal{X}_\chi(s) = 2^s \pi^{s-1} \text{Sin}_\chi(\pi s/2) \Gamma_\chi(1-s),$$

where $2^s = E_\chi(s \log 2)$ and $\pi^{s-1} = E_\chi((s-1) \log \pi)$. On $\mathcal{T}_h$ these powers are phase-free, because $\text{Im } s$ is finite. (03 writes χ_ϕ for $\mathcal{X}_\chi$; the letter χ is the character here.)

Lemma 16.1 (The reflection factor [03]). *On $\mathcal{T}_h$, as meromorphic germ identities, $\mathcal{X}_\chi(s)\mathcal{X}_\chi(1-s) = 1$ and $\mathcal{X}_\chi(s) = \pi^{s-1/2} \Gamma_\chi(\frac{1-s}{2})/\Gamma_\chi(\frac{s}{2})$.*

Proof. In the product, the powers give $2/\pi$, and $\text{Sin}_\chi(\pi s/2) \text{Sin}_\chi(\pi(1-s)/2) = \frac{1}{2} \text{Sin}_\chi(\pi s)$. Reflection cancels the rest, in the field of germs. For the second identity use duplication, $\Gamma_\chi(1-s) = \frac{2^{-s}}{\sqrt{\pi}} \Gamma_\chi(\frac{1-s}{2}) \Gamma_\chi(1-\frac{s}{2})$ ([Theorem 15.2](#) with $m=2$), and reflection at $s/2$. $\square$

Define

$$\zeta_\chi(s) = \begin{cases} \zeta^\#(s), & s \in \mathcal{O}, \\ \zeta_D(s), & s \in \mathcal{T}_+, \\ \mathcal{X}_\chi(s) \zeta_D(1-s), & s \in \mathcal{T}_-. \end{cases}$$

This is not circular: for left-infinite s , the Gamma factor lies on the independent positive branch.

Theorem 16.2 (Zeta on $\mathcal{T}_h$ [03]). *ζ_χ is locally Hahn-meromorphic on $\mathcal{T}_h$, with a single pole at 1, of residue 1. It satisfies $\zeta_\chi(s) = \mathcal{X}_\chi(s)\zeta_\chi(1-s)$ throughout $\mathcal{T}_h$ and commutes with conjugation. Its zeros are the ordinary zeros of ζ with their multiplicities, together with the simple real zeros $\{x < -\mathbb{R} : x \in 2\mathcal{I}(\Psi_\chi)\}$. At such a zero,*

$$\zeta'_\chi(x) = 2^{x-1} \pi^x \text{Cos}_\chi(\pi x/2) \Gamma_\chi(1-x) \zeta_D(1-x) \neq 0.$$

Proof. On $\mathcal{O}$ the claims are the germ claims, with no displaced zeros by [Theorem 6.3](#). On $\mathcal{T}_+$, ζ_D is holomorphic and zero-free ([Theorem 8.5](#)). On $\mathcal{T}_-$ every factor except $\text{Sin}_\chi(\pi s/2)$ is holomorphic and nonzero, so the zeros are those of $\text{Sin}_\chi(\pi s/2)$, real and simple by [Theorem 5.7](#). Differentiating the product at such a zero leaves only the sine derivative. The functional equation is the definition on $\mathcal{T}_-$. On $\mathcal{T}_+$ it follows from the definition at $1-s$ and [Lemma 16.1](#), and on $\mathcal{O}$ it is the lifted classical equation. Conjugation is preserved by each ingredient, including the strong Dirichlet sum with real coefficients. $\square$

Corollary 16.3 (Canonical trivial zeros [03]). *For Exp the zeros of ζ_{Exp} on $\mathcal{T}_h \cap \mathbb{R}$ below $-\mathbb{R}$ are exactly $2\mathbf{Oz}_{<0}$ restricted to $x < -\mathbb{R}$. Together with $-2, -4, \dots$ they are $2\mathbf{Oz}_{<0}$. For example $\zeta_{\text{Exp}}(-\omega) = 0$, since $-\omega/2 \in \mathbf{Oz}$. But $-\omega - 1 \notin 2\mathbf{Oz}$, since $(-\omega - 1)/2$ has finite part $-\frac{1}{2}$. Moreover $\zeta'_{\text{Exp}}(-2\omega) = 2^{-2\omega-1} \pi^{-2\omega} \Gamma_{\text{St}}(1+2\omega) \zeta_D(1+2\omega)$.*

“Even” means membership in the additive group $2\mathbf{Oz}$. No recurrence is iterated a surreal number of times, and no product with an omnific number of factors occurs [03].

Theorem 16.4 (Completed-divisor rigidity [03]). $\xi_\chi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma_\chi(s/2)\zeta_\chi(s)$ extends to a locally Hahn-holomorphic function on $\mathcal{T}_h$ with $\xi_\chi(s) = \xi_\chi(1-s)$. It has no zeros on $\mathcal{T}_+ \cup \mathcal{T}_-$ and equals $\xi^\#$ on $\mathcal{O}$. So its zero divisor is exactly the classical nontrivial divisor, independent of χ , with no zeros at nonstandard finite points and none at infinite real part.

Proof. With Lemma 16.1, $\pi^{-s/2}\Gamma_\chi(s/2)\mathcal{X}_\chi(s) = \pi^{-(1-s)/2}\Gamma_\chi(\frac{1-s}{2})$, and $s(s-1)$ is symmetric, which proves the functional equation for germs. On $\mathcal{T}_+$ every factor is holomorphic and nonzero. On $\mathcal{T}_-$ the functional equation expresses ξ_χ through the nonzero function on $\mathcal{T}_+$, so every apparent singularity is removable with a nonzero value. On $\mathcal{O}$ the expression is the classical identity for the entire ξ , so it equals $\xi^\#$, whose zeros are given by Theorem 6.3. $\square$

Corollary 16.5 (Horizontal RH [03]). For every phase, “ $\xi_\chi(s) = 0$ with $s \in \mathcal{T}_h$ implies $\operatorname{Re} s = \frac{1}{2}$ ” is equivalent to classical RH.

The two statements quantify over the same zeros. The content is the cancellation theorem, and nothing is said at $\frac{1}{2} + iT$ with infinite T , which lies outside $\mathcal{T}_h$ [03]. At a nontrivial zero ρ of order m , $\xi_\chi(\rho + \varepsilon) = \frac{\xi^{(m)}(\rho)}{m!}\varepsilon^m(1 + \eta) \neq 0$, unconditionally and for every m . Simplicity is not smuggled in.

Proposition 16.6 (Explicit cancellation [03]). Let $A > \mathbb{R}$ with $A \in \mathcal{I}(\Psi_\chi)$. For h small enough for the local germs (possibly much smaller than $1/\log A$),

$$\Gamma_\chi(-A + h/2) = \frac{2 \operatorname{Cos}_\chi(\pi A)}{\Gamma_\chi(1 + A)} h^{-1} + O(1),$$

$$\zeta_\chi(-2A + h) = 2^{-2A-1}\pi^{-2A} \operatorname{Cos}_\chi(\pi A)\Gamma_\chi(1 + 2A)\zeta_D(1 + 2A)h + O(h^2),$$

and, since $\operatorname{Cos}_\chi(\pi A)^2 = 1$, $\xi_\chi(-2A) = A(2A + 1)2^{-2A}\pi^{-A}\frac{\Gamma_\chi(1+2A)}{\Gamma_\chi(1+A)}\zeta_D(1 + 2A) = \xi_\chi(1 + 2A)$. The last equality follows from duplication, $2^{-2A}\Gamma_\chi(1 + 2A)/\Gamma_\chi(1 + A) = \Gamma_\chi(A + \frac{1}{2})/\sqrt{\pi}$.

The computation keeps the local multiplicities and gives the removable value; no symbol $0 \cdot \infty$ is used. At $A = \omega$ with Exp, the residue $1/\Gamma_{\text{St}}(1 + \omega)$ and the zero derivative at -2ω live on very different scales, but their product is fixed by duplication.

16.1 Phase changes move the infinite divisors

Proposition 16.7 (A movable infinite zero [01, 03]). Fix a positive infinite monomial M and consider the phases $\chi_{\lambda,M}$ of Proposition 5.5(b). They all give the same finite Gamma and zeta, the same Gamma on the positive real axis, and the same zeta on $\mathcal{T}_+$. But the left-infinite real zeros of $\zeta_{\chi_{\lambda,M}}$ in $-M + \mathcal{O}_{\mathbb{R}}$ are exactly $-M + \lambda + 2k$ ($k \in \mathbb{Z}$), so $\zeta_{\chi_{\lambda,M}}(-M) = 0$ if and only if $\lambda \in 2\mathbb{Z}$. The completed ξ divisors are identical.

For $M = \omega$ this is 01’s one-sided prescription: ζ_{χ_λ} on $\mathcal{T}_-$ has the zero set $\{\sigma < -\mathbb{R} : P_\lambda(\sigma) \in 2\mathbb{Z}\}$, all real and simple, and

$$\zeta_{\chi_\lambda}(-\omega) = -A_\omega \sin(\pi\lambda/2), \quad A_\omega = 2^{-\omega}\pi^{-\omega-1}\Gamma_{\text{St}}(1 + \omega)\zeta_D(1 + \omega) > 0.$$

In particular $\zeta_{\text{Exp}}(-\omega) = 0$ with $\zeta'_{\text{Exp}}(-\omega) = \frac{\pi}{2}A_\omega$, while $\zeta_{\chi_1}(-\omega) = -A_\omega \neq 0$.

Proof. $-M + r \in 2\mathcal{I}(\Psi_{\chi_{\lambda,M}})$ if and only if $r - \lambda \in 2\mathbb{Z}$, by Proposition 5.5(b); apply Theorem 16.2, and Theorem 16.4 for ξ . For $M = \omega$, 01’s $Z_{-, \lambda}(s) = 2^s\pi^{s-1}\sin_\lambda(\pi s/2)\Gamma_{\text{St}, \lambda}(1 - s)Z_\infty(1 - s)$ is literally $\mathcal{X}_{\chi_\lambda}(s)\zeta_D(1 - s)$. At $-\omega$, $\operatorname{Sin}_{\chi_\lambda}(-\pi\omega/2) = \sin(\lambda c_\omega(-\pi\omega/2)) = -\sin(\pi\lambda/2)$. The derivative is Theorem 16.2 with $\operatorname{Cos}(-\pi\omega/2) = 1$. $\square$

Remark 16.8 (01’s remark after its left-zero theorem, corrected). 01 says its result “does not say that a canonical surreal zeta must have a zero at $-\omega$, or that it must not”, and that a simultaneous functional equation on a larger Gamma domain “would require an additional construction and proof”. That construction is 03’s ([Theorems 15.1 and 16.2](#)), and it contains 01’s prescription. With Ehrlich and Kaplan’s canonical Exp ($\lambda = 0$) the zero at $-\omega$ is present. No reflection or functional-equation requirement forces it, since every λ gives an FE-compatible package on $\mathcal{T}_h$ (error 2). The canonical initial-integer-part convention is a meaningful way to choose, and it is not a consequence of reflection [03]. 01’s non-claim, that its one-sided prescription is not a proved global functional equation, stays true of that prescription taken alone. 01’s third research direction, a coherent continuation deciding which left zeros survive, is thereby answered on $\mathcal{T}_h$ relative to a named phase: every phase gives such a package, with left zeros $2\mathcal{I}(\Psi_\chi) \cap \{x < -\mathbb{R}\}$. At infinite imaginary height it remains open ([Question 23.1](#)).

Agreement on all ordinary arguments, on the positive-real Gamma and on the right-infinite zeta therefore does not determine the negative-infinite divisor. Every phase has infinite poles [03].

Part V

Infinite imaginary parts: obstructions and underdetermination

The genuinely new points for RH are $\sigma + iT$ with $0 < \sigma < 1$ and T infinite, the strip $\mathcal{V}$. None of the constructions of Parts II–IV assigns zeta values there. The finite lift covers $\mathcal{O}$. The Dirichlet algebra and the wing have positive infinite real part. The tube has finite imaginary part. This part shows what goes wrong in the obvious attempts [02, 03, 04, 05, 06].

17 Reflection versus the Stirling modulus on the vertical strip

Theorem 17.1 (The reflection defect [05] for $\sigma = \frac{1}{2}$; all σ [merge]). *Let $T > \mathbb{R}$ and let σ be finite with $0 < \sigma < 1$, and put $z = \sigma + iT$. Then $z, 1 - z \in \mathcal{S}_\infty$, and:*

- (a) $L_{\text{St}}(z) + L_{\text{St}}(1 - z) = \log 2\pi - \pi T + i\pi(\sigma - \frac{1}{2})$ exactly: every inverse-power term cancels. In particular $2 \operatorname{Re} L_{\text{St}}(\frac{1}{2} + iT) = \log 2\pi - \pi T$.
- (b) For every homomorphism E extending E_{fp} , and so for every phase, $\operatorname{Sin}_E(\pi z) = (E(i\pi z) - E(-i\pi z))/(2i)$ is phase-free, and $|\pi / \operatorname{Sin}_E(\pi z)| = 2\pi e^{-\pi T} / |1 - e^{2\pi iz}|$ with $e^{2\pi iz} = e^{-2\pi T} \operatorname{cis}(2\pi\sigma)$.
- (c) Let G be defined at z and $1 - z$ with Stirling moduli, $|G(w)| = \exp_{\mathbf{No}} \operatorname{Re} L_{\text{St}}(w)$ for $w = z, 1 - z$. Then $G(z)G(1 - z) = \pi / \operatorname{Sin}_E(\pi z)$ fails unless $|1 - e^{2\pi iz}| = 1$. For each T this happens for exactly two abscissae, $\sigma_-(T) = \arccos(e^{-2\pi T}/2)/2\pi$ (the arccosine taken as the unique finite angle in $[0, \pi]$ with the given cosine, which exists by [Proposition 5.1](#)) and $\sigma_+(T) = 1 - \sigma_-(T)$, which are infinitesimally close to $\frac{1}{4}$ and $\frac{3}{4}$.
- (d) Any G satisfying reflection at z has the total log-modulus defect

$$\log |G(z)| + \log |G(1 - z)| - \operatorname{Re}(L_{\text{St}}(z) + L_{\text{St}}(1 - z)) = -\frac{1}{2} \log(1 - 2e^{-2\pi T} \cos 2\pi\sigma + e^{-4\pi T}),$$

which is smaller than every T^{-N} and nonzero unless $\sigma = \sigma_\pm(T)$. At $\sigma = \frac{1}{2}$ with $G(\bar{z}) = \overline{G(z)}$ [05], $\log |G(\frac{1}{2} + iT)| = \operatorname{Re} L_{\text{St}}(\frac{1}{2} + iT) - \frac{1}{2} \log(1 + e^{-2\pi T})$, and the correction is $-\frac{1}{2}e^{-2\pi T} + \frac{1}{4}e^{-4\pi T} - \frac{1}{6}e^{-6\pi T} + \dots$.

(e) [merge] For every additive-to-multiplicative $E : \mathbf{SC} \rightarrow \mathbf{SC}^\times$ extending E_{fp} (no modulus law is assumed), and so for every phase and for 06's $\Gamma_{E,\text{St}} = E \circ L_{\text{St}}$,

$$\Gamma_{E,\text{St}}(z) \Gamma_{E,\text{St}}(1-z) = (1 - e^{2\pi iz}) \frac{\pi}{\text{Sin}_E(\pi z)}.$$

Since $e^{2\pi iz} \neq 0$, $\Gamma_{E,\text{St}}$ violates reflection at z for every finite $\sigma \in (0, 1)$, including $\sigma_\pm(T)$: there the factor $1 - e^{2\pi iz}$ has modulus 1 but argument $\pm e^{-2\pi T}(1 + \varepsilon)$ at $\sigma_\pm(T)$, with $\varepsilon \in \mathfrak{m}$. The same holds at $\sigma - iT$, so $\Gamma_{E,\text{St}}$ violates reflection at every point of $\mathcal{V}$.

Proof. (a) Apply [Theorem 11.2](#) twice. First take $Z = iT$ with $\text{Log } Z = \log T + i\pi/2$ and $a = \sigma$. The principal $\text{Log}(\sigma + iT)$ equals $\log T + i\pi/2 + \text{Log}_{\mathfrak{m}}(1 + \sigma/(iT))$: both are logarithms of $\sigma + iT$, and both imaginary parts lie in $(-\pi/2, \pi/2)$, since $\text{Im } \text{Log}_{\mathfrak{m}}(1 - i\sigma/T) = -\sigma/T + \dots < 0$. Then take $Z' = -iT$ with $\text{Log } Z' = \log T - i\pi/2$ and $a' = 1 - \sigma$. The sum of the two elementary parts is $(iT + \sigma - \frac{1}{2})(\log T + \frac{i\pi}{2}) - iT + (-iT + \frac{1}{2} - \sigma)(\log T - \frac{i\pi}{2}) + iT + \log 2\pi = \log 2\pi - \pi T + i\pi(\sigma - \frac{1}{2})$. The n th inverse-power terms are $\frac{(-1)^{n+1}}{n(n+1)}(B_{n+1}(\sigma)(iT)^{-n} + B_{n+1}(1-\sigma)(-iT)^{-n}) = 0$, because $B_{n+1}(1-\sigma) = (-1)^{n+1}B_{n+1}(\sigma)$ and $(-iT)^{-n} = (-1)^n(iT)^{-n}$. (b) $i\pi z = -\pi T + i\pi\sigma$ has finite imaginary part, so $E(\pm i\pi z) = e^{\mp\pi T} \text{cis}(\pm\pi\sigma)$ and $\text{Sin}_E(\pi z) = -\frac{e^{\pi T} \text{cis}(-\pi\sigma)}{2i}(1 - e^{2\pi iz})$. (c) By (a) the Stirling moduli give $|G(z)G(1-z)| = 2\pi e^{-\pi T}$. With $q = e^{-2\pi T}$, $|1 - q \text{cis } 2\pi\sigma|^2 = 1 - 2q \cos 2\pi\sigma + q^2$ equals 1 exactly when $\cos 2\pi\sigma = q/2$. For finite $\sigma \in (0, 1)$ this has exactly the two solutions stated. (d) Take logarithms of the moduli in (b). At $\sigma = \frac{1}{2}$, $1 - z = \bar{z}$ and $|1 - e^{2\pi iz}| = 1 + e^{-2\pi T}$. Conjugation splits the defect equally. (e) By (a), $w_0 = L_{\text{St}}(z) + L_{\text{St}}(1-z) = \log 2\pi - \pi T + i\pi(\sigma - \frac{1}{2})$ has finite imaginary part, so $E(w_0) = E_{\text{fp}}(w_0) = -2\pi i e^{-\pi T} \text{cis}(\pi\sigma)$. Since E is a homomorphism, $\Gamma_{E,\text{St}}(z)\Gamma_{E,\text{St}}(1-z) = E(L_{\text{St}}(z))E(L_{\text{St}}(1-z)) = E(w_0)$. By (b), $\pi/\text{Sin}_E(\pi z) = -2\pi i e^{-\pi T} \text{cis}(\pi\sigma)/(1 - e^{2\pi iz})$, and comparing the two gives the identity. Here $e^{2\pi iz} = E_{\text{fp}}(2\pi iz)$ has modulus $e^{-2\pi T} \neq 0$, so $1 - e^{2\pi iz} \neq 1$. With $q = e^{-2\pi T}$ and $\cos 2\pi\sigma_\pm = q/2$, $1 - e^{2\pi iz} = 1 - \frac{q^2}{2} - iq \sin 2\pi\sigma_\pm$, where $\sin 2\pi\sigma_\pm = \mp(1 - q^2/4)^{1/2}$; this has modulus 1 and argument $\pm q(1 + \varepsilon)$. Finally, for $z' = \sigma - iT$ put $w = 1 - z' = (1 - \sigma) + iT$. Then $\Gamma_{E,\text{St}}(z')\Gamma_{E,\text{St}}(1-z') = \Gamma_{E,\text{St}}(w)\Gamma_{E,\text{St}}(1-w)$, and $\text{Sin}_E(\pi z') = \text{Sin}_E(\pi - \pi w) = \text{Sin}_E(\pi w)$ because $E(\pm i\pi) = -1$, so the identity at w gives the failure at z' . $\square$

05 proves the case $\sigma = \frac{1}{2}$ (its [Theorem 7.1](#) and [Corollary 7.2](#)), including $\sin(\pi(\frac{1}{2} + iT)) = \cosh \pi T$. It uses only a finite real phase and the real exponential. Classically the correction is encoded in $|\Gamma(\frac{1}{2} + it)|^2 = \pi/\cosh \pi t$. The extension to every finite $\sigma \in (0, 1)$, and the two exceptional abscissae where the modulus test is silent, were established for this merge. The reconciliation plan had claimed the obstruction for every σ . The exceptions $\sigma_\pm(T)$ were found while the proof was rechecked. At them the classical identity $\text{Re}(\log \Gamma(z) + \log \Gamma(1-z)) = \log 2\pi - \pi T$ holds as well (checked numerically at ordinary T , [Appendix B](#)), so the modulus test alone claims nothing there for a general G with Stirling moduli; for $\Gamma_{E,\text{St}}$ itself, (e) still gives failure. Clause (e) was added after a post-merge audit ([Appendix A.5](#)). The symbolic check of (a) through T^{-14} and the numerical check of (d) are in the reconciliation suite; the check of (e) is described in [Appendix B](#).

Reconciliation. $\frac{1}{2} \pm iT$ lies in $\mathcal{S}_\infty$ but not in $\mathcal{T}_{\text{h}}$, so the theorem does not touch 03's reflection on the tube ([Theorem 15.1](#)). It does apply to 06's $\Gamma_{E,\text{St}} = E \circ L_{\text{St}}$ on $\mathcal{S}_\infty$ for every E extending E_{fp} (by (e) no modulus law is needed), in particular to Γ_{Exp} and every Γ_χ . These satisfy the recurrence on $\mathcal{S}_\infty$ ([Proposition 11.7](#)) but violate reflection at every point of $\mathcal{V}$, by (e) (the remark missing from 06, error 48). The exceptional abscissae concern only an arbitrary G with Stirling moduli. A conjugation- and reflection-compatible Gamma on the line $\frac{1}{2} + i\mathbb{R}_{>\mathbb{R}}$ must add the flat correction, and even then it has fixed only a modulus: no phase and no global Gamma function [05]. The theorem is not an objection to asymptotic Stirling expansions, to positive-real surreal Gamma, or to sectorial resummation. It is an exact incompatibility test for

a proposed surreal prescription [05]. It matters for zeta directly, since $\Gamma(s/2)$ in ξ lies in exactly this regime when $s \in \mathcal{V}$. A global ξ cannot use bare Stirling values, keep every classical Gamma identity and expect compatibility. Phase data and beyond-all-orders corrections must both be accounted for [05].

18 Resonant galaxies

18.1 A phase does not give strong summability

At $s = \sigma + iT$ with ordinary σ and infinite T , and for any unit phase, the Dirichlet terms $n^{-\sigma}\Psi_{\chi}(-T \log n)$ are finite units with nonzero standard parts. The family is not strongly summable, even for $\sigma > 1$ [04]. This obstruction is separate from the phase question: prescribing every phase does not turn the Dirichlet series at infinite height into a Hahn sum. Similarly, the Riemann–Siegel cutoff $\sqrt{T/2\pi}$ at infinite T does not give an ordinary finite sum. In a bare surreal field, “ $n \leq \sqrt{T}$ ” means all ordinary n , and in an elementary model it means a hyperfinite internal sum, which is additional structure [04]. With Exp , $n^{-iT} = \text{cis}(-\text{fin}(T) \log n)$, so the terms at infinite height reduce to terms at finite height. They are ordinarily summable in $\mathbb{C}$ but not strongly summable.

18.2 Coherent continuation on a resonant galaxy

Let $T > 0$ be a resonance of Ψ_{χ} (Definition 5.9). Then $-T$ is one too. On the galaxy $\pm iT + \mathcal{O}$ write $s = \pm iT + w$ with w finite. For every ordinary n ,

$$E_{\chi}(-s \log n) = E_{\text{fp}}(-w \log n) \Psi_{\chi}(\mp T \log n) = n^{-w}. \quad (18.1)$$

So coefficientwise ordinary Dirichlet summation on $\text{Rest } w > 1$ reproduces $\zeta^{\#}(w)$, not a new high-frequency function.

Definition 18.1 (Coherent meromorphic continuation [06]). A *coherent meromorphic continuation in the galaxy coordinate* is a section with one set-sized well-ordered Hahn support and ordinary meromorphic coefficient functions on the connected ordinary w -plane, evaluated at finite w by Taylor–Laurent lifting. It is stronger than fine analyticity at each point separately.

06 calls this “the common-domain notion employed in the repository”. It *extends* the analysis report’s coherent sections (`c:p4:def-HU`), which have holomorphic coefficients on a connected open set, to meromorphic coefficients on the whole plane. The identity theorem still applies off the discrete pole sets (error 45).

Theorem 18.2 (Resonant-galaxy obstruction [06]; general form written out [merge]). *Let $T > 0$ be a resonance of Ψ_{χ} . Suppose Z uses E_{χ} for its Dirichlet powers, agrees with coefficientwise ordinary Dirichlet summation on $\text{Rest } w > 1$ in both galaxies $\pm iT + \mathcal{O}$, and has coherent meromorphic continuations on their ordinary coordinate planes. Then $Z(iT + w) = \zeta^{\#}(w)$ and $Z(-iT + w) = \zeta^{\#}(w)$ as meromorphic functions of finite w . In particular:*

- (i) Z has poles at $1 \pm iT$, so it cannot have the pole at 1 alone;
- (ii) $Z(3 + iT) = \zeta(3) \neq 0$, whereas $Z(-2 - iT) = 0$;
- (iii) the pointwise functional equation $Z(s) = 2^s \pi^{s-1} \text{Sin}_{\chi}(\pi s/2) G(1-s) Z(1-s)$, with E_{χ} powers, fails at $s = 3 + iT$ for every Gamma factor G that is field-valued at $-2 - iT$.

Proof. By (18.1) and the specified summation, $Z(iT + w) = \zeta^{\#}(w)$ on $\text{Rest } w > 1$. Subtract the canonical section of ζ from the coherent continuation. On that half-plane every Hahn

coefficient of the difference vanishes at every ordinary w , so by the ordinary identity theorem each coefficient vanishes on the connected plane. The same holds on $-iT + \mathcal{O}$. (i) and (ii) are ordinary zeta facts. For (iii), at $s = 3 + iT$ the right side is a product of field elements whose last factor is $Z(-2 - iT) = \zeta(-2) = 0$, while the left side is $\zeta(3)$. $\square$

The Gamma qualification in (iii) is not vacuous. The shifted Stirling value

$$G(-2 - iT) = \frac{\Gamma_\chi(1 - iT)}{(-2 - iT)(-1 - iT)(-iT)}$$

is field-valued and nonzero, since $1 - iT \in \mathcal{S}_\infty$ [06]. Introducing an actual Gamma pole at $-2 - iT$ would change the pole prescription and require a meromorphic cancellation, not a pointwise product of field values.

For Exp every nonzero $T \in \Pi$ is a resonance; this is the case 06 states as its theorem. 06's Remark 7.4 already notes that every phase with a common resonance “encounters the same argument”; the proof above writes it out. By Proposition 5.10(a), every phase with ordinary-circle character values has resonances of arbitrarily large modulus. This includes E_λ (with $T = \omega - \lambda$ by part (b)), $E_{\lambda,M}$, E_θ and every explicitly specified phase in the sources (error 40). 03's statements for arbitrary phases, 04's prescribed prime phases and exponentials with character values in $1 + \mathfrak{m}$ are covered only when the phase in question has a resonance. A “chosen phase” of this kind does not escape the obstruction; the galaxy replica is then $\zeta^\#(w)$ on the resonant galaxy. The phase of Proposition 5.10(c) has no resonance, so the theorem does not apply to it.

Reconciliation with 03. The galaxies $\pm iT + \mathcal{O}$ lie outside $\mathcal{T}_h$. 03's “single pole at 1” (Theorem 16.2) is a statement on $\mathcal{T}_h$ and is untouched. Both statements express the $i\Pi$ -invariance of Exp, and the wing periodicity of Proposition 13.1 is its shadow on the right.

Remark 18.3 (Scope [06]). The theorem rules out a conjunction of explicit axioms, not every global surcomplex zeta function. A phase without resonance, a different summation rule, or a failure of coherence avoids its hypotheses. The phrase “trivial zeros” also needs care in a global theory. Extra zeros of a chosen global sine are not automatically trivial zeros of a completed zeta. Gamma poles and zeta zeros must cancel in a specified meromorphic construction before a completed function and its RH can be formulated. On $\mathcal{T}_h$, Theorem 16.4 is such a construction.

19 What weak global analyticity does not decide

Theorem 19.1 (Symmetric completions with different infinite zeros [05, 06]). *Let $T > \mathbb{R}$ and let $0 < a < \frac{1}{2}$ be ordinary, and put $c_\pm = \frac{1}{2} \pm iT$. In (a) assume moreover that T is purely infinite [06]; (b) holds for every positive infinite T [05].*

- (a) [06] *The class functions $F_0 = \xi^\#$ on $\mathcal{O}$, $F_0 = 1$ off $\mathcal{O}$, and $F_1 = F_0$ except $F_1(s) = (s - c_\pm)^2 - a^2$ on $\pm iT + \mathcal{O}$, are locally Hahn-analytic at every point of $\mathbf{SC}$, agree with $\xi^\#$ on $\mathcal{O}$, and satisfy $F(1 - s) = F(s)$ and $F(\bar{s}) = \overline{F(s)}$. F_0 has no infinite zeros. F_1 has exactly four, $c_\pm \pm a$, all simple and off the line.*
- (b) [05] *Let $F : \mathbf{SC} \rightarrow \mathbf{SC}$ be monadwise analytic (on every monad it has a strongly evaluable local power series) with the same two symmetries. Let $\mathcal{Q} = \{\frac{1}{2} \pm a \pm iT\}$, let U be the union of the four monads, and let $P_U = \prod_{b \in \mathcal{Q}} (s - b)$ on U and $P_U = 1$ off U . Then $\tilde{F} = P_U F$ has the same symmetries and local property, agrees with F wherever $\text{Im } s$ is finite, and vanishes on $\mathcal{Q}$. It differs from F unless F vanishes at every point of U where $P_U \neq 1$.*

Proof. (a) Finite translation and conjugation preserve $\mathcal{O}$, and every point has a fine neighbourhood inside its galaxy, so local analyticity holds. Under $s \mapsto 1 - s$ the two galaxies are

exchanged and the centred coordinate changes sign; conjugation also exchanges them. The zeros are $c_{\pm} \pm a$, with real parts $\frac{1}{2} \pm a$. (b) The four monads are disjoint: their centres differ by a nonzero ordinary real part or an infinite imaginary part. $\mathcal{Q}$ and U are stable under $s \mapsto 1 - s$ and conjugation, so the polynomial $\prod_{b \in \mathcal{Q}} (s - b)$ has real coefficients and is invariant under $s \mapsto 1 - s$, as its degree is even and its root set is reflection-stable. Hence P_U has both symmetries. On each monad P_U is a polynomial or 1, and U contains no point of finite imaginary part. $\square$

05's statement uses $a = \frac{1}{6}$, with zeros at $\frac{1}{3} \pm iT$ and $\frac{2}{3} \pm iT$, and says "there is another such function". That is false when F vanishes on U (for example $F \equiv 0$), hence the qualifier in (b) (error 32). The functions are not proposed as zeta completions; no Euler product or Gamma factorization is asserted [06]. They are not counterexamples to RH, and they are not an obstruction to a more rigid global category [05]. They show that finite agreement, local analyticity and the two completed symmetries do not settle RH at infinite height. A global RH must name its function and the extra axioms that select it.

19.1 Shortcuts that do not work, and several infinite RH statements

The Taylor series of ξ at 0 cannot be evaluated at an infinite s : infinitely many nonzero standard coefficients multiply powers of decreasing valuation, so the support condition fails [05]. A Hadamard product over the ordinary zeros is not automatically strongly summable at infinite s . Transferring an asymptotic formula while dropping its remainder leaves room for a flat term, and Theorem 17.1 shows that an exact identity can force such a term [04, 05].

For a specified global or partial function one can ask for the exact-line statement $\text{Re } \rho = \frac{1}{2}$ at all nontrivial zeros, for the standard-part statement $\text{st } \text{Re } \rho = \frac{1}{2}$ (which allows $\frac{1}{2} + \varepsilon + iT$), or for an infinite-height-only statement. These are different [05]. Density theorems need not exclude even rare zeros at a fixed distance from the line, and excluding infinite-height exceptions alone does not exclude a bounded set of ordinary exceptions. RH equivalences in this report use the full universal statement. "Trivial zeros" depend on the construction as well. The finite lift has exactly $-2, -4, \dots$. A tube phase has $2\mathcal{I}(\Psi_{\chi})_{<0}$, which is $2\mathbf{Oz}_{<0}$ for Exp (Corollary 16.3). An elementary model has the transferred negative even *internal* integers, which are not automatically $2\mathbf{Oz}_{<0}$ (Section 20.2). A finite-step recurrence cannot reach a negative infinite integer [03, 05].

Part VI

Elementary extensions and transfer

20 Transported structures

20.1 The structure that is transferred

Start from a set-sized first-order expansion $\mathfrak{A}$ of the real field. It names the real and imaginary coordinate functions of the entire functions $\xi(s)$, $(s-1)\zeta(s)$ and $1/\Gamma(s)$, with complex inputs as pairs of reals, and meromorphic functions are recovered by division off the zero set of a denominator or by domain predicates. It names the real exponential, sin and cos, *a constant for every real number*, and, as needed, counting functions $(N, N_0, N_0^s, N(\sigma, \cdot))$, an integer predicate with finite sums, Chebyshev's ψ , and the heat family H [04, 05, 06]. The real constants are used as parameters in the finite-zero and continuity arguments below. 06 names only the functions and leaves this implicit; Ehrlich and Kaplan's $T_{\text{trig,exp}}$ does the same (error 43).

A proper elementary extension $\mathfrak{M}^* \succ \mathfrak{A}$ exists, as a nonprincipal ultrapower [04, 06] or by compactness [05]. Its real field ${}^*\mathbb{R}$ is real closed and contains infinite elements, and it carries transferred functions ${}^*\zeta, {}^*\Gamma, {}^*\exp, \dots$. Transfer (Łoś) applies to first-order formulas in the named symbols. No transfer principle for all surreals is assumed.

Proposition 20.1 (Realization inside the surreals [04, 05, 06]). *Every set-sized real closed ordered field $F \supseteq \mathbb{R}$ has an ordered-field embedding into $\mathbf{No}$ fixing $\mathbb{R}$. Hence $F[i]$ embeds in $\mathbf{SC}$, and every named function of $\mathfrak{M}^*$ can be transported to the image.*

Proof. Well-order a transcendence basis of F over $\mathbb{R}$. At a stage with real closed domain E and embedding j , the next basis element α determines the set-sized sets $L = \{j(e) : e < \alpha\}$ and $R = \{j(e) : e > \alpha\}$. A surreal $\{L \mid R\}$ lies strictly between them, and it is not in $j(E)$, since every element of $j(E)$ lies on one side of its own cut. As $j(E)$ is real closed, it is transcendental over $j(E)$. The signs of polynomials over a real closed field are determined by the cut, so this extends j to an ordered embedding of $E(\alpha)$ and then of its real closure inside $\mathbf{No}$. At limit stages take unions, which are real closed. At the end the embedding is defined on the real closure of the rational-function field of the whole basis. F is algebraic over that real closed field, hence equal to it, so the embedding is defined on all of F . $\square$

This is the standard universality of $\mathbf{No}$ [2, 4], in a relative form. It supplies surcomplex-valued models at infinite height, but no unique function on all of $\mathbf{SC}$.

Warning 20.2 (Transported operations are extra structure [04, 05, 06]). An ordered-field embedding need not preserve the Gonshor exponential, strong Hahn sums, restricted analytic functions, a scalar derivation, or Exp . The transported ${}^*\exp$ is not automatically $\exp_{\mathbf{No}}$. ${}^*\zeta$ must not be identified with $\zeta^\#$ or ζ_D where their domains overlap, and transported values are not automatically the strong Taylor–Stirling values. Stars must not be erased at infinite arguments. The two-frequency test (Proposition 5.8) is a first-order sentence true in every elementary extension and false for Exp , so the transported full phase is not Exp on any domain containing a resonance. Transferred finite-order Taylor estimates allow, in a non-Archimedean field, discrepancies smaller than every power of an infinitesimal. Equality with a specified strong Hahn sum is an additional compatibility requirement. It would need an embedding preserving a restricted-analytic structure, and the relevant results [4, 5, 6] have hypotheses that would have to be checked for the chosen language.

20.2 Divisors in transported models

Include $G = 1/\Gamma$ and the integer sort. The classical facts $G(z) = 0 \iff z \in -\mathbb{N}$ and $G'(-n) = (-1)^n n!$ transfer. So ${}^*\Gamma$ has no zeros, and its poles are exactly $-{}^*\mathbb{N}_{\geq 0}$, simple, with $\text{Res}({}^*\Gamma, -n) = (-1)^n / {}^*n!$, which is infinitesimal at infinite n . Likewise ${}^*\zeta(-2n) = 0$ for every positive internal n , with the transferred derivative formula (6.5), and ${}^*\xi$ cancels these zeros and poles. Recurrence, reflection and the functional equation hold wherever the transported expressions are defined. A Gauss product with an internal number of factors needs the internal product operation, not a Hahn product [04].

Reconciliation. These poles are at the internal integers, which depend on the ultrafilter and the embedding. They are not every negative omnific integer, and ${}^*\mathbb{N}$ is not $\mathbf{Oz}$ [04, 05, 06]. The tube construction (Theorem 15.1) puts the poles on the period group of a phase, $\mathbf{Oz}_{\leq 0}$ for Exp . The two are different constructions. Both choose the alternative that Theorem 14.1 allows, extra infinite poles [06]; inside the model the transported sine has zero set $\pi {}^*\mathbb{Z}$. The positive-infinite Stirling chart and the finite lift do not by themselves specify an internal lattice of negative poles [04].

20.3 The transferred Riemann hypothesis

Theorem 20.3 (Transferred RH [04, 05, 06]). *Let $F = {}^*\mathbb{R}$ with the transported structure.*

- (a) $\forall s \in F[i] \ ({}^*\xi(s) = 0 \Rightarrow \operatorname{Re} s = \frac{1}{2})$ *is equivalent to classical RH.*
- (b) *The same with the added conclusion ${}^*\xi'(s) \neq 0$ is equivalent to RH together with simplicity of every nontrivial zero [04].*
- (c) *The model has zeros of ${}^*\xi$ at infinite heights.*
- (d) *Every finite zero of ${}^*\xi$, of ${}^*\zeta$, or of any transported nonzero ordinary entire function is an exact ordinary zero.*

Proof. (a) and (b) are transfers of first-order sentences about two real coordinates, in both directions. (c) “For every real T there is a zero with imaginary part greater than T ” is first order and transfers to infinite $T \in F$. (d) Let z be finite with standard part a . If $f(a) \neq 0$, an ordinary disk about a is zero-free, and that statement transfers. If $f(a) = 0$, an ordinary disk contains only the zero a , and “every zero in this disk equals a ” transfers, with a and the radius named by constants. Alternatively, an ordinary disk of radius R contains a finite list of zeros $\rho_1, \dots, \rho_m$, and “every zero in the disk is one of them” transfers. For zeta use the pole predicate as well. $\square$

This is a global RH at finite and infinite heights, but not a new conjecture. Its extra instances follow from transfer, and a proof of (a) independent of classical RH would prove classical RH [04, 05]. The zeros of the model are cofinal in the set-sized field F , not in **No** [05]. By contrast, an independently constructed global surcomplex zeta could have a genuinely stronger RH unless its connection with the classical function were proved. Finite-zero agreement with $\xi^\#$ does not give equality of finite values [04, 06].

20.4 Internal counting and quantifiers

${}^*N(T)$ is a named internal function. At infinite T it is an infinite internal integer, not the external cardinality of the set of ordinary zeros below T , which is countable [04, 05, 06]. It is not a Hahn sum and not a proper-class contour integral. An estimate $A(T) = O(B(T))$ transfers only in quantified form, with ordinary witnesses C, T_0 such that $|A(T)| \leq CB(T)$ for $T \geq T_0$. A bound $T^{\alpha+o(1)}$ transfers as “for each ordinary $\eta > 0$ there are ordinary C, T_0 with $A(T) \leq CT^{\alpha+\eta}$ ”. The exponent $o(1)$ cannot be suppressed at infinite T , since an infinitesimal exponent times $\log T$ can be large [04, 05]. A theorem for each fixed ordinary σ or η is not uniform in an infinitesimal perturbation of the parameter unless its constants are tracked. A bounded statement transfers with its ordinary endpoints, and $N(T)$ must not be replaced by $N(\operatorname{st} T)$ at an endpoint without a separate argument [05].

21 Classical results at the new scales

This section gives selected benchmarks with their exact scopes and statuses. It does not claim that any constant is the best currently known, and it improves no classical bound. Each classical result has one meaning for the finite lift (the zero set is unchanged, by [Theorem 6.3](#)) and another in an elementary model (quantified transfer), and the proofs are not rerun over Hahn coefficients [04, 05, 06].

21.1 Zero-free regions

By [Corollary 6.7](#), every ordinary zero-free region lifts exactly to its halo, boundary included. Mossinghoff, Trudgian and Yang prove zero-freeness in

$$\sigma \geq 1 - \frac{1}{55.241(\log |t|)^{2/3}(\log \log |t|)^{1/3}} \quad (|t| \geq 3), \quad \sigma \geq 1 - \frac{1}{5.558691 \log |t|} \quad (|t| \geq 2),$$

and with 48.1588 in place of 55.241 for all sufficiently large $|t|$ [\[20\]](#) [\[04, 05, 06\]](#).

Proposition 21.1 (Zero-free layers at infinite height [\[04, 05, 06\]](#)). *In an elementary model containing zeta and the logarithm, both explicit regions exclude zeros of ${}^*\zeta$ for internal σ, t with the transported ${}^*\log$. At every infinite $|T|$ the asymptotic region does too. The width $w(T) = 1/(55.241({}^*\log |T|)^{2/3}({}^*\log {}^*\log |T|)^{1/3})$ is a positive infinitesimal, and the layer $\sigma \geq 1 - w(T)$ is exactly zero-free.*

Proof. Each region is a first-order implication with ordinary constants. The asymptotic one has an ordinary threshold, exceeded by every infinite $|T|$. $\square$

This is stronger than $\text{st } \sigma \leq 1$ and far from $\sigma = \frac{1}{2}$. It holds for the transported structure. Substituting the Gonshor logarithm after an arbitrary embedding would need a compatibility that the embedding does not supply. At positive infinite real part, ζ_D is zero-free on $\mathcal{T}_+$ without transfer ([Theorem 8.5](#)), a different sector.

21.2 Verified heights

Platt and Trudgian verified RH rigorously for all zeros with $0 < \text{Im } \rho \leq T_{\text{PT}} = 3 \cdot 10^{12}$, using interval arithmetic [\[19\]](#). The published abstract adds that all these zeros are simple; arXiv v1 lacks that sentence [\[05\]](#). 06 declines to infer simplicity from the verification statement it quotes. 06's argument does not need simplicity. [Corollary 21.2](#) (from 05) does use it, through the published abstract's statement, whose computation this merge did not recheck. T_{PT} is a height, not a number of zeros, and it is not claimed to be the latest record [\[04, 05, 06\]](#). The canonical finite statement is immediate: every finite zero of $\zeta^\#$ with ordinate in $(0, T_{\text{PT}}]$ lies on the line.

Corollary 21.2 (Protection in verified monads [\[05\]](#)). *Assume the published simplicity statement of [\[19\]](#). For every $F \in \mathcal{D}_J$ and every real infinitesimal τ , every finite zero s of F_τ with $0 < \text{Im } s \leq 3 \cdot 10^{12}$ lies on $\text{Re } s = \frac{1}{2}$. Each verified zero has exactly one deformed zero in its monad.*

Proof. $\text{st } s$ is a zero of ξ in the verified range, simple and on the line by the published statement. Apply [Theorem 7.7](#). $\square$

The range refers to standard parts: a zero at a boundary ordinate may move infinitesimally above or below it. The corollary says nothing at infinite height. Platt and Trudgian also give $|\psi(x) - x| \leq \frac{\sqrt{x}}{8\pi} \log^2 x$ for $59 < x \leq 2.169 \cdot 10^{25}$ [\[04\]](#). This transfers with the same ordinary range. It does not become an all-infinite- x theorem, and the raw sum over ordinary primes below an infinite surreal is not the transported ${}^*\psi$.

21.3 Counting and Hardy’s theorem

The Riemann–von Mangoldt formula $N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$ [18, 16] holds exactly for the finite lift at every ordinary height. In an elementary model there is an ordinary C with $|*N(T) - \frac{T}{2\pi} \log \frac{T}{2\pi} + \frac{T}{2\pi}| \leq C \log T$ at all large internal T , including every infinite T . Hence $*N(T)/(\frac{T}{2\pi} \log \frac{T}{2\pi}) \in 1 + \mathfrak{m}$, equivalently $\text{st}(*N(T)/(T \log T)) = 1/(2\pi)$ [04, 05, 06]. (05 cites MTY for the formula, but MTY only quotes an explicit version; Titchmarsh is the primary source.) 04 reports that Rodgers and Tao record the version in the doubled heat variable; this was not rechecked. Hardy’s theorem, in the form “for every real bound there is a critical-line zero above it”, transfers to zeros above every internal bound. For the finite lift it gives only the original countable set [04, 05].

21.4 Proportions

Let $\kappa_0 = \liminf N_0(T)/N(T)$, where N_0 counts critical-line zeros with multiplicity, and $\kappa_s = \liminf N_0^s(T)/N(T)$, which counts simple critical-line zeros. These are different quantities.

- *Published* [21]: $\kappa_0 \geq 0.417293$ and $\kappa_s \geq 0.407511$. “More than five-twelfths” concerns zeros on the line, not simple zeros on the line. 05 uses only $\kappa_0 > 5/12$, which is consistent, since $0.417293 > 5/12 > 0.407511$. Whether PRZZ’s N_0 counts multiplicities was not checked by 05.
- *Preprints, August and September 2026*. Alpöge and Furman [23]: at least $2/3$ of the zeros are simple and on the line, and at least $5/6$ are distinct. With a Montgomery–Taylor window the constants become 0.6725 and 0.8362. Lamzouri [24], Theorem 1.1: $\kappa_s \geq C_0 = \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} = 0.6725007\dots$, distinct-zero proportion at least $(1 + C_0)/2 = 0.8362503\dots$, and 88.76% simple or on the line. The closed form C_0 , the Montgomery–Taylor constant, is stated in Lamzouri’s Theorem 1.1; AF’s 0.6725 is their Montgomery–Taylor-window variant (a precision added to 04, error 28; 06 was correct in substance). These are preprints. Their proofs were not verified by any source or by this merge, and AF’s reported formal certificates were not run. Every conclusion below is parametric in the accepted bound.

For the finite lift both sets of estimates apply unchanged. Theorem 7.14 gives an unconditional protected proportion at least 0.407511 for symmetric deformations with the published input, and at least C_0 with the preprint input. Using 0.417293 there would be wrong: a multiple critical-line zero may split off the line [04, 06].

Corollary 21.3 (Stable proportions [04, 05, 06]). *If $\liminf N_0^s(T)/N(T) \geq c_*$ for an ordinary c_* , the centre-indexed proportion of uniformly stable centres (Section 7.5) is at least c_* . In any elementary model naming the counts, for every ordinary $c < c_*$ and every infinite T , $*N_0^s(T) \geq c \log T$; hence $\text{st}(*N_0^s(T)/\log T) \geq c_*$. The same holds for N_0 under $\liminf N_0(T)/N(T) \geq c_*$.*

Proof. The halo statement is Theorem 7.7 at every simple critical-line centre. The $\liminf$ gives an ordinary T_c with $N_0^s(T) \geq c \log T$ for $T \geq T_c$, and this transfers. Taking every $c < c_*$ gives the standard-part form. $\square$

The conclusion is not the exact endpoint inequality with $c = c_*$. A positive proportion does not show that every zero is on the line or simple, and an infinite exceptional count of smaller order remains possible. With the unbounded total count it does give critical-line zeros at infinite heights in the model [04, 05, 06].

21.5 Zero density

Ingham's exponent is $A_I(\sigma) = 3(1 - \sigma)/(2 - \sigma)$. Guth and Maynard's Theorem 1.2 gives $A_{GM}(\sigma) = 15(1 - \sigma)/(3 + 5\sigma)$, and with Ingham in the complementary range their (1.4) gives $N(\sigma, T) \leq T^{30(1-\sigma)/13+o(1)}$ [22]. The two exponents cross at $\sigma = 7/10$, and $A_{GM}(3/4) = 5/9$. The paper is published in *Ann. of Math.* 203(2) (2026). 04 cites only arXiv v2 and 06 gives no status (errors 29, 46). Guth–Maynard count zeros with $|\operatorname{Im} \rho| \leq T$, as 04's $N_{\geq \sigma}^{\pm}$ does. The one-sided count $0 < \operatorname{Im} \rho \leq T$ of 05 and 06 is smaller, so the same bounds hold for it (05's remark, added to 06's text, error 47).

Proposition 21.4 (Fixed-parameter density transfer [04, 05, 06]). *Fix ordinary $\frac{1}{2} < \sigma < 1$ and $\eta > 0$. For each applicable exponent there are ordinary constants with ${}^*N(\sigma, T) \leq C_{\sigma, \eta} T^{A(\sigma) + \eta}$ at every infinite internal T , with the transported power. Unconditionally, ${}^*N(\sigma, T)/{}^*N(T)$ is infinitesimal for every fixed ordinary $\sigma > \frac{1}{2}$ and every infinite T .*

Proof. The first claim is the quantifier rule of Section 20.4. For the second, $A_I(\sigma) < 1$ for $\sigma > \frac{1}{2}$. Choose η with $A_I(\sigma) + \eta < 1$ and divide by the transferred counting formula: the ratio is at most $CT^{A_I + \eta - 1}/{}^*\log T \in \mathbf{m}$. For $\sigma \geq 1$ the count is zero. $\square$

By reflection and conjugation the same holds to the left of the line, so the fraction of zeros outside every fixed ordinary-width neighbourhood of the line is infinitesimal. This is not a proof that all zeros lie on the line. An infinite population can have infinitesimal relative density, and fixed- σ bounds do not reach $\sigma = \frac{1}{2} + \varepsilon$ with infinitesimal ε . For the finite lift the count is exactly the ordinary one, and no density statement for an arbitrary additive Hahn perturbation follows from its standard-part coefficient [04].

21.6 Primes and finite positivity tests

RH is equivalent to $\psi(x) = x + O(x^{1/2} \log^2 x)$ [18]. Its quantified consequences transfer once ψ and the arithmetic are named. In a bare surreal field there is no specified function “number of primes below an infinite x ”. Ordinary integers, omnific integers and transferred integers are different data, and a sum over unspecified “surreal primes” defines nothing [06]. Classical RH equivalences involving prime-counting errors, derivatives of zeta or positivity conditions transfer only after every function, sum and index set they use has been named in the structure. They do not become statements about a separately chosen Hahn-valued prime-counting function. A pair-correlation statement likewise needs its counting measure, normalization, test functions and limit quantifiers; it cannot be replaced by an unspecified “surreal distribution of zeros” [05].

Proposition 21.5 (Finite Hermitian tests [04, 06]). *An ordinary Hermitian matrix A is positive semidefinite (definite) over $\mathbb{C}$ if and only if $u^*Au \geq 0$ (> 0 for $u \neq 0$) for all $u \in \mathbf{SC}^n$. A positive semidefinite ordinary Hermitian form stays so after algebraic scalar extension to $\mathbf{SC}$.*

Proof. Diagonalize $A = UDU^*$ over $\mathbb{C}$. Then $u^*Au = \sum d_j |(U^*u)_j|^2$ in the ordered real field. A negative d_j already fails at an ordinary vector. Every vector of an algebraic scalar extension is a finite combination of ordinary vectors, whose Gram matrix is an ordinary positive semidefinite matrix. $\square$

Proposition 21.6 (A real margin protects positivity [04]). *If A is positive definite and Δ is Hermitian with infinitesimal entries, then $A + \Delta$ is positive definite over $\mathbf{SC}$. This fails for positive semidefinite A : take $A = \operatorname{diag}(0, 1)$ and $\Delta = \operatorname{diag}(-\eta, 0)$.*

Proof. With $u^*Au \geq a \sum |u_j|^2$ for an ordinary $a > 0$ and $e = \max |\Delta_{jk}|$, Cauchy–Schwarz gives $|u^*\Delta u| \leq de \sum |u_j|^2$, and $de < a$. $\square$

Surreal coefficients in a fixed finite family of Weil-type tests therefore add no condition. A test over all Hahn-supported infinite families needs a summation theorem, and a test over all infinitesimal deformations needs control of null directions [04, 06]. RH criteria built from fixed-order derivatives of ξ at ordinary points are preserved exactly by lifting. A hyperinteger index requires transfer, and a new Hahn test family is a third operation. Keeping the three apart prevents false “stronger RH” conclusions [04].

22 Heat flow at infinitesimal time

Use the normalization of Rodgers and Tao [25] [04, 05, 06]:

$$H_t(z) = \int_0^\infty e^{tu^2} \Phi_{\text{RT}}(u) \cos(zu) du, \quad \Phi_{\text{RT}}(u) = \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}},$$

$$H_0(z) = \frac{1}{8} \xi\left(\frac{1}{2} + \frac{iz}{2}\right) = \frac{1}{8} \Xi(z/2).$$

The last form reconciles 04’s stability variable $\Xi(w) = \xi(\frac{1}{2} + iw)$ with the doubled heat variable; there is no other factor of 2. The de Bruijn–Newman constant Λ_{DN} is characterized by: all zeros of H_t are real if and only if $t \geq \Lambda_{\text{DN}}$. RH is equivalent to $\Lambda_{\text{DN}} \leq 0$, and Rodgers and Tao proved $\Lambda_{\text{DN}} \geq 0$. So every negative ordinary t has a nonreal zero. (The upper bound $\Lambda_{\text{DN}} \leq 0.2$ is not used.) The super-exponential decay of Φ_{RT} makes H jointly entire in $(t, z) \in \mathbb{C}^2$, since each derivative of $e^{tu^2} \cos(zu)$ is bounded on compacta by a polynomial times $e^{Cu^2 + Cu}$. Differentiation under the ordinary integral gives $\partial_t H = -\partial_z^2 H$ and $H_t = \sum_n \frac{(-t)^n}{n!} H_0^{(2n)}$. No surreal integral is used: these are ordinary germs before lifting or transfer. In the variable $s = \frac{1}{2} + iz/2$ the formal family $8H_t$ is a deformation in $\mathcal{D}_J$ with both symmetries, since each $H_0^{(2n)}$ is real and even, and t is real [05].

Lemma 22.1 (Local real-root persistence [04, 05, 06]). *Let a be a simple real zero of H_0 . There are ordinary $r, \delta > 0$ such that for every real $|t| < \delta$, H_t has exactly one zero in $|z - a| < r$, and it is real, with $z(t) = a + t \frac{H_0''(a)}{H_0'(a)} + O(t^2)$. The same holds after canonical infinitesimal evaluation and after transfer.*

Proof. The analytic implicit-function theorem at $(0, a)$ gives one root with a unit factorization. Shrink r, δ until it is the only root in the disk. For real t , conjugation preserves the disk and the function, so uniqueness makes the root real. Differentiating $H_t(z(t)) = 0$ with the heat equation gives $z'(0) = -\partial_t H / \partial_z H = H_0''(a) / H_0'(a)$. Hahn evaluation preserves the factorization. The existence, uniqueness and reality statement, with a, r, δ named by constants, is first order and transfers. $\square$

Theorem 22.2 (Escape at negative infinitesimal time [04, 05, 06]). *Let $\mathfrak{M}^*$ be an elementary extension naming H as in Section 20.1, with real field F . The hypotheses are stated clause by clause.*

- (i) Assume RH and simplicity. *For every real infinitesimal τ , of either sign, every finite zero of the canonical joint Taylor lift $H^\#(\tau, \cdot)$ is real and simple.*
- (ii) Unconditionally, *for every negative infinitesimal $\tau \in F$, the transferred ${}^*H_\tau$ has a nonreal zero.*
- (iii) Assume RH and simplicity. *For every negative infinitesimal $\tau \in F$, every nonreal zero of ${}^*H_\tau$ has infinite modulus.*
- (iv) Unconditionally, *a finite nonreal zero of ${}^*H_\tau$ at an infinitesimal τ has as standard part either a nonreal zero of H_0 or a multiple real zero of H_0 .*

At a simple real zero c , $c(-\eta) = c - \eta H_0''(c)/H_0'(c) + O(\eta^2)$.

Proof. (i) Under the assumptions every zero of H_0 is real and simple. A finite zero of the joint lift has as standard part a zero c of H_0 . In the monad of c the evaluated unit factorization of [Lemma 22.1](#) leaves exactly one zero, which is real and simple. (ii) The ordinary sentence $\forall t < 0 \exists z (H_t(z) = 0 \wedge \text{Im } z \neq 0)$ follows from $\Lambda_{\text{DN}} \geq 0$ and the definition of Λ_{DN} , with no hypothesis, and it transfers. (iv) If z is finite with $a = \text{st } z$, then ordinary continuity of H near $(0, a)$, with each ordinary tolerance transferred, gives $H_0(a) = 0$. If a were a simple real zero, the transferred [Lemma 22.1](#) would make z real. (iii) Under RH and simplicity every zero of H_0 is simple and real, so (iv) leaves no possible standard part for a finite nonreal zero. $\square$

04 states (ii) under the RH hypothesis of its theorem, although (ii) needs no hypothesis (error 27). 06 places the canonical-lift clause (i) after its conditional statement in a way that reads as unconditional; it holds under RH and simplicity (error 44). 05 notes that only its part (b), which is (iii) here, uses the hypotheses, and it isolates (i) as a consequence of the deformation theorem. 06 states (iv).

The theorem explains why infinitesimal stability at every ordinary zero is compatible with $\Lambda_{\text{DN}} \geq 0$. Each fixed simple zero has a real time interval protecting it in a fixed disk, but no common interval protects all zeros at all heights. An infinitesimal time lies inside every individual interval, so the nonreal zeros forced by the global theorem must escape every finite disk. It proves infinite modulus only: no direction, rate, leading monomial or asymptotic scale of the escaping zeros is given [04, 05, 06]. Simplicity matters, because a multiple zero can split nonreally inside its own finite monad ([Corollary 7.13](#)). Clauses (i) and (iii) are conditional, and the theorem proves neither assumption.

Part VII

Synthesis

23 Synthesis and open questions

There is no single unqualified “RH over the surcomplex numbers”. There are precise statements attached to precise extensions.

- The finite RH ([Theorem 7.2](#)), the horizontal RH on $\mathcal{T}_h$ for every phase ([Corollary 16.5](#)) and the transferred RH ([Theorem 20.3\(a\)](#)) are each equivalent to classical RH. The first two hold for the reason that the divisors are rigid, and the third because of elementarity.
- The robust finite RH ([Theorem 7.9](#)) and the transferred RH with nonvanishing derivative ([Theorem 20.3\(b\)](#)) are each equivalent to RH together with simplicity.
- For a separately chosen global completion on $\mathcal{V}$, RH is an additional assertion whose content depends on the construction. Weak axioms do not determine it ([Theorem 19.1](#)). Natural combinations of a phase, coefficientwise summation and coherent continuation are inconsistent for every resonant phase ([Theorem 18.2](#)), and reflection conflicts with the Stirling modulus there except at two exceptional abscissae $\sigma_{\pm}(T)$ for each height, while $E \circ L_{\text{St}}$ fails reflection at every point of $\mathcal{V}$ ([Theorem 17.1](#)).

The finite theory is fully specified on its domain. The theory at positive infinite real part is a second exact construction, with an injective arithmetic algebra, exact Euler products, complete inverse fibres, independence theorems and exact Stirling and Hurwitz identities. A named phase

extends Gamma, zeta and ξ to the horizontal tube, with divisors that move with the phase while the ξ divisor stays classical. At infinite height the global problem needs additional choices, and several natural ones are inconsistent.

Question 23.1 (Nonresonant global completion [02, 06]; re-scoped [merge]). Specify a phase with no resonance, a summation rule in a finite-real-part region at infinite height, and a meromorphic continuation class. Can one keep a single zeta pole, the completed functional equation and a prescribed Gamma normalization? [Theorem 18.2](#) excludes every phase with a resonance, and in particular every phase with ordinary-circle character values. 02 asks for a chosen continuation or summation mechanism near infinite imaginary height compatible with the Gamma factors and phases, and 06 asked for a phase satisfying the two-frequency test. [Proposition 5.10\(c\)](#) supplies one, and whether a coherent galaxy zeta exists for it is open.

Question 23.2 (Combined independence). Are dilation-and-shift families $\zeta_D^{(j)}(k(X + a))$ algebraically independent over E_X^Γ , or over $E_X^\Gamma(\log X)$, which also contains $L_{\text{St}}(X)$? Neither [Theorem 10.8](#) nor [Theorem 10.14](#) implies this ([Section 10.4](#)).

Question 23.3 (Canonical versus transported values [04, 05, 06]). An elementary model has exactly the right finite zeros, but an arbitrary embedding does not identify its values with Hahn lifts. Which hypotheses on the embedding, or on a restricted analytic substructure, force equality, and how far into infinite sectors does it extend? The restricted-analytic results of [4, 5, 6] provide structure but do not settle this. A first version [04]: characterize extensions that preserve restricted analytic germs, the surreal exponential, the Dirichlet chart on $\mathcal{T}_+$ and a global functional equation. Existence alone would be meaningful, and uniqueness should not be assumed, in view of flat Gamma and Hurwitz modifications.

Question 23.4 (Quantitative escape [04, 06]). Under RH and simplicity, define in a transferred model the least, or an infimum, scale at which nonreal zeros appear for $\tau = -\eta$. How does it relate to η , to classical small gaps and to the collision mechanism? [Theorem 22.2](#) proves only that every ordinary bound is exceeded.

Question 23.5 (Selecting a phase [01, 02, 03]). Which compatibility conditions select a preferred phase, or exclude all the coefficient-functional phases, beyond the group law and the local Taylor law? Candidates include compatibility with a scalar derivation, with automorphisms or with simplicity structure. For which classes of infinite imaginary arguments does $(E_X(-s \log n))_n$ have a provable joint support certificate ([Propositions 13.1](#) and [13.3](#)), and is such a phase compatible with a scalar-derivation chain rule [03]? Any proposed uniqueness theorem must explain why it excludes the explicit families of [Proposition 5.5](#) [01]. For the robustness questions of Ehrlich and Kaplan, see `trigonometry:per:sec:ek`.

Question 23.6 (Arithmetic restrictions on deformations [04]). Which arithmetic restrictions on coherent perturbations of ξ exclude off-line splitting without assuming simplicity? Constant perturbations show that symmetry alone is not enough. An Euler product, an explicit formula or compatible Dirichlet data are candidates, and each must be formulated on a domain where the operation exists.

Question 23.7 (Extending the constructions [01, 02, 03]). (a) Inverse theorems for other Dirichlet series $1 + a_N N^{-s} + \dots$, with weights $\log_N(n/N)$, complex leading coefficients, missing indices and cancellations [01]. (b) Generalized Dirichlet series with well-ordered real frequency supports, keeping a leading-form classification; arbitrary frequency sets can destroy well-ordering or finite convolution fibres [02]. (c) Dirichlet characters and Euler factors of finite degree, with a completed functional equation compatible with their infinite divisors and phase factors [03]. (d) Coefficient fields built from several Gamma or Barnes scales, adjoining a new monomial only when its support coordinate is separated from the arithmetic one [03].

Question 23.8 (Gamma on the vertical strip [05]). Is there a Gamma prescription on $\mathcal{V}$ with conjugation symmetry, reflection, the recurrence and a flat modulus correction as in [Theorem 17.1\(d\)](#), together with a specified phase? At the two exceptional abscissae $\sigma_{\pm}(T)$ the modulus test is silent, and nothing is claimed there for a general prescription with Stirling moduli; $E \circ L_{\text{St}}$ itself fails reflection there too ([Theorem 17.1\(e\)](#)), so such a prescription cannot be $E \circ L_{\text{St}}$.

Formalization. No theorem of this report is formalized. Only imported infrastructure has Lean-proved rows in `docs/FORMALIZATION.md`: the substitution clauses of `a:cor:complexsub`, near-one binomial powers, and fine derivatives of evaluated ordinary power series. The sources propose similar orders. First comes the support layer: Neumann’s lemma, strong substitution, evaluation of weighted series and the divided-difference unit lemma. Then the valuation-unit and divisor lemmas, the local perturbation factorization with the involution J , and the arithmetic algebra with finite-prime projection and the Jacobian argument. Then the formal Bernoulli identities for Stirling and Hurwitz. Last come the phase-period, reflection and completed-divisor theorems as abstract consequences, and transfer in a specified first-order language. The analytic implicit-function theorem, the surreal exponential and every imported number-theoretic theorem should remain named inputs, not be hidden in a computational certificate [01, 02, 03, 04, 05, 06].

24 What this report does not claim

24.1 For the report as a whole

- It does not prove, disprove or advance the classical Riemann hypothesis, and it improves no classical numerical bound.
- It constructs no canonical, unique or maximal global zeta, ξ or Gamma function on all of $\mathbf{No}[i]$. Every global object here depends on a named phase, domain, summation rule and continuation class.
- No theorem of the report is verified in Lean or any proof assistant; only the imported infrastructure noted above has Lean-proved rows. The report is unrefereed and AI-assisted.
- Priority is not certified for any result. “Derived theorem” and [merge] describe proof status, not historical novelty. No named published open problem is claimed to be solved.
- The 2026 proportion results are cited as preprints, not verified, and nothing local depends on them.
- The finite checks of the six suites and of the merge verify finite algebra and ordinary numerics. They do not verify strong summability, class quantification, independence, continuation, RH or any cited paper.
- The repository audits of the sources were targeted, and so was the merge’s comparison with the collection. An empty search is not evidence of absence.

24.2 Non-claims of the sources

Every limitation, disclaimer and priority caveat stated by a source is kept, grouped by source. Symbols are renamed as in [Table 1](#).

01.

- Unrefereed, not verified in Lean, priority not certified.
- The Taylor–Stirling baseline and the finite-phase locus are repository precedents; the finite lifts and the Stirling extension have precedents (Costin–Ehrlich, Berarducci–Mantova).
- Lagrange inversion is classical and inverse zeta has a literature (Kawalec). Only the package with its surreal domain and support assertions is claimed.
- The lift is canonical only within germ evaluation, not by uniqueness among finely differentiable functions.
- Divisor rigidity does not apply to Hahn-coefficient deformations.
- The finite-halo RH adds nothing toward RH and says nothing at infinite height.
- The finite-phase criterion is not a maximal-domain or impossibility theorem.
- Stirling uniqueness holds only among normalized formal series, and flat modifications are not excluded.
- E_λ is not asserted canonical or preferred, and its fine derivative is not the Berarducci–Mantova derivation.
- 01’s own construction gives no Gamma reflection across the negative infinite half-plane. The tube construction of 03 now supplies one under a named phase, and the non-claim stays true of 01’s construction.
- The Dirichlet embedding is not a classical analytic-continuation claim.
- ζ_D and $\zeta^\#$ are glued only on disjoint domains, with no fine-topological identification.
- The fibre theorem is exhaustive only in $\mathcal{T}_+$, and there is no covering-map claim.
- The integer-power sector is a projection of the formal u -expansion, not of the evaluated normal form.
- The left prescription $Z_{-, \lambda}$ is one-sided and not by itself a proved global functional equation.
- No contour integral, proper-class sum or transfinite recurrence is used, and Gauss indices are ordinary.
- The normalization at infinity is not forced by the finite function.
- Numerics are ordinary-complex illustrations, not error bounds.
- The Lean plan is a dependency plan only.
- The repository audit was targeted.

02.

- No unique Gamma or zeta on $\mathbf{No}[i]$: no canonical global continuation, no unique phase at infinite imaginary part, no maximal domain.
- No zeta and no functional equation in the infinite-height strip.
- The $\xi^\#$ statement is equivalent to RH, not a proof.
- Independence concerns infinite values, not $\zeta(2), \zeta(3), \dots$, and standard parts destroy the information.
- Derivatives are coordinate derivatives. No mean value or identity theorem across fine components is used.
- The finite lift is canonical only as a germ lift.

- The injectivity of $\mathcal{D}_s$ is non-effective.
- Gamma is not defined at negative infinite arguments by 02.
- ζ_H is not a rearrangement of $\sum(A+n)^{-s}$. The Lerch identity does not exclude flat modifications, and $-1/12$ is not the strong sum $1+2+3+\dots$.
- The E_θ are exponential homomorphisms, not preferred ones. The list of phase prescriptions is not exhaustive, and Ω_∞ is not claimed maximal.
- $\zeta_{\chi_\theta}^{\mathcal{W}}$ depends on the phase and is not a canonical continuation.
- Pong and the classical literature are acknowledged. No novelty of the functional-independence principle and no priority are claimed. The repository reading was targeted to three README files; the gamma-functions article source was not reviewed.
- No proof assistant was used. The 189 checks are finite identities. The normal-form and exponential facts are imported inputs.
- All supports are set-sized, and no basis or choice is used for proper-class vector spaces. The phases E_θ are stated for a given real-linear θ ; the explicit ones are the coefficient functionals $\theta = r c_M$, which need no choice.

03.

- No RH and no resolution of a named problem. Unrefereed, no proof assistant, priority not certified.
- No Gamma or zeta at arbitrary infinite imaginary height, and no value at $\frac{1}{2} + i\omega$.
- Zeta on $\mathcal{T}_-$ is defined by the functional equation, and the exact Stirling sum excludes flat terms by definition. Neither is a continuation or uniqueness theorem.
- [Proposition 15.3](#) is not a surreal Bohr–Møllerup theorem.
- No connected continuation between $\mathcal{T}_-$, $\mathcal{O}$ and $\mathcal{T}_+$, no global identity theorem, no single entire series.
- The Gamma values are not claimed mutually independent. Coefficients from all of **SC** are not allowed, nor is a Hahn completion containing the zeta series.
- No compatibility with the Berarducci–Mantova derivation. A phase that commutes with a scalar derivation is flagged as an extra, possibly restrictive, hypothesis.
- Hurwitz uniqueness holds only in $A^{1-s}\mathbb{C}(s)[[A^{-1}]]$; $-1/12$ is not a strong sum; flat Hurwitz sectors are not excluded.
- The checks do not verify summability, quantification, independence or novelty.
- The canonical phase is a convention, not forced by reflection. Every phase has infinite poles.

04.

- No unique meromorphic zeta or Gamma on **No**[i]. The constructions are not identified with each other; ζ_D is not a continuation of $\zeta^\#$; no surcomplex GRH.
- The Stirling and Hurwitz values are a selected normalization. Flat alternatives exist, and no uniqueness, convexity or Gauss claim is made for the correction.
- No values at infinite critical heights from local data. Phases are underdetermined, and they do not give strong summability. Riemann–Siegel at infinite T is not an ordinary sum.

- The transported model depends on the ultrafilter and the embedding. ${}^*\mathbb{N} \neq \mathbf{Oz}$, and the poles are at internal integers only. The transferred RH is not a new conjecture.
- Only quantified estimates transfer, and bounded verifications do not reach infinite heights. No density statement at infinitesimal $\sigma - \frac{1}{2}$.
- [Theorem 7.14](#) proves neither RH nor simplicity, and 0.417293 must not be used in it.
- The 2026 preprints are marked, unverified and not needed locally.
- The escape clause of the heat theorem is conditional on RH and simplicity, and it gives infinite modulus only, not direction or rate.
- Priority not established; no resolution of named open problems; unrefereed; no Lean. The code checks finite identities. The repository audit covered the catalogue and summaries only.

05.

- No proof of RH, no numerical improvement, no canonical zeta or ξ on $\mathbf{No}[i]$.
- The lift is canonical only on finite arguments and undefined at actual poles; punctured pole monads are in the domain.
- The ratio cocycle is formal and phase-free. No claim that exponentially small corrections vanish.
- Bare Stirling is not an exact global log-Gamma. E_{fp} does not define $\exp(iT)$ for infinite T . The flat correction fixes only a modulus on one line, and supplies neither a phase nor a global Gamma.
- The gauge $\lambda e^{-\text{pp}}$ carries no convexity or multiplication claim.
- $\mathcal{T}_+$ misses the strip at infinite height, and there is no uniqueness of continuation between the finite and $\mathcal{T}_+$ regions.
- [Theorem 19.1\(b\)](#) is not a counterexample to RH and not an obstruction to a more rigid category.
- Elementary realizations are noncanonical; ${}^*\exp \neq \exp_{\mathbf{No}}$ in general; internal counts are not external cardinalities or Hahn sums; zeros are cofinal in the set-sized field F (the image of ${}^*\mathbb{R}$), not in $\mathbf{No}$.
- Transfer holds for fixed ordinary parameters only, with no shortcut at the endpoint $\sigma = \frac{1}{2} + \varepsilon$ or $N(T) \rightarrow N(\text{st } T)$.
- A proportion is not an all-zeros assertion.
- Benchmarks are selected published results, not records.
- In 05's heat theorem only the infinite-modulus clause uses RH and simplicity, and no rate is given.
- RH equivalences through prime counting, derivatives, positivity or pair correlation transfer only after every function, sum and index set they use is named; no unspecified “surreal distribution of zeros” is used.
- No exhaustive priority claim; targeted audit; no Lean. Finite checks do not certify the support lemmas, quantifiers or transfers.

06.

- No proof of RH and no unique global zeta. Unrefereed, no proof assistant, no priority for every consequence. The repository audit was targeted.
- The Gamma nonuniqueness, the convexity classification and the finite-phase tube are not new.
- “Canonical” means the explicit Taylor–Laurent rule.
- A strong sum is not a fine limit. The formal, local-analytic, fine and Berarducci–Mantova derivatives are distinct, and no chain rule for the last is asserted.
- Infinitesimal residuals are not zeros.
- The robust criterion proves neither RH nor simplicity.
- Stable proportions are centre-indexed.
- The results on $\mathcal{T}_+$ say nothing about the infinite-height strip.
- ζ_H is not a strong sum of $(A + n)^{-s}$. There is no global zeta by substituting $A = 1$, and no unrestricted complex-parameter prescription without a named phase.
- $E_{\text{fp}}(L_{\text{St}})$ is meaningful only on Ω_∞ ; elsewhere a phase must be chosen.
- The Gamma obstruction does not rule out every global Gamma, and the resonance theorem does not exclude every global zeta.
- The functions of [Theorem 19.1\(a\)](#) are not proposed completions.
- Transported operations are not canonical ones, and finite-zero agreement is not equality of values.
- $*N$ is internal; big-O needs ordinary witnesses; no statement at $\sigma = \frac{1}{2} + \varepsilon$.
- Platt–Trudgian is cited, not recomputed, and is not the latest record. 06 does not infer simplicity from it. This report cites the published simplicity statement separately, as 05 does, and uses it only in [Corollary 21.2](#).
- The proportion preprints are unverified. The conclusions are parametric, and the $\liminf$ gives only every ordinary c below the accepted bound c_* .
- No surreal primes or prime counting in a bare surreal field.
- Finite Hermitian tests gain nothing from surreal scalars.
- The heat escape has no scale, and its escape clause is conditional.
- The checks are finite algebra only.
- Workspaces are set-sized, and no single Hahn field is closed under all operations.

The merge.

- [Proposition 5.10\(c\)](#) and [Proposition 13.3](#) use the axiom of choice, for a basis of $\mathbb{R}$ over $\mathbb{Q}(\log 3/\log 2)$ and for a Hamel basis.
- Whether a coherent galaxy zeta exists for a phase without resonance is open ([Question 23.1](#)).
- The class $\mathcal{D}_J$ is not claimed to be maximal for [Theorem 7.9](#).
- The combined dilation-and-shift independence is open.

- The content of Pong’s 2020 Addendum [10] was not checked. Whether other classical work covers 02’s dilation family was not searched.
- At the two exceptional abscissae $\sigma_{\pm}(T)$ of [Theorem 17.1](#), for a general prescription with Stirling moduli, no reflection obstruction and no compatibility is claimed. For $E \circ L_{St}$ itself reflection fails there as well ([Theorem 17.1\(e\)](#)).
- [Proposition 5.10\(a\)](#) covers only phases with ordinary-circle character values. 03’s statements for arbitrary phases, 04’s prescribed prime phases and exponentials with character values in $1 + \mathfrak{m}$ fall under [Theorem 18.2](#) only when they have a resonance.
- The page range of the Guth–Maynard article is as cited by source 05; the Crossref record consulted shows the volume, issue and DOI.
- The merge checks corroborate finite algebra and ordinary numerics. They are not shipped and are not proofs.

A Provenance and reconciliation record

A.1 The six manuscripts

01 and 03 were prepared with ChatGPT, 02 and 06 call themselves AI-assisted, and 04 and 05 call themselves research articles; 01, 02, 04, 05 and 06 name Vladimir Reshetnikov as the person they were prepared for. Five are dated 22 September 2026 and 06 September 2026. None names a human author. Each describes itself as unrefereed and not verified in a proof assistant. Each cites a pinned commit of this repository and a targeted reading of it.

No.	Package	Pin; reading	Shipped here (code and data)
01	31-page article, <code>verification.py</code> , recorded results, CSV of inverse coefficients, source audit, build record	804c63c; catalogue, root README, gamma-functions README and introduction, opening of the analysis article	01-inverse-zeta-*: suite, results, CSV (CRLF line ends, kept byte-identical), requirements, audit, build record
02	29-page article, <code>checks/verify.py</code> , <code>checks/results.json</code> , <code>build.sh</code> , audit, validation	804c63c; catalogue, main README, gamma-functions README	02-scale-separation-*: suite, build script, results, audit, validation, requirements
03	34-page article, <code>code/verify.py</code> , verification and build records	804c63c; root README, catalogue, gamma-functions README and introduction	03-horizontal-tube-*: suite, verification, build record, requirements

No.	Package	Pin; reading	Shipped here (code and data)
04	31-page article, <code>code/verify.py</code> , verification, audit, build review	905dd19; catalogue, gamma-functions and analysis READMEs	04-rh-transfer-*: suite, verification, audit, build review
05	28-page article, <code>verify_identities.py</code> , verification	905dd19; catalogue, analysis opening, gamma-functions README	05-surcomplex-fields-*: suite, verification, requirements
06	28-page article, <code>code/checks.py</code> , <code>build.sh</code> , checks, proof status, audit, build record	905dd19; catalogue, gamma-functions, analysis and trigonometry READMEs	06-phases-and-rh-*: suite, build script, checks, proof status, audit, build record

No checksum or manifest file was delivered with any package. The manuscripts themselves (their L^AT_EX sources, PDFs and delivery READMEs) are not shipped. The shipped `article.tex` is this report, rewritten from 06's.

A.2 Where each result is printed

Every numbered result of every source is printed here, once, with a second route where marked. Source numbers are the sources' own.

Src	Source results	Here
01	Thm 1.1; Def 2.1; Lem 2.2, 2.3, 2.4; Ex 2.5; the fine-topology paragraph of Section 2.4	Theorem 9.1 , Definition 4.1 , and Lemmas 4.3, 4.5 and 9.2 ; the paragraph after Definition 4.1 ; Remark 4.7
01	Prop 3.1; Thm 3.2; Cor 3.3; eqs. 3.3–3.9; Warn 3.4	Proposition 6.2 and Theorems 6.3 and 7.2 ; Sections 6.2 and 6.3 ; Warning 6.6
01	Prop 4.1, 4.2	Proposition 5.1 ; Proposition 5.5(a) with Proposition 5.3
01	Prop 5.1; Warn 5.2; Thm 5.3, 5.4	Proposition 12.1 , Warning 12.2 , and Theorems 12.4 and 12.6 (second route after it)
01	Thm 6.1; eqs. 6.4–6.5; Thm 6.2; Thm 6.3, eq. 6.11	Proposition 11.1 and Remark 12.7 ; (11.2), Theorem 12.6 ; Propositions 11.5 and 11.6 ; Proposition 11.7 and Example 11.8 , Section 11.4
01	Thm 7.1; Def 7.2; Thm 7.3, 7.4	Theorems 8.1 and 8.5 , Definition 8.4 , and Proposition 8.3
01	Lem 8.1; Thm 8.2; Cor 8.3, 8.4	Lemma 9.3 , Theorem 9.4 , and Corollaries 9.5 and 9.6
01	Lem 9.1; Thm 9.2, 9.3; Cor 9.4; Ex 9.5; numerics	Lemma 9.7 , Theorems 9.8 and 9.9 , and Corollary 9.10 ; end of Section 9

Src	Source results	Here
01	Thm 10.1 and remark; Sections 11–12	Proposition 16.7 and Remark 16.8; Questions 23.1, 23.5 and 23.7, Section 24; Appendix B
02	Thm 1.1; Def 2.1, Lem 2.2, Rem 2.3	Theorem 10.8; Definition 4.1 and Lemmas 4.3 and 4.4; after Definition 4.1
02	Thm 3.1, Cor 3.2, Rem 3.3; eqs. 4.1–4.9; Thm 4.1; eqs. 4.10–4.12; Warn 4.2	Proposition 6.2, Theorem 6.3, and Remark 6.4; Sections 6.2 and 6.3; Corollary 6.5 and Theorem 7.2, (6.5); Section 6.4 and Warning 6.6
02	Thm 5.1, Rem 5.2, Prop 5.3; Thm 6.1, Cor 6.2, Prop 6.3, Warn 6.4	Theorems 8.1 and 8.5, Remark 8.2, and Propositions 8.3 and 8.6; Section 13
02	Lem 7.1, 7.2; Thm 7.3; Cor 7.4	Lemmas 10.1 and 10.2 and Theorem 10.3
02	Prop 8.1; eq. 8.6; Prop 8.2; Ex 8.3; Section 8.4	Propositions 11.1 and 11.5 and Theorem 11.2; after Proposition 11.5; after (11.2)
02	Lem 9.1; Thm 9.2; Cor 9.3; Lem 9.4; Thm 9.5; Cor 9.6	Lemmas 10.4 and 10.6 and Theorems 10.5 and 10.7
02	Warn 10.1; Thm 10.2, 10.3; Cor 10.4	Warning 12.2 and Theorems 12.4 and 12.6
02	Prop 11.1; Thm 11.2; Prop 11.3; Section 12; Section 13, Appendix A	Proposition 5.5(c), Proposition 5.3 and Definition 5.4; Proposition 11.7 and Example 11.8(b); Proposition 13.1 and Corollary 13.2; Example 8.8, Section 11.4, Section 6.2, Appendix B; Sections 23 and 24, and for Appendix A (dependency audit) the proofs of Theorems 8.1, 10.3, 10.5 and 10.7 and Lemma 10.6
03	Thm 1.1; Thm 1.2	Part IV; Theorem 10.13
03	Def 2.1, Lem 2.2, Warn 2.3; Section 2.3	Definition 4.1, Lemma 4.3, and Remark 4.7; Section 4.2
03	Prop 3.1; Thm 3.2; eq. 3.1; eqs. 3.2–3.4	Proposition 6.2 and Theorem 6.3; Definition 6.1; Section 6.2
03	Lem 4.1; Def 4.2; eqs. 4.2–4.3; Prop 4.3; Def 4.4; Thm 4.5; Thm 4.6	Propositions 5.1 and 5.3 and Definitions 5.2 and 5.4; Theorem 5.7 (definition of $\mathcal{I}$ and parts (a)–(b)); Theorem 14.1(b)
03	Prop 5.1; Thm 5.2; eqs. 5.12–5.14; Thm 5.3	Proposition 11.7 and Theorem 11.2; (11.2), Example 11.8(c); Propositions 11.1 and 11.7
03	Thm 6.1, 6.2; Prop 6.3	Theorems 15.1 and 15.2 and Proposition 15.3
03	Thm 7.1; Lem 7.2; eq. 7.6; Thm 7.3	Theorem 8.1; Lemma 4.4 with the proof of Theorem 8.1; Proposition 8.3 and Definition 8.4; Theorem 8.5
03	Lem 8.1; Thm 8.2; Section 8.2	Lemma 16.1, Theorem 16.2, and Corollary 16.3
03	Thm 9.1; Cor 9.2; eqs. 9.3–9.5; eqs. 9.6–9.7; Prop 9.3	Theorem 16.4, Corollary 16.5, and Proposition 16.6; Proposition 5.5(b); Proposition 16.7
03	Lem 10.1, 10.2; Thm 10.3; Lem 10.4; Thm 10.5, 10.6	Lemmas 10.9, 10.10 and 10.12 and Theorems 10.11, 10.13 and 10.14
03	Prop 11.1; Thm 11.2, 11.3; Cor 11.4; Thm 11.5	Warning 12.2, Theorems 12.3, 12.4 and 12.6, and Corollary 12.5
03	Section 12; Prop 13.1; Sections 13–14	Example 8.8, after Example 11.8 and Theorem 10.14; Example 15.4 and Corollary 16.3, after Proposition 16.6 and Corollary 16.5; Theorem 12.6; Proposition 13.1; Sections 23 and 24
04	Section 1; Lem 2.1; Def 2.2; Prop 2.3; Warn 2.4	Sections 1.1 and 2.4; Lemma 4.3, Definition 6.1, and Proposition 6.2; Section 4.2 and Remark 6.4
04	Thm 3.1; Cor 3.2; Ex 3.3; Cor 4.1; eqs. 4.1–4.11	Theorem 6.3; Section 6.2; Corollary 6.5; Sections 6.2 to 6.4, (6.5)

Src	Source results	Here
04	Def 5.1, Thm 5.2; Lem 5.3; Thm 5.4; Cor 5.5; Section 5.3	Definition 7.1, Theorems 7.2 and 7.8, and Lemma 7.4; Theorem 7.9, Corollary 7.10(a); Remark 7.12 and Corollary 7.13
04	Prop 6.1; Cor 6.2; Thm 6.3	Proposition 7.5 and Theorem 7.7; Remark 7.11, Corollary 7.10(a); Theorem 7.14
04	Thm 7.1; Prop 7.2; Thm 7.3	Theorems 8.1 and 8.5, Remark 8.2, and Proposition 8.3
04	eq. 8.1; Thm 8.1; eq. 8.8; Warn 8.2; Prop 8.3; eq. 8.9	Section 11; Theorems 12.4 and 12.6 and Remark 12.7; Theorem 12.6; Warning 12.2 and Propositions 11.5 and 12.8
04	Prop 9.1; Sections 9.2–9.3	Proposition 5.6; Sections 18 and 19
04	Lem 10.1; Section 10.2; Thm 10.2; Prop 10.3; Sections 10.4–10.5	Proposition 20.1; Section 20.2; Theorem 20.3(a),(b); Theorem 20.3(d); Warning 20.2 and Section 20.4
04	Section 11	Sections 20.4 and 21.2 to 21.5 and Proposition 21.1
04	eqs. 12.1–12.4; Thm 12.1; Prop 13.1, 13.2; Section 14; Appendices A, B	Section 22; Theorem 22.2; Propositions 21.5 and 21.6; Section 23; Theorem 7.7, Corollary 7.13, and Example 8.8, Section 11.4; Appendix B
05	Lem 2.1; Prop 2.2; Prop 3.1; eqs. 3.2–3.9	Lemma 4.3, Warning 6.6, Remark 4.7, and Proposition 6.2; Sections 6.2 and 6.3
05	Thm 4.1; Cor 4.2; Thm 4.3; Section 4.1	Theorems 6.3 and 7.2 and Corollary 6.5; Section 6.2
05	Def 5.1; Lem 5.2; Thm 5.3, 5.4; Rem 5.5	Definition 7.6, Lemma 7.3, and Theorems 7.7 and 7.9; Corollary 7.10(b); after Corollary 7.13
05	Section 6.1; Prop 6.1; eq. 6.4; Section 6.3, eq. 6.5; Thm 6.2, eqs. 6.6–6.12	Propositions 5.1, 11.1 and 11.5; Section 11.5(2); Theorems 11.2 and 11.3 and Remark 11.4
05	Thm 7.1; Cor 7.2	Theorem 17.1
05	Thm 8.1; Def 8.2; Cor 8.3 and the expansion after it; Warn 8.4	Theorems 8.1 and 8.5, Definition 8.4, Proposition 8.3, and Remark 8.7
05	Section 9.1; Thm 9.1; Section 9.3	Part V (opening) and Section 19; Theorem 19.1(b); Section 19 (second subsection)
05	Section 10.1; Prop 10.1; Warn 10.2; Thm 10.3; Prop 10.4; Section 10.3	Section 20.1; Proposition 20.1 and Warning 20.2; Theorem 20.3(a); Theorem 20.3(d); Sections 20.4 and 21.3, Theorem 20.3(c)
05	Section 11.1; eqs. 11.1–11.2, Prop 11.1; eq. 11.3, Cor 11.2; eqs. 11.4–11.7; eqs. 11.8–11.10, Prop 11.3; Section 11.7	Section 20.4; Proposition 21.1; Section 21.2 and Corollary 21.2; Sections 21.3 and 21.4 and Corollary 21.3; Section 21.5 and Proposition 21.4; Section 21.6
05	eqs. 12.1–12.4; Thm 12.1; Section 13; Appendix A	Section 22; Theorem 22.2; Section 23; Section 6.2 and Example 8.8
06	Lem 2.1; Section 2.3	Lemma 4.3; Section 2.2
06	Def 3.1; Prop 3.2; Thm 3.3; Cor 3.4; Prop 3.5; eq. 3.8	Definition 6.1, Proposition 6.2, Theorem 6.3, and Corollary 6.5; Sections 6.3 and 6.4
06	Def 4.1; Thm 4.2; Thm 4.3; Thm 4.4; Cor 4.5, Ex 4.6; Section 4.4	Definition 7.1; Theorem 7.2; Theorem 7.7; Theorem 7.9 with Corollary 7.10(c); Corollary 7.13; Section 7.5, Corollary 21.3
06	Thm 5.1; Def 5.2; Thm 5.3; Prop 5.4; eq. 5.13	Theorems 8.1 and 8.5, Definition 8.4, Proposition 8.3, and Remark 8.7
06	eq. 6.1; Prop 6.1; eqs. 6.4–6.5; Thm 6.2; Ex 6.3; Warn 6.4; eq. 6.12	(11.1); Propositions 11.1 and 11.7, 06's route Remark 12.7; Proposition 11.5; Theorems 12.4 and 12.6; Theorem 12.6; Warning 12.2; Section 11.5(3)

Src	Source results	Here
06	eqs. 7.1–7.2; Thm 7.1; Prop 7.2; eq. 7.5; coherent continuation; Thm 7.3; eq. 7.8; Rem 7.4	Definition 5.4; Theorem 14.1(a); Proposition 5.8; (18.1); Definition 18.1 and Theorem 18.2; after it; Remark 18.3 and Proposition 5.10
06	Thm 8.1; Prop 9.1; Thm 9.2 with the remark on $^*\Gamma$; Section 9.3	Theorem 19.1(a); Proposition 20.1; Theorem 20.3, Section 20.2; Section 20.4
06	Prop 10.1; eqs. 10.1–10.9; Cor 10.2; Prop 10.3; Section 10.5	Corollary 6.7; Section 21; Corollary 21.3 and Proposition 21.5; Section 21.6
06	Lem 11.1; Thm 11.2; Section 12; Appendices A–B	Lemma 22.1 and Theorem 22.2; Section 23; Appendices B and C

A.3 Where the merge had to choose

- *Base text and notation.* 06 is the base; 03 is the base of Part IV. Notation follows Section 2. $\mathbf{SC}$, Π , Exp and $\mathcal{I}(\Psi)$ are taken from the collection, and every source symbol that collided is renamed (Table 1).
- *Duplicates printed once.* Substitution (all six); divisor rigidity (all six, with the analysis report); finite RH (01, 02, 04, 05, 06); the Dirichlet realization and Euler identities (all six); Stirling recurrence and Gauss (01, 02, 03, 05, 06, the recurrence also by 04, and the gamma-functions report); the finite-phase criterion (all six, and the gamma-functions report); the Hurwitz identities (01, 02, 03, 04, 06); the embedding lemma and transferred RH (04, 05, 06); forced poles (03, 06); weak completions (05, 06); the heat theorem (04, 05, 06); the MTY, counting, proportion and density transfers (04, 05, 06).
- *Second routes kept.* The Hurwitz route to the Stirling identities (Remark 12.7); 01's operator proof of distribution after Theorem 12.4; 06's sign choice for the stability converse (Corollary 7.10); local conservation (04) beside the involution argument (05, 06); the two independence mechanisms, which concern different families. The germ-level proofs of the Hurwitz shift and derivative identities in 01, 02 and 06 are not reprinted; they differ from the printed proof only in working with evaluated germs instead of $\mathbb{C}(s)$ -formal series.
- *Statements chosen.* For forced poles, 06's statement about arbitrary functions as part (a), with 03's local form as part (b). For stability, the class $\mathcal{D}_J$ contains all three source classes, with 04's one- η converse. On Platt and Trudgian, the published simplicity statement is cited, with 06's abstention kept as a non-claim. Journal versions are cited where they exist.
- *Results added.* Lemma 4.6, Proposition 5.10, the strip-wide form of Theorem 17.1, Propositions 11.6 and 13.1 (general form), Proposition 13.3 and Corollary 13.4, the Gauss failure in Section 11.5(2), and Theorem 7.9 as one theorem. After the post-merge audit: clause (e) of Theorem 17.1 and the constant-term step in the proof of Proposition 11.1. Further [merge] steps complete proofs inside Lemma 9.3, Lemma 10.10 and Question 23.1.
- *A correction to the reconciliation plan.* The plan stated the strip defect as an obstruction at every $\sigma \in (0, 1)$. The proof shows two exceptional abscissae $\sigma_{\pm}(T)$ where the modulus test is silent, and the theorem is stated with them. Clause (e) shows that the Stirling Gamma $E \circ L_{\text{St}}$ itself fails reflection at every point of the strip, so the exceptions concern only a general prescription with Stirling moduli.

A.4 Errors found in the sources and how they were fixed

No source contains a fatal error or a false main statement. Severity codes: G, a proof gap in a true statement; S, a statement needing a hypothesis or qualifier; W, wording; A, attribution or bibliography; F, framing that becomes false in the merged context.

Table 3: Errors and fixes.

No.	Source, location	Sev.	Issue	Fix, and where
1	01 Prop 4.2 and §4	A/F	E_0 is Ehrlich–Kaplan’s Exp and E_λ is <code>e:thm-twisted</code> ; 01 calls E_{fp} “the canonical exponential”	identified and credited; E_{fp} renamed (Sections 2.3 and 5.3)
2	01 remark after Thm 10.1	F	“nothing canonical decided at $-\omega$ ”; “a simultaneous FE would require an additional construction”	with Exp the zero is present; 03 supplies the construction; no FE forces it (Remark 16.8)
3	01 Lem 8.1, existence	G	evaluation of weighted series used as a strongly additive ring homomorphism without a lemma	Lemma 4.6 , used in Lemma 9.3
4	01 proof of Thm 6.2	W	remainder omits $y^3/(6x^2)$ and $-y/(12x^2)$	corrected expansion in Proposition 11.5
5	01 Thm 6.2 “defined iff”	F	reads as “undefined elsewhere” once Exp exists	Proposition 11.6
6	01 Prop 5.1; text after Thm 6.3	W	“ $1/(s-1)$ interpreted separately”; “agreement is exact” on disjoint domains	reworded (after Proposition 12.1 ; after Proposition 11.7)
7	01 bibliography	A	Costin–Ehrlich cited as forthcoming; Gessel published; Ehrlich–Kaplan missing	journal data in the references
8	02 Lem 2.2(b)	G	the “words” argument treats ε^α as monomials	$\text{supp } \varepsilon^\alpha \subseteq \langle S \rangle$ (Lemma 4.4)
9	02 Prop 5.3	W	“ h sufficiently small” is unnecessary	every infinitesimal h (Proposition 8.3)
10	02 Lem 9.1	G	positivity of the tail valuation needs one line	$q_n/q_{n_0} \prec x^{-D}$ (Lemma 10.4)
11	02 eq. 8.6	G	joint summability in $(1/x, \delta)$ asserted	joint evaluation (Theorem 11.2)
12	02 Thm 11.2	S	the functional ℓ depends on x	stated (Example 11.8(b))
13	02 Warn 6.4	W	“zero-freeness confined to $\mathcal{T}_+$ ” against its own Prop 11.3	“for the raw ζ_D ” (Section 13)
14	02 throughout	A/F	$E_0 = \text{Exp}$ not identified; “no canonical choice”	canonical in the integer-part sense, not unique (Section 2.3)
15	02 notation	W	Γ_0 and ℓ collide with the collection	renamed (Table 1)
16	02, 03 bibliographies	A	Pong and Costin–Ehrlich as preprints	journal data; Pong’s Addendum listed
17	03 Lem 10.2	G	works in $F(x)[[t]]$ although the objects live in $F[[x, t]]$	the local ring $R = F(x) \cap F[[x]]$ (Lemma 10.10)
18	03 Lem 10.2 framing	A	presented as self-contained; it is Pong’s Thm 5.5 for $F = \mathbb{C}$	credited; new content identified (after Lemma 10.10)
19	03 §2.3	G	fine derivative of germs argued in outline	<code>b:realization</code> , <code>b:germalgebra</code> (Section 4.2)
20	03 Thm 4.5, phases, ϕ_λ , Z_ϕ	A	repository results not credited	credited (Section 5)

No.	Source, location	Sev.	Issue	Fix, and where
21	03 bibliography	A	Ehrlich–Kaplan as arXiv	JSL 86 (2021) and erratum
22	04 §5.3, §14.1	W	exact and shadow RH “inequivalent in strength”	“formally stronger” (Remark 7.12)
23	04 Prop 8.3; 05 §6.3	S	flat gauges break Gauss, unstated	stated and proved (Section 11.5)
24	04 §9 and domain table	S	omits the extension of the Dirichlet chart under Exp	Proposition 13.1
25	04 proof of Thm 8.1	G	operator regrouping sketched	finite coefficient sums (Theorem 12.4)
26	04 Lem 2.1, Def 2.2	G	transfer from the workspace to <code>No[i]</code> asserted	Lemma 4.3 stated on SC ; <code>a:cor:complexsub</code>
27	04 Thm 12.1(ii)	S	the nonreal zero is unconditional	hypotheses clause by clause (Theorem 22.2)
28	04 §11.4	A	AF’s headline is 2/3; 0.6725 is their Montgomery–Taylor-window variant; the closed form C_0 (the Montgomery–Taylor constant) is stated in Lamzouri’s Theorem 1.1	Section 21.4
29	04 bibliography; 04 §7	A/W	Guth–Maynard and MTY as preprints; “divergent-at-most- ordinary-points”	journal data; phrase dropped
30	05 §6.2	W	“usually leaves”	“always leaves” (Proposition 11.5)
31	05 Cor 8.3	W	ζ'_D used before definition	defined first (Proposition 8.3)
32	05 Thm 9.1	S	“another such function” fails if F vanishes on U	qualified (Theorem 19.1(b))
33	05 Thm 5.3	G	purely imaginary displacement coefficients unproved	the involution $d^* = -\bar{d}$ (Theorem 7.7)
34	05 Thm 6.2, eq. 6.12	S	the Gamma-ratio gloss holds only off the negative real direction	Remark 11.4
35	05 Prop 3.1	W	conjugation stated without $\bar{f}$	Proposition 6.2
36	05 §6 versus §8	W	“right infinite” for two sets	$\mathcal{S}_\infty$ and $\mathcal{T}_+$ (Section 2.4)
37	05 README	W	“no LaTeX warnings”; its build has one duplicate-destination warning	disclosed here; this build uses <code>pageanchor=false</code> around the title page
38	05	A	Exp and repository labels not mentioned; Thm 4.1 duplicates <code>c:p4:liftzeros</code> ; MTY cited for the counting formula, which MTY only quotes	credited (Sections 1.5 and 21.3 and Theorem 6.3)
39	06 Rem 7.4, last sentence	G	“an infinite period does not imply a resonance” unproved	proved (Proposition 5.10(c))
40	06 Thm 7.3; Rem 7.4	W	the theorem is stated for Exp and extended to resonant phases only in a remark; which phases have resonances is left open	general statement with proof (Theorem 18.2); existence and failure of resonances (Proposition 5.10)
41	06 Thm 7.1	A	cites the trigonometry report for infinite periods but not <code>trigonometry:thm:</code> <code>infiniteperiods</code> ; the explicit pole requirement is not stated	label cited; explicit pole requirement (Theorem 14.1)
42	06 §5.3	W	“a genuine new domain result”	“a result on a new domain; no priority” (after Theorem 8.5)

No.	Source, location	Sev.	Issue	Fix, and where
43	06 proofs of Thm 9.2, 11.2	G	real constants not in the language	constants for all reals (Section 20.1)
44	06 Thm 11.2, second paragraph	W	canonical-lift clause reads as unconditional	“under RH and simplicity” (Theorem 22.2(i))
45	06 §7.3	W	coherent continuation called “the” repository notion	it extends <code>c:p4:def-HU</code> (Definition 18.1)
46	06 bibliography	A	Guth–Maynard without status	<i>Ann. of Math.</i> 203 (2026)
47	06 §10.4	W	one-sided count against Guth–Maynard’s two-sided	05’s sentence added (Section 21.5)
48	06 Prop 6.1	S	$\Gamma_{E,\text{St}}$ satisfies the recurrence but not reflection on $\mathcal{V}$, unstated; by Theorem 17.1(e) it fails at every point of $\mathcal{V}$, for every E extending E_{fp} , including $\sigma_{\pm}(T)$	Theorem 17.1(e) and the remark after it
49	03, 06 (external)	–	Costin–Ehrlich eqs. (82)–(83) print $B_{2n}n!/((2n)!x^{n+1})$ in the extracted text	not used; only the framework is cited
50	01, 02, 04, 06 bibliographies	A	Rodgers–Tao, MTY, Platt–Trudgian, Costin–Ehrlich partly as preprints	journal data in the references
51	06 §6.1 (a doubt raised in the review of 06)	–	L_{St} prescribed at points $\varepsilon + iT$	stated as a prescription only (Section 11)
52	03 (framing, §14.1)	–	the reflection obstruction is listed as a contribution; forced poles follow at once from the existence of infinite periods, and tube uniqueness from the definition of g_{χ} on $\mathcal{T}_{-}$	said so (after Theorem 14.1 ; Part IV ; after Proposition 15.3)
53	05 §3.1; 06 after Def 3.1	W	a nonzero infinitesimal displacement from a pole has a value “usually” of infinite modulus	“always”, since the value is $g(0)\varepsilon^m(1+\eta)$ with $m \leq -1$ (after Definition 6.1)

The sources’ statements about the collection were checked against the current tree and are accurate at their pins ([Section 1.5](#)). Item 53 was found by the post-merge audit below.

A.5 Post-merge audit corrections

After the merge, three independent read-only audits checked the report: one for fidelity to the six sources, one for non-claims, notation and citations, and one for mathematical correctness. They found no false theorem, no counterexample to a [merge] result and no lost source result. Their findings were applied as follows.

- *Mathematics.* Clause (e) of [Theorem 17.1](#): for every E extending E_{fp} , $E \circ L_{\text{St}}$ fails reflection at every point of $\mathcal{V}$, including $\sigma_{\pm}(T)$; the statements that “nothing is claimed” there are restricted to a general prescription with Stirling moduli. The relation between exact and shadow stability is stated correctly ([Remark 7.12](#)). The source classes are subclasses of $\mathcal{D}_J$ but not nested. The constant term in the proof of Gauss in [Proposition 11.1](#) is computed.

The scope of the resonance theorem is narrowed to phases with a resonance; the phase of [Proposition 5.10\(c\)](#) is built with choice, not explicitly; its character values are described exactly; the intersection describing resonances runs over $n \geq 2$. The comparison of the independence theorems ([Section 10.4](#)) is corrected. “Four criteria” became the criteria of 04, 05 and 06, and the arccosine in [Theorem 17.1\(c\)](#) is defined.

- *Fidelity to the sources.* Restored in their source generality: 04’s Proposition 6.1 and Corollary 6.2 for every real entire f ([Proposition 7.5](#), [Remark 7.11](#)); 05’s Theorem 9.1 for every positive infinite T ([Theorem 19.1\(b\)](#)); 02’s Warning 10.1 on all of $\mathcal{S}_A$ ([Warning 12.2](#)); 02’s Lemma 9.4 with all its conditions and proof steps ([Lemma 10.6](#)). Printed: 05’s baseline recurrence, 01’s multi-index formula and the derivation of its headline terms, 01’s third research direction, the converse clause of 02’s Remark 2.3, 05’s pair-correlation caveat and 01’s reworded Gamma gluing sentence (item 6).
- *Credits and provenance.* 06 cites the trigonometry report, and its Remark 7.4 anticipates the extension of its Theorem 7.3. The proof routes of the Stirling identities, the two parts of [Theorem 14.1](#), the values in [Example 15.4](#) not printed by 03, 05’s gloss of $R_{1/2,0}$, the Montgomery–Taylor constant, and further credits to 02, 04 and 06 are attributed correctly. The dates, authorship and reading lists of the sources are stated as they are. A dropped second route is disclosed. References to internal review notes, an unused phantom label, a quotation that no source wrote and two paraphrases printed as quotations were removed or replaced by verbatim quotations. Two collection labels were corrected (`found:eq:omnific`; the holomorphic scope of `c:p4:def-lift`).
- *Pointers.* Equation numbers taken from review notes were checked against the source PDFs and corrected (02’s (8.6); 05’s (6.4), (6.11) and (6.12); 03’s (9.3)–(9.7); 04’s (4.1)–(4.11) and (8.8)). Several rows of [Appendix A.2](#) were corrected or completed, and rows 6, 11, 17, 28, 34, 40, 41, 46, 48, 51 and 52 of [Table 3](#) were corrected.
- *Notation.* Symbols reserved in [Section 2](#) now keep one meaning; the local objects that had reused them were renamed, as listed in the last row of [Table 1](#). In particular κ_0 is the line proportion, the de Bruijn–Newman constant is Λ_{DN} , and [Theorem 7.14](#) and [Corollary 21.3](#) use their own constants c , c_* .
- *Non-claims and caveats.* Restored at the point of use and in [Section 24](#): the two exceptions and clause (e) of the strip theorem at headline level; that tube reflection and the functional equation are definitions on $\mathcal{T}_-$; that forced poles lie on the real axis; that [Corollary 21.2](#) uses the published Platt–Trudgian simplicity statement; that only two clauses of [Theorem 22.2](#) are conditional; that the Pong Addendum was not checked; and the dropped non-claims of 02, 03, 04 and 05.

The corrected statement of [Theorem 17.1\(e\)](#) was checked numerically for this revision ([Appendix B](#)).

B Verification suites and checks

The six delivered suites are shipped byte-identical in `code/`, with their recorded outputs in `data/`. They check finite polynomial, rational and truncated-series identities, and in 01 also ordinary-complex numerics. They do not check strong summability, class quantification, independence of infinite families, continuation, RH, or any cited paper.

Suite	Checks	Content	Output behaviour
01	61	inverse sector (37 multisets, 23 support values, the V equation, log formula, 9 table entries, 30 integer-sector identities); Hurwitz and Stirling coefficients; 5 branches at $u = 10^{-8}$, 100 digits	rewrites verification-results.json and inverse-coefficients.csv (unprefixed) next to itself
02	189	Stirling recurrence and Gauss; finite shift; Hurwitz difference; Lerch coefficients; Bernoulli values; Möbius and Euler-log coefficients; Jacobian samples	rewrites results.json next to itself
03	1670	Bernoulli translation and distribution; Gauss coefficients; Hurwitz identities and values; log-Gamma bridge; arithmetic identities to $n \leq 256$; a Jacobian block; sine products	default verification-rerun.json in the working directory; refuses an existing path
04	350	Stirling shift defect; Lerch coefficients; Bernoulli values; Möbius and Euler-log to 128; root motion; splitting; heat polynomials	-output is mandatory
05	296	Stirling coefficients; recurrence defect; ratio cocycle; $B_{\text{odd}}(\frac{1}{2}) = 0$; shifted real parts; convolution identities; Ingham checks	default verification-rerun.json in the working directory
06	221	Stirling shift; Hurwitz shift and values; Hurwitz–Stirling bridge; root deformation; Dirichlet identities; quartet symmetry	prints only, unless -output is given

All six were rerun on a copy of this directory for the merge (Python 3.14.4, SymPy 1.14.0, mpmath 1.3.0), and all passed with the recorded counts. 01's and 02's rewritten files agreed with the shipped records except for the recorded interpreter version (recorded runs: Python 3.13.5). The two build scripts, **02-scale-separation-build.sh** and **06-phases-and-rh-build.sh**, were written for the source manuscripts, which are not shipped; the README says how to build this report instead. Some checks are near-tautological, for example the Lerch coefficients through $(d/ds)(s)_m|_0 = (m-1)!$ and the von Mangoldt convolution.

For the reconciliation, a separate suite of 29 checks passed. It checks the strip identity of [Theorem 17.1\(a\)](#) symbolically through T^{-14} and the flat defect numerically. It also checks the value of 01's left zeta at $-\omega$ against the formula for $\sin \lambda$, the two Hurwitz distribution forms, the Lerch identity, 03's (5.13) and 05's cocycle leading term, the literature constants (C_0 , $(1+C_0)/2$, the crossover $7/10$, $A(3/4) = 5/9$, $0.417293 > 5/12 > 0.407511$), the trivial-zero slopes, and $\xi(0) = \xi(1) = \frac{1}{2}$. For this report a further 22 checks passed. They check the Bernoulli reflection behind [Theorem 17.1\(a\)](#) (to $n = 29$), the Stirling sum at $T = 200$ to 10^{-40} , the two exceptional abscissae, and the agreement of 04's and 06's second-order root motion. They also check the J-symmetry of the Taylor coefficients of ξ at its first zero, the decreasing exponents of [Proposition 13.3](#), the cocycle identities, and the three Hurwitz normalizations. For clause (e) of [Theorem 17.1](#), added after the post-merge audit ([Appendix A.5](#)), a further check passed. It confirms symbolically the cancellation of the inverse powers through $n = 24$ and the elementary sum of (a). With $P = \exp(\log 2\pi - \pi T + i\pi(\sigma - \frac{1}{2}))$, the common value of $E(L_{\text{St}}(z) + L_{\text{St}}(1-z))$, it confirms $P \sin(\pi z)/\pi = 1 - e^{2\pi iz}$ to 10^{-50} at $T \in \{0.7, 2, 3, 7, 15\}$ and eight abscissae including $\sigma_{\pm}(T)$, where the factor has modulus 1 and argument $\pm e^{-2\pi T}(1+\varepsilon)$. It also checks the truncated Stirling sum at $T = 30$ against (a) to $3 \cdot 10^{-83}$, and $P/(\Gamma(z)\Gamma(1-z)) = 1 - e^{2\pi iz}$ with the

classical Gamma. These suites are corroboration only. They are not shipped and are not proofs.

C Notation index

Symbol	Meaning
$\mathbf{SC} = \mathbf{No}[i]$	surcomplex field; class-sized, with set-sized supports in every calculation
$\mathcal{O}, \mathbf{m}; \text{st}$	finite elements and infinitesimals; standard part on $\mathcal{O}$
$\text{pp}, \text{fin}; \Pi; \mathbf{Oz}$	purely infinite and finite parts (fin keeps infinitesimals); purely infinite reals; omnific integers $\Pi \oplus \mathbb{Z}$
$U^\#, c + \mathbf{m}, b + \mathcal{O}$	halo, monad, galaxy
$\sum^H, \prod^H, \sum^{\text{coef}}$	strong sum, strong product, coefficientwise sum
$\text{Exp}_\mathbf{m}, \text{Log}_\mathbf{m}; \text{cis}; E_{\text{fp}}$	small exponential and logarithm; finite angle; finite-phase exponential
$\Psi_\chi, E_\chi, \text{Sin}_\chi, \text{Cos}_\chi$	phase from a character $\chi : \Pi \rightarrow \mathbb{T}(\mathbf{No})$ and its functions
Exp, Φ	canonical (Ehrlich–Kaplan) exponential and phase, $\chi = 1$
$E_\lambda, E_{\lambda, M}, E_\theta$	explicit phase families (Proposition 5.5)
$\mathcal{I}(\Psi)$	normalized period group $\{x : \Psi(2\pi x) = 1\}$
resonance	$T \neq 0$ with $\Psi(T \log n) = 1$ for all ordinary n
$f^\#$	canonical Taylor–Laurent lift
ξ, Ξ, H_t	completed zeta; $\Xi(w) = \xi(\frac{1}{2} + iw)$; heat family, $H_0(z) = \Xi(z/2)/8$
$\Phi_{\text{RT}}, \Lambda_{\text{DN}}$	Rodgers–Tao kernel of H_t ; de Bruijn–Newman constant
μ, Λ	Möbius and von Mangoldt functions
H_m	harmonic numbers $\sum_{j \leq m} 1/j$
κ_0, κ_s	lim inf proportions of critical-line zeros, and of simple ones
$K_\Gamma, \mathcal{A}, \mathfrak{A}, \mathfrak{M}^*$	set-sized Hahn workspace; arithmetic functions; first-order structure and its elementary extension
$\mathcal{D}_J, \mathcal{S}_{\text{sym}}$	J-symmetric formal deformations; entire h with both symmetries
$\mathcal{T}_+, \mathcal{T}_-, \mathcal{T}_h$	positive and negative infinite real part with finite imaginary part; horizontal tube
$\mathcal{W}, \mathcal{S}_\infty, \Omega_\infty, \mathcal{V}$	wing $\text{Re } s > \mathbb{R}$; Stirling half-plane; finite-phase locus; vertical strip of infinite height
$\mathcal{D}_s, \zeta_D$	Dirichlet evaluation and Dirichlet zeta on $\mathcal{T}_+$
$L_{\text{St}}, S_{\text{St}}, \Gamma_{\text{St}}, \psi_{\text{St}}$	Stirling logarithm, its inverse-power part, real Stirling Gamma, digamma
$\Gamma_\chi, \Gamma_{\text{fp}}, \Gamma_{E, \text{St}}$	phase Gamma; finite-phase Gamma on Ω_∞ ; 06’s Gamma for any homomorphic E
$Q_a, Q_{a,b}, R_{a,b}$	translation remainder, cocycle, exponentiated cocycle
$\zeta_\chi, \xi_\chi, \mathcal{X}_\chi$	zeta and ξ on $\mathcal{T}_h$; reflection factor
$\zeta_\chi^{\mathcal{W}}$	wing sum
$\zeta_\chi, \hat{\zeta}_\chi$	
$\zeta_H, \hat{\zeta}_H, \mathcal{S}_A$	Hurwitz zeta at infinite parameter, normalized form, phase-free parameter set
$*f, *N$	transported function, internal count

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